

Surcomplex Analysis

Germes, coherent Hahn sections, and infinitesimal zero geometry
a theory of holomorphic functions on $\mathbb{K} = \mathbf{No}[i]$

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1 Foundations: the field $\mathbb{K} = \mathbf{No}[i]$

All nine source manuscripts begin by fixing the same field, the same monomial scale, the same valuation and the same class-size discipline. This section states that material once. Nothing in it is new, and nothing in it depends on any analytic notion; every later part is entitled to use it without further comment.

1.1 Class-size discipline

We work in NBG with global choice. The symbols $\mathbf{No}$ and $\mathbb{K}$ denote proper classes. Any specified set of coefficient data can be placed in a set-sized Hahn field, but assertions quantifying over *all* surreal radii or scales retain their ambient class meaning; replacing that ambient field by a fixed working field changes those assertions. The following standing rule is a hypothesis of the whole theory and is never relaxed.

Remark 1.1. In any single construction, every support, every coefficient family, every polynomial and every formal series is *set*-sized. A notation such as $\mathbb{K}[[X]]$ denotes a class of set-indexed coefficient sequences, not a set of all of them. No genuinely proper-class family of nonzero terms is ever summed. Every integer indexing a power series, a derivative or a multiplicity is an *ordinary finite* integer, and every polynomial has ordinary finite degree with a set of coefficients. The rule is what makes the Hahn machinery of Section 3 applicable at all: Neumann [3]’s lemma is a statement about well-ordered *sets*, and the valuation constructed below takes values in a proper class. Assertions about that valuation are assertions about explicitly displayed normal forms; they are not assertions about a class-sized object in the codomain position of a set-valued function.

1.2 Algebra, conjugation and the modulus

The class $\mathbf{No}$ of surreal numbers is a real closed ordered field in the class-sized sense. Adjoining a square root of -1 gives

$$\mathbb{K} = \mathbf{No}[i] = \{x + iy : x, y \in \mathbf{No}\}, \quad i^2 = -1,$$

and the usual algebraic proof that a real closed field acquires an algebraically closed quadratic extension applies to each individual (finite-degree, set-coefficient) polynomial. Hence $\mathbb{K}$ is algebraically closed. Conjugation and the modulus are

$$\overline{x + iy} = x - iy, \quad |x + iy| = (x^2 + y^2)^{1/2} \in \mathbf{No}_{\geq 0},$$

the square root being supplied by real closedness.

Proposition 1.2 (Modulus identities over an ordered field). *For all $z, w \in \mathbb{K}$,*

$$|zw| = |z| |w|, \quad |z + w| \leq |z| + |w|, \quad |\operatorname{Re} z|, |\operatorname{Im} z| \leq |z|.$$

Every nonzero z is invertible, with $z^{-1} = \bar{z}/|z|^2$.

Proof. Multiplicativity follows from $z\bar{z} \cdot w\bar{w} = (zw)\overline{(zw)}$. For the triangle inequality, write $z = x + iy$, $w = u + iv$ and use the Lagrange identity

$$(xu + yv)^2 + (xv - yu)^2 = (x^2 + y^2)(u^2 + v^2),$$

valid in any commutative ring. Since $(xv - yu)^2 \geq 0$ in the ordered field $\mathbf{No}$, this gives the Cauchy–Schwarz inequality $(xu + yv)^2 \leq (x^2 + y^2)(u^2 + v^2)$, hence $xu + yv \leq |z||w|$. Adding $|z|^2 + |w|^2$ to twice this inequality and taking the positive square root of both sides — legal because squaring is order preserving on $\mathbf{No}_{\geq 0}$ — yields $|z + w| \leq |z| + |w|$. The coordinate bounds follow from $x^2 \leq x^2 + y^2$, and the inverse formula from $z\bar{z} = |z|^2$. $\square$

No completeness, no limit and no archimedean property is used; only the order and the existence of positive square roots. This is why the estimate survives verbatim over a proper-class field.

1.3 Complex normal forms and the monomial scale

Combining the Conway [7] normal forms of the real and imaginary parts, and using that a union of two well-ordered subsets of a linear order is well ordered, every surcomplex number has a unique representation

$$z = \sum_{\gamma \in S} c_\gamma t^\gamma, \quad c_\gamma \in \mathbb{C} \setminus \{0\}, \quad t^\gamma := \omega^{-\gamma}, \quad (1.1)$$

with $S = \text{supp}(z) \subset \mathbf{No}$ a well-ordered *set*. Zero has empty support. Conversely every such set-supported expression is a surcomplex number.

Remark 1.3 (What t^γ is, and what it is not). The notation $t^\gamma = \omega^{-\gamma}$ refers to *Conway's monomial map* $\gamma \mapsto \omega^{-\gamma}$ from the additive ordered group $\mathbf{No}$ onto the multiplicative group of monomials. It is *not* the Gonshor [1] surreal exponential evaluated at $-\gamma \log \omega$, and the two must never be silently identified for arbitrary surreal exponents. In particular the law

$$t^{\gamma+\delta} = t^\gamma t^\delta$$

is a normal-form *convention* about the monomial map, not a theorem about surreal exponentiation. This warning is attached here once and is in force throughout the article.

1.4 The natural valuation, finite elements, standard part

For $z \neq 0$ with normal form (1.1) put

$$v(z) = \min \text{supp}(z), \quad \text{lc}(z) = c_{v(z)} \in \mathbb{C} \setminus \{0\}, \quad v(0) = +\infty.$$

Then for all z, w

$$v(zw) = v(z) + v(w), \quad v(z + w) \geq \min\{v(z), v(w)\}, \quad v(|z|) = v(z). \quad (1.2)$$

The last identity is the useful one and deserves its reason: if $z = t^\gamma(c + \eta)$ with $c \in \mathbb{C}^*$ and $v(\eta) > 0$, then $z\bar{z} = t^{2\gamma}(|c|^2 + \eta_0)$ with $v(\eta_0) > 0$, and the positive square root of a finite element with nonzero standard part has the same shape; so

$$|z| = t^\gamma(|c| + \eta_1), \quad v(\eta_1) > 0. \quad (1.3)$$

Thus v is an additive valuation with a *proper-class* value group. By Rule 1.1 this is harmless: each displayed statement quantifies over explicitly given normal forms.

Set

$$\begin{aligned} \mathcal{O} &= \{z \in \mathbb{K} : v(z) \geq 0\} = \{z \in \mathbb{K} : |z| < n \text{ for some } n \in \mathbb{N}\}, \\ \mathfrak{m} &= \{z \in \mathbb{K} : v(z) > 0\} = \{z \in \mathbb{K} : |z| < 1/n \text{ for every } n \geq 1\}. \end{aligned}$$

$\mathcal{O}$ is the valuation ring of finite elements and $\mathfrak{m}$ its maximal ideal of infinitesimals. Extraction of the coefficient at exponent zero is a surjective ring homomorphism

$$\text{st} : \mathcal{O} \longrightarrow \mathbb{C}, \quad \mathcal{O}/\mathfrak{m} \cong \mathbb{C}, \quad (1.4)$$

so every finite surcomplex number is uniquely $z = a + \varepsilon$ with $a = \text{st}(z) \in \mathbb{C}$ and $\varepsilon \in \mathfrak{m}$. We call $a + \mathfrak{m}$ the *monad* of a , and more generally $b + r\mathfrak{m}$ a *scaled monad* for $b \in \mathbb{K}$ and $r \in \mathbb{K}^*$. Note that $c + \mathfrak{m}$ makes sense for every $c \in \mathbb{K}$, even when c has no standard part.

Lemma 1.4 (Crude valuation bound). *If $v(z) \geq \beta$ then $|z| < t^{\beta-1}$.*

Proof. $t^{-\beta}z$ is finite, hence of absolute value less than the positive infinite number $\omega = t^{-1}$. Multiply by $t^\beta > 0$. $\square$

The factor ω is deliberately generous: it makes the bound independent of any ordinary real size of a leading coefficient, which is exactly what is needed when the leading coefficient function is only known to be holomorphic. The bound is used again in the Liouville hierarchy of Section 11, where it is the reason a single surreal bound on the finite halo is vacuous.

Definition 1.5 (Halo over an ordinary domain). For an ordinary open $U \subseteq \mathbb{C}$ its *halo* is

$$U^\# = \text{st}^{-1}(U) = \{a + \varepsilon : a \in U, \varepsilon \in \mathfrak{m}\} \subseteq \mathcal{O}.$$

For $c \in \mathbb{K}$ and $r \in \mathbb{K}^*$, $c + rU^\#$ is a *scale chart*, with normalized coordinate $w = (z - c)/r$.

Two warnings belong here, at the point of definition, and are not repeated later.

Remark 1.6 (The halo is not a surcomplex ball). Writing $\mathbb{D}$ for the ordinary open unit disk, $\mathbb{D}^\#$ is *not* the surcomplex ball $\{z \in \mathbb{K} : |z| < 1\}$. The ball contains $1 - t$, whose standard part 1 lies on the ordinary boundary; the halo omits the entire infinitesimal layer sitting just inside every standard boundary point. Every Schwarz-type statement in Section 11 is a statement about $\mathbb{D}^\#$ and not about the ball, and the counterexample $(1 + t)z$ there turns on precisely the omitted layer.

Remark 1.7 (Puncturing removes a monad). $(U \setminus \{0\})^\# = U^\# \setminus (0 + \mathfrak{m})$ deletes a whole monad, not a point. The class $U^\# \setminus \{0\}$ is a different object. Removable-singularity statements in Section 11 concern the former; keeping the two puncturing operations distinct is what prevents a false analogy with the classical punctured disk.

1.5 Set-sized working fields $\mathbb{K}_\Gamma$

Occasionally a statement must locate objects inside a set-sized subfield — for instance to say that the roots produced by preparation lie in an algebraically closed *set*, or that a zero set is countable. The following device, which costs nothing, is used for that and only for that.

For a set-sized divisible ordered subgroup $\Gamma \subseteq \mathbf{No}$ put

$$\mathbb{K}_\Gamma = \mathbb{C}((t^\Gamma)), \quad R_\Gamma = \mathbb{R}((t^\Gamma)),$$

the Hahn fields of set-supported series with exponents in Γ . The normal-form embeddings place them inside $\mathbb{K}$ and $\mathbf{No}$. By the Hahn-field theorem, $\mathbb{K}_\Gamma$ is algebraically closed [39, Corollary 4, p. 10 of the author PDF]. Since $\mathbb{K}_\Gamma = R_\Gamma[i]$ and R_Γ is ordered, R_Γ is real closed; in particular every root of a polynomial over $\mathbb{K}_\Gamma$ already lies in $\mathbb{K}_\Gamma$.

Proposition 1.8 (Containment of set-sized data). *Every set of surcomplex numbers is contained in $\mathbb{K}_\Gamma$ for some set-sized divisible ordered subgroup $\Gamma \subseteq \mathbf{No}$.*

Proof. Take the union of the supports of the members of the set; it is a set. Pass to the divisible additive subgroup of $\mathbf{No}$ that it generates; that is again a set, being a countable union of sets. Every original normal form has support inside it. $\square$

Proposition 1.8 contains each specified set of data and the set of algebraic outputs generated from it. It does not replace the full ambient class in statements about every fine radius or every scale. In particular the topology induced from $\mathbb{K}$ on $\mathbb{K}_\Gamma$ must be distinguished from the intrinsic valuation topology of $\mathbb{K}_\Gamma$ (Remark 4.7).

2 Conventions: one symbol per concept

This article is assembled from nine independent developments of the same subject. Six of its load-bearing notions are genuinely inequivalent across those developments, in the strong sense that a theorem proved for one is *false* for another and the sources exhibit the counterexamples. The present section fixes one symbol per concept and, more importantly, records which concepts must be kept apart. It proves nothing; each pointer names the part where the corresponding theory is developed. The historical reconciliation in `MERGE_NOTES.md` predates the corrections recorded in the maintained `README.md`; use the statements below for their present mathematical scope.

2.1 The notation table

Table 1: Fixed notation. A symbol appearing here has this meaning throughout and no other.

Concept	Symbol	Remark
The surcomplex field	$\mathbb{K} = \mathbf{No}[i]$	The letter S never denotes the field; S is a support.
Well-ordered set support	S ; $\text{supp}(z)$, $\text{supp}(F)$	Of a normal form or of a coefficient family.
Conway monomial scale	$t^\gamma = \omega^{-\gamma}$, $t = \omega^{-1}$	Monomial map, not Gonshor exponentiation; see Remark 1.3.
Valuation ring; its maximal ideal	$\mathcal{O}$; $\mathfrak{m}$	$\mathcal{O} = \{v \geq 0\}$, $\mathfrak{m} = \{v > 0\}$.
Ordinary holomorphic functions	$\text{Hol}(U)$	Written Hol , never $\mathcal{O}$, which is reserved for the valuation ring.
Halo over an ordinary domain	$U^\# = \text{st}^{-1}(U)$	Definition 1.5; not a surcomplex ball.
Evaluation of a coherent section; canonical lift	$F^\#$; $f^\#$	One symbol, because a lift <i>is</i> a section supported at exponent 0.
Coherent algebra; positive-support part; meromorphic coefficients	$\mathcal{H}(U)$; $\mathcal{H}_{>0}(U)$; $\mathcal{M}(U)$	Section 7 and Section 10.
Family valuation; leading coefficient <i>function</i>	$v_{\mathbf{H}}(F)$; $f_{v_{\mathbf{H}}(F)}$	Not the pointwise valuation or leading coefficient of a <i>value</i> .
Derivative; the operator itself	F' ; D_z	See Remark 2.7.
Neumann monoid of a positive well-ordered set	$\langle S \rangle$	Definition 3.3.
Coefficientwise contour integral	$\int^{\mathbf{H}}$, $\oint^{\mathbf{H}}$	Superscript permanent: not a limit of Riemann sums.
Formal residue	$\text{Res}_{X=0}$	In $\mathbb{K}((X))$; Section 6 and Section 10.
Cluster residue at an <i>ordinary</i> centre a	$\text{Res}_a^{\mathbf{H}}$	Section 10.
Actual local residue at a surcomplex point b	Res_b	Section 10. Never conflate with $\text{Res}_a^{\mathbf{H}}$.
Standard part; finite decomposition	st ; $z = a + \varepsilon$	$a = \text{st}(z) \in \mathbb{C}$, $\varepsilon \in \mathfrak{m}$.
Canonical exponential; twisted family	Exp ; E_λ , $E_0 = \text{Exp}$	λ an ordinary real parameter; Section 12.
Omnific integers; purely infinite surreals	$\mathbf{Oz} = \mathbf{J} + \mathbb{Z}$; $\mathbf{J}$	$\mathbf{J}$ is a real vector space; Section 12.
Ordinary unit disk; bounded Jordan domain; its boundary; a neighborhood of its closure	$\mathbb{D}$; Ω ; $\Gamma = \partial\Omega$; V	Fixing $\mathbb{D}$ and Ω separately removes the clash with the derivative operator D_z .
Hermite division data	P_0 , $u_0 = f_0/P_0$, T , Q_0 , $d = \deg P_0$	Section 9.

Concept	Symbol	Remark
The topology	fine topology	Definition 4.1.

2.2 Four function classes and their proved inclusions

Definition 2.1 (Names of the classes). The following four adjectives are fixed, and are not synonyms.

- (a) **Germ-analytic** (the *local* class). A function that is, near each point, the Hahn sum of a power series $\sum_{n \geq 0} a_n h^n$ with *arbitrary* coefficients $a_n \in \mathbb{K}$, the family $(a_n h^n)_n$ being required to be Hahn summable: no ordinary domain, no growth condition, no radius of convergence. Developed in Section 6, where the germ algebra at any point is shown to be exactly $\mathbb{K}[[X]]$. The sources give this class six names — microholomorphic, surcomplex analytic germ, Hahn-analytic germ, germ-analytic, fine-analytic, microanalytic — all with the same meaning; *germ-analytic* is the one adopted here.
- (b) **Coherent**, the algebra $\mathcal{H}(U)$. A series $F = \sum_{\gamma \in S} f_\gamma t^\gamma$ with well-ordered set support S and every $f_\gamma \in \text{Hol}(U)$ for *one common connected ordinary domain* U , realized as a function $F^\#$ on the halo $U^\#$. Developed in Section 7.
- (c) **Canonical lift** $f^\#$ of an ordinary $f \in \text{Hol}(U)$: the sub-class of $\mathcal{H}(U)$ supported at exponent 0 alone.
- (d) **Differentiable**, reserved for the bare ε - δ difference-quotient condition over all positive surreal radii, with *no* power-series requirement. Section 5 and Section 6.

The superscript $\#$ is used for (b) and (c) alike, because a lift *is* a section supported at exponent zero; the local class (a) carries no superscript. The inclusions are

$$\{f^\#\} \subsetneq \mathcal{H}(U) \subsetneq \{\text{germ-analytic on } U^\#\} \subseteq \{\text{differentiable on } U^\#\}.$$

The first two inclusions are strict. Strictness of the last inclusion is not established here; see Remark 5.3.

Remark 2.2 (The non-transfer rule). *Every theorem in this article names the class it is proved for, and no theorem is imported across classes. Concretely:*

- (i) The identity theorem, the maximum principle in global form, uniqueness of primitives and the open mapping theorem hold in (b) under the domain hypotheses stated below, and their unrestricted global forms fail in (a). The coefficientwise Liouville theorem additionally requires bounded entire coefficient functions. The witnesses are the locally constant L of Section 5 and its sharper variant $b(z) = 1$ exactly when $v(z)$ is an ordinal.
- (ii) The sharp Schwarz and Schwarz–Pick inequalities hold in (c) and *fail* for general coherent self-maps of the halo, i.e. in (b). The witness is $F = (1+t)z$, which fixes 0, maps $\mathbb{D}^\#$ into itself, and has $|F'(0)| = 1+t > 1$ (Section 11).
- (iii) Results in (a) do not descend to (b): $\sum_{n \geq 0} n! \varepsilon^n$ is a legitimate germ on $\mathfrak{m}$ but is not the restriction of any element of $\mathcal{H}(U)$ on an ordinary neighborhood of 0 having those derivatives, since its exponent-zero coefficient function would need Taylor coefficients $n!$ (Section 7).
- (iv) Class (d) supports *only* the calculus rules — uniqueness of the derivative and the sum, product and chain rules — together with the counterexamples. No theorem deriving a power-series representation from bare differentiability is proved here.

The one bridge that *is* proved, and that must be cited whenever a statement about one derivative is used for the other, is the theorem of Section 7 that for a coherent section the coefficientwise derivative $F' = \sum f'_\gamma t^\gamma$ equals the fine ε - δ derivative of $F^\#$.

2.3 Three inequivalent residues

Definition 2.3 (The three residue symbols). (R1) $\text{Res}_{X=0}$, the **formal** residue: the coefficient of X^{-1} of an element of the Laurent field $\mathbb{K}((X))$, which by definition has a *finite* principal part. Purely algebraic, valid at any surcomplex centre.

(R2) Res_a^H , the **cluster** residue at an *ordinary* centre $a \in \mathbb{C}$: for $A = \sum_s a_s t^s$ with meromorphic coefficient functions, $\text{Res}_a^H A = \sum_s (\text{Res}_a a_s) t^s$.

(R3) Res_b , the **actual** local residue of a meromorphic germ at a genuinely surcomplex point $b \in \mathbb{K}$.

Remark 2.4 (Why three symbols). (R1)/(R3) and (R2) *disagree numerically at the same point*. For $A(\zeta) = 1/(\zeta - \varepsilon)$ with $\varepsilon \in \mathfrak{m}$ nonzero, the cluster residue at the ordinary centre 0 is $\text{Res}_0^H A = 1$, while the local residue of A at the exact point 0 is 0 — A has no pole at 0 at all, its pole sitting at ε . Writing a single symbol Res_a for both would convert a correct statement into a false one. The two bridge theorems that connect them, and the standing example worked out in full, are in Section 10.

2.4 Three inequivalent Liouville hypotheses

Three different boundedness hypotheses circulate under the name “bounded entire function”, and they give three different theorems. They are stated as a hierarchy in Section 11; the discipline imposed here is that no conclusion drawn under one may be restated under another.

(L1) A single *surreal* bound on the finite halo. This hypothesis is **vacuous**: as Section 11 proves from Lemma 1.4, every nonzero $F \in \mathcal{H}(\mathbb{C})$ already satisfies $|F^\#| < t^{v_H(F)-1}$ there, so the hypothesis holds for every such F and selects nothing.

(L2) A single *ordinary real* bound. This does **not** give constancy; it gives an exact characterization (Section 11). The standing counterexample is $F = tz$, which is bounded by 1 in ordinary real terms on the whole finite plane and is not constant.

(L3) *Coefficientwise* bounds, one ordinary real constant per exponent. This is the hypothesis that does force constancy, and whose degree- d version forces a polynomial of degree $\leq d$.

2.5 Two Cauchy estimates that are not in conflict

Two estimates appear, and they look like a clash. They are about different quantities and both are kept.

- The **coefficientwise majorant** $B_r(F) = \sum_s (\max |f_s|) t^s$, with its Cauchy–Schwarz majorant lemma. Its coefficientwise definition is part of its statement: $B_r(F)$ is *not* in general the least upper bound of $|F^\#|$ on a surreal circle, and it is not claimed to be a norm.
- The **pointwise surreal** estimate $|(D_z^n F)(c)| \leq n! M/R^n$, obtained from a single pointwise bound $|F^\#| \leq M$ on an ordinary circle via the positivity lemma for the coefficientwise integral.

The standing warning is that a single coarse surreal bound does *not* control every coefficient function uniformly: for $F = w(1 + it)$ on the unit circle one has the pointwise bound $M = 1 + t^2$, while the coefficientwise majorant is $1 + t > M$. That warning forbids a statement about *coefficients*; it does not forbid the estimate above, which is about the surreal *modulus*, and it must not be used to suppress it. Both live in Section 8.

2.6 Two logarithms

Two distinct functions are called “the” logarithm in the sources, and they differ by exactly $2\pi i$ on an explicit class of points. Both are kept, with distinct names, in Section 12:

- the **global** fine-analytic logarithm $\mathcal{L}$ on all of $\mathbb{K}^\times$, whose branch is chosen from the fine-locally constant standard part of the unit direction — this is the primary object, and it is the one consumed by the contour no-go theorem of Section 5;
- the **principal** logarithm Log on the cut plane $\mathbb{K} \setminus \mathbf{No}_{\leq 0}$, defined by $\text{Log } z = \log_{\mathbf{No}} |z| + i \text{Arg } z$ with $-\pi < \text{Arg } z < \pi$.

At $z = -1 - i\varepsilon$ with $\varepsilon \in \mathbf{m}_{>0}$ the two values differ by $2\pi i$. Neither is a coherent primitive of $1/z$ on $(\mathbb{C}^\times)^\#$.

2.7 The derivative operator

Remark 2.5. D_z differentiates a *function of a variable* and holds every surcomplex coefficient fixed. In particular $D_z(\omega) = 0$, ω being a constant function. It is *not* the Berarducci [9]–Mantova [11] field derivation on surreals viewed as transseries, whose normalization has $\partial\omega = 1$. Section 6 upgrades this notational warning into a theorem: no field derivation on $\mathbb{K}$ fixing $\mathbb{C}$ with $\partial\omega = 1$ can satisfy $\partial(\text{Exp } z) = \text{Exp}(z) \partial z$. A reader importing surreal differential algebra or transseries results would silently change the meaning of every derivative in this article.

3 Hahn summability over the monomial scales

This is the one load-bearing notion on which all nine sources agree exactly, clause for clause. Every infinite operation in the article — every power series, every geometric series, every preparation recursion and every reversion tree — uses this operation over Hahn exponents. Ordinary Taylor expansions and contour integrals of the complex coefficient functions are performed in $\mathbb{C}$ first; their values then enter the Hahn sum. These are distinct operations.

3.1 The definition

Definition 3.1 (Hahn summable family). A *set-indexed* family $(z_j)_{j \in J}$ of normal forms is **Hahn summable** if

- (i) (**well-ordering clause**) $\bigcup_{j \in J} \text{supp}(z_j)$ is well ordered, and
- (ii) (**finiteness clause**) each exponent $\gamma \in \bigcup_j \text{supp}(z_j)$ belongs to $\text{supp}(z_j)$ for only finitely many j .

Its *Hahn sum* is the normal form obtained by performing, at each exponent, the finite complex addition supplied by clause (ii). The same definition, verbatim, applies to families with coefficients in any commutative ring, in particular in a ring $\text{Hol}(U)$ of ordinary holomorphic functions. A Hahn summable family is also called *strongly summable*; the two terms are synonyms and the first is used throughout.

The definition is invariant under reindexing, and a jointly summable family may be regrouped arbitrarily, each regrouping being a finite rearrangement at each exponent. Joint summability of a double family, rather than the bare existence of two iterated sums, is what licenses an interchange of summation; that distinction is used at every interchange below and is never assumed.

Remark 3.2 (The finiteness clause is never dropped). Clause (ii) is not a technicality. Verifying only clause (i) proves nothing: a family whose supports are all $\{0\}$ has well-ordered union of supports and is not summable unless all but finitely many of its members vanish. Every summability verification in this article checks both clauses.

3.2 Neumann's support lemma

Definition 3.3 (Neumann monoid). For a well-ordered set S of strictly positive elements of an ordered abelian group, write

$$\langle S \rangle = \{0\} \cup \bigcup_{n \geq 1} \underbrace{(S + \cdots + S)}_n$$

for the additive monoid it generates.

Lemma 3.4 (Neumann support lemma). *Let G be an ordered abelian group.*

- (a) *If $A, B \subseteq G$ are well ordered, then $A + B$ is well ordered, and every element of $A + B$ has only finitely many representations $a + b$ with $a \in A$, $b \in B$.*
- (b) *If $S \subseteq G_{>0}$ is well ordered, then $\langle S \rangle$ is well ordered, and every element of G is the sum of only finitely many finite nondecreasing words over the alphabet S — counting all word lengths together, not merely finitely many for each fixed length.*

Proof sketch. For (a): an infinite strictly decreasing sequence in $A + B$, or infinitely many pairs with a fixed sum, can be thinned so that the A -coordinates are nondecreasing, which forces a strictly decreasing sequence in B and contradicts well-ordering. For (b), the combinatorial mechanism is the well-quasi-ordering of finite words over a well-ordered positive alphabet: a subsequence embedding of one word into another cannot decrease the sum, since the alphabet is positive. A decreasing sequence of sums would give a sequence of words with no earlier word embedding coefficientwise into a later one, contrary to that well-quasi-ordering. For distinct nondecreasing words, such an embedding strictly increases the sum unless the words are equal: either a letter strictly increases or an extra positive letter is inserted. Thus infinitely many distinct words with a fixed sum are also impossible. This is the classical lemma of Neumann; we import it as a foundational input. All its applications below take place inside the set-sized ordered group generated by the displayed supports, even though the ambient value class is **No**. $\square$

Remark 3.5 (The all-lengths clause is the load-bearing one). Part (a) alone licenses only the product of *two* Hahn series. What licenses a geometric series $\sum_n (-H)^n$, the homogeneous preparation recursion of Section 9, Hahn reversion, and the implicit-function tree of Section 11 is the all-lengths clause of part (b): the number of words of *every* length summing to a given exponent is finite in total. Dropping the clause silently invalidates those four arguments and no others replace it.

3.3 Substituting an arbitrary complex sequence

Corollary 3.6 (No growth condition). *Let $(c_n)_{n \geq 0}$ be an arbitrary sequence of complex numbers and $\varepsilon \in \mathfrak{m}$. Then $(c_n \varepsilon^n)_{n \geq 0}$ is Hahn summable. The same holds for a formal series in finitely many infinitesimal variables. Evaluation is compatible with addition, multiplication, formal differentiation, and composition $A(B(X))$ when $B(0) = 0$.*

Proof. Let $S = \text{supp}(\varepsilon) \subseteq \mathbf{No}_{>0}$; if $\varepsilon = 0$ the claim is trivial. Then $\text{supp}(c_n \varepsilon^n) \subseteq \langle S \rangle$, well ordered by Lemma 3.4(b); a complex coefficient has support $\{0\}$ and does not enlarge it. A fixed exponent $\gamma \in \langle S \rangle$ is the sum of only finitely many finite words over S by the all-lengths clause, hence receives a contribution from only finitely many n . Both clauses of Definition 3.1 hold. The stated identities are verified exponent by exponent, where each verification is a finite computation, and the regroupings they require are exactly the ones Lemma 3.4 permits. For finitely many variables use the union of their positive supports and note that a fixed total degree contributes only finitely many monomials. $\square$

Thus $\sum_{n \geq 0} (n!)^2 \varepsilon^n$ is a perfectly legitimate element of $\mathbb{K}$ for every surcomplex infinitesimal ε , although the associated complex series has radius of convergence zero. No radius-of-convergence formula in terms of the ordinary magnitudes of the coefficients is asserted anywhere in this article, and none is available.

Remark 3.7 (What this corollary does *not* say). Corollary 3.6 is stated for *complex* coefficients. For arbitrary surcomplex coefficients $a_n \in \mathbb{K}$ the family $(a_n \varepsilon^n)_n$ need *not* be summable: the union $\bigcup_n \text{supp}(a_n)$ need not even be well ordered. What is true is that after a single monomial rescaling the family becomes summable, by the separated-degree-block lemma of Section 6. The rescaling is not cosmetic; it changes the chart, and Section 7 states the resulting obstruction beside the positive result.

3.4 The separating examples

The following two examples are the ones on which all later summability discipline rests. They are stated once, here.

Example 3.8 (A convergent complex series that is not a Hahn sum). Regard $\sum_{n \geq 0} 2^{-n}$ as a family of constant surcomplex numbers. Every member has support $\{0\}$, so the well-ordering clause holds trivially while the finiteness clause fails at the single exponent 0: infinitely many members contribute there. The family is therefore *not* Hahn summable. This is not a defect to be repaired. The correct procedure is to compute the ordinary complex sum first, in $\mathbb{C}$, and to use the resulting number 2 as a *single coefficient*. Classical summation in the coefficient field and Hahn summation over scales are different operations, and the article never substitutes one for the other.

Example 3.9 (A Hahn identity whose partial sums do not converge). With $t = \omega^{-1}$, the family $(t^n)_{n \geq 0}$ has supports $\{n\}$, so both clauses hold and

$$\sum_{n \geq 0} t^n = (1 - t)^{-1}. \quad (3.1)$$

This is an identity between normal forms, established exponent by exponent. It is *not* the convergence of the sequence of finite partial sums. Indeed

$$(1 - t)^{-1} - \sum_{n=0}^N t^n = \frac{t^{N+1}}{1 - t},$$

whose valuation is $N + 1$; since $\omega > N + 1$ for every ordinary finite N , this remainder exceeds t^ω . Hence the ball $B((1 - t)^{-1}, t^\omega)$ of the fine topology contains *no* partial sum at all, and (3.1) cannot be read as a limit.

4 Why summability is not convergence

The previous section supplied an infinite operation. This section proves that no topological limit notion can supply one in its place: the natural topology on $\mathbb{K}$ is so fine that set-indexed limits

collapse entirely. Everything here is a statement about the field $\mathbb{K}$; no function class is involved, and nothing proved here is a theorem about germ-analytic, coherent or lifted functions.

4.1 The fine topology

Definition 4.1 (Fine topology). The **fine topology** on $\mathbb{K}$ is generated by the balls

$$B(a, r) = \{z \in \mathbb{K} : |z - a| < r\}, \quad a \in \mathbb{K}, r \in \mathbf{No}_{>0},$$

where the radius ranges over *all* positive surreal numbers. Equivalently it is generated by the valuation balls $\{z : v(z - a) > \alpha\}$, $\alpha \in \mathbf{No}$: indeed $t^{v(r)+1} \prec r$ gives one comparison and a smaller monomial ball the other.

The sources call this the full surreal topology, the neighborhood topology and the all-radii topology; all three name this one topology.

4.2 Uniform discreteness of every set

Theorem 4.2 (Set-sized convergence collapses, with a uniform radius). *Let $A \subseteq \mathbb{K}$ be a set. Then:*

- (i) *A is closed and discrete in the fine topology;*
- (ii) *if A has at least two elements, there is a single $r \in \mathbf{No}_{>0}$ with $|a - b| > r$ for every pair of distinct $a, b \in A$;*
- (iii) *every net indexed by a set which converges in the fine topology is eventually equal to its limit. In particular every convergent ordinary sequence is eventually constant.*

Proof. The mechanism is the surreal cut property (the simplicity theorem), not an infimum and not an order limit. For any set $D \subseteq \mathbf{No}_{>0}$, the cut $\{0 \mid D\}$ is a surreal number r with $0 < r < d$ for every $d \in D$.

For (ii), let $D = \{|a - b| : a, b \in A, a \neq b\}$, a set of positive surreals because A is a set, and take r as above; then $|a - b| > r$ for all distinct $a, b \in A$ simultaneously. Hence $B(a, r) \cap A = \{a\}$ for each $a \in A$, which is discreteness with one radius serving all points.

For closedness in (i), fix $x \notin A$ and apply the same construction to $\{|x - a| : a \in A\}$, obtaining a ball about x missing A entirely.

For (iii), let $(x_j)_{j \in J}$ be a net with J a set and proposed limit x . The values form a set, so $D = \{|x_j - x| : j \in J, x_j \neq x\}$ is a set of positive surreals; choose r below all of it. Convergence forces the net to lie eventually in $B(x, r)$, which contains no value other than x itself. The case $D = \emptyset$ is immediate. $\square$

Clause (ii) is the sharpest available form: the separating radius does not merely depend on the point, it is uniform over the whole set. It is what makes the total disconnectedness of the fine topology quantitative.

4.3 Clopen cosets, total disconnectedness, no paths

Proposition 4.3 (Clopen infinitesimal cosets). *Every coset $c + \mathfrak{m}$ is both open and closed, as are $\mathcal{O}$ and every affine image $b + r\mathcal{O}$, $b + r\mathfrak{m}$ with $r \neq 0$. The fine topology is totally disconnected.*

Proof. A ball of infinitesimal radius about a point of $c + \mathfrak{m}$ stays inside $c + \mathfrak{m}$, so the coset is open; its complement is a union of cosets and is open for the same reason, so it is also closed. For $\mathcal{O}$: a point $z \notin \mathcal{O}$ has $|z| \geq 1$, and the ball $B(z, 1/2)$ misses $\mathcal{O}$; a point of $\mathcal{O}$ has a ball of radius 1 inside $\mathcal{O}$ after translating by its standard part. Affine images of clopen classes under $z \mapsto b + rz$

are clopen because that map is a bijection with continuous inverse. Finally, given $z \neq w$, the class $z + (z - w)\mathbf{m}$ is clopen, contains z , and does not contain w , since $(w - z)/(z - w) = -1 \notin \mathbf{m}$. So no connected subclass contains both. $\square$

Corollary 4.4 (No nonconstant continuous paths). *Every map $\varphi : [0, 1] \rightarrow \mathbb{K}$ from the ordinary real unit interval which is continuous for the fine topology is constant.*

Proof. $[0, 1]$ is a set, so $\varphi([0, 1])$ is a set, hence a closed discrete subspace by Theorem 4.2(i). The continuous image of an ordinary connected space is connected. A connected discrete space is a singleton. $\square$

Remark 4.5 (Consequence for contours). An ordinary parametrized complex circle, viewed inside $\mathbb{K}$, is therefore *not* a nonconstant continuous path for the fine topology. The coefficientwise contour functional $\int^H$ introduced in Section 8 is consequently not a limit of Riemann sums along a fine-continuous path, and its parameterization is nowhere asserted to be fine-continuous. The superscript H is permanent for exactly this reason.

4.4 What the discreteness theorem does not say

Two non-consequences must be recorded, or a later reader will over-apply Theorem 4.2.

Remark 4.6 ($\mathbb{K}$ itself is not discrete). Theorem 4.2 concerns *sets*. It does not make the class $\mathbb{K}$ discrete: every ball $B(a, r)$ contains points other than a — for instance $a + r/2$ — and in fact contains a proper class of them. There is no contradiction, because the class of points of a ball is not a set and the cut construction in the proof does not apply to it.

Remark 4.7 (The induced topology is not a valuation topology). Theorem 4.2 does not identify the fine topology induced on a set-sized Hahn subfield $\mathbb{K}_\Gamma \subseteq \mathbb{K}$ with that subfield's own valuation topology. The valuation topology of $\mathbb{K}_\Gamma$ has balls of radii indexed by Γ ; the fine topology supplies, in addition, radii smaller than t^γ for every $\gamma \in \Gamma$, by Proposition 1.8 and the cut property. The induced topology is discrete; it is strictly finer when $\Gamma \neq \{0\}$. For $\Gamma = \{0\}$ the intrinsic valuation is trivial and its topology is discrete as well. Example 3.9 exhibits the difference concretely: the partial sums of $\sum_n t^n$ converge to $(1 - t)^{-1}$ in the valuation topology of $\mathbb{K}_\mathbb{Q}$ and do not converge to it in the fine topology. A Hahn sum need not give convergence even in an intrinsic valuation topology: in $\mathbb{K}_\mathbb{Q}$ the summable family $(t^{1-1/(n+1)})_{n \geq 1}$ has all its exponents below 1, so its partial-sum remainders never enter the valuation ball $\{v > 1\}$. Thus the topology and a separate convergence argument must always be specified; the existence of a Hahn sum alone does not supply one.

4.5 What the fine topology is used for

The collapse proved above is not a reason to discard the fine topology. It is used, genuinely and essentially, in three places, and in none of them does it appear as a source of limits of sequences.

- (1) **The ε - δ derivative.** A derivative is defined by: $F'(a) = L$ iff for every $\eta \in \mathbf{No}_{>0}$ there is $\delta \in \mathbf{No}_{>0}$ with

$$0 < |h| < \delta \implies \left| \frac{F(a+h) - F(a)}{h} - L \right| < \eta.$$

Both radii range over *all* positive surreals, and h ranges over a proper class of increments; no sequence is chosen. Uniqueness of L follows from Proposition 1.2 alone. Theorem 4.2 says nothing against this definition: it forbids nontrivial set-indexed limits, and this quantifier is not one.

- (2) **Open classes.** The open mapping theorem and the strong maximum modulus principle of Section 11 are statements about actual fine-open classes, at arbitrary surcomplex points; they are not statements about standard parts, and they would be false if weakened to such.
- (3) **Fine-open hypotheses in the identity theorem.** One of the four equivalent hypotheses of the identity theorem in Section 7 is vanishing on a nonempty fine-open subclass. That this is a strong hypothesis, and not a vacuous one, is exactly Remark 4.6: a fine neighborhood of a point contains a proper class of points.

Remark 4.8 (The accumulation hypothesis lives in $\mathbb{C}$). Wherever an accumulation hypothesis appears later — in the identity theorem of Section 7, in the zero geometry of Section 9 — it is an accumulation hypothesis on *standard parts*, in the ordinary complex topology of $\mathbb{C}$. By Theorem 4.2 no nontrivial set-sized sequence accumulates in the fine topology, so no theorem in this article asserts, or may assert, the existence of one.

5 What fails, and the contour no-go theorem

The constructions of Sections 6 and following carry hypotheses that look, on a first reading, like caution: one common ordinary domain for all coefficient functions, a contour integral defined coefficientwise rather than as a limit, a fundamental theorem restricted to coherent primitives. None of them is caution. Each is forced by an explicit counterexample or by a no-go theorem, and this part collects those obstructions before any of the positive theory is built, so that every later hypothesis can be read against the thing it excludes.

5.1 Four function classes, and the rule against transfer

Four classes of functions occur in this article. They are genuinely different, and a theorem proved for one of them is false for another. We fix their names and symbols here and never use one name for another object.

Definition 5.1 (The four classes). Let $U \subseteq \mathbb{C}$ be a nonempty connected ordinary domain.

- (a) A function on a fine-open class is *germ-analytic* at a if on some fine ball about a it is the Hahn sum of a formal series $\sum_{n \geq 0} a_n(z - a)^n$ with *arbitrary* coefficients $a_n \in \mathbb{K}$. There is no growth condition on the coefficients, no ordinary domain, and no relation between the expansions chosen at different points. The theory of this class is Section 6.
- (b) $\mathcal{H}(U)$, the *coherent* algebra, consists of the Hahn series $F = \sum_{\gamma \in S} f_\gamma t^\gamma$ with $S \subseteq \mathbf{No}$ a well-ordered *set* and every $f_\gamma \in \text{Hol}(U)$ an ordinary holomorphic function on the *one* domain U . Its evaluation $F^\#$ lives on the halo $U^\# = \text{st}^{-1}(U)$.
- (c) The *canonical lift* $f^\#$ of an ordinary $f \in \text{Hol}(U)$ is the element of $\mathcal{H}(U)$ supported at the single exponent 0; this is why lifts and evaluations carry the same superscript.
- (d) A function F is *differentiable* at a (01's bare class) if some $L \in \mathbb{K}$ satisfies

$$\forall \eta \in \mathbf{No}^{>0} \exists \delta \in \mathbf{No}^{>0} \forall h \in \mathbb{K} : \quad 0 < |h| < \delta \implies \left| \frac{F(a+h) - F(a)}{h} - L \right| < \eta,$$

all radii ranging over the positive surreals. No power-series representation is required. We write $F'(a) = L$; uniqueness of L is the triangle inequality.

Proposition 5.2 (The inclusions, and what witnesses them). *Restricted to a halo $U^\#$,*

$$\begin{aligned} \{f^\# : f \in \text{Hol}(U)\} &\subsetneq \mathcal{H}(U) \\ &\subsetneq \{\text{germ-analytic functions on } U^\#\} \\ &\subseteq \{\text{differentiable functions on } U^\#\}. \end{aligned}$$

The first inclusion is strict: the constant section $t \in \mathcal{H}(U)$ is not a lift, since its support is $\{1\}$. For the second, choose $a \in U$ and set $q_a(z) = 0$ on $a + \mathfrak{m}$ and $q_a(z) = 1$ on the other monads of $U^\#$. This is locally constant, hence germ-analytic. If it were coherent, its values at ordinary points would force its exponent-zero coefficient to be 0 at a and 1 everywhere else in U , which is not holomorphic. There is also the stronger local obstruction that a germ-analytic function need not extend to any coherent section at all: $\sum_{n \geq 0} n! \varepsilon^n$ is a legitimate germ-analytic function on $\mathfrak{m}$ by Theorem 6.4, but it is not the restriction of any element of $\mathcal{H}(U)$ on an ordinary neighbourhood U of zero having those same derivatives, because such a section's exponent-zero coefficient function would be an ordinary holomorphic function with Taylor coefficients $n!$ at zero, hence of zero radius of convergence. The third inclusion is Theorem 6.9.

Remark 5.3 (Two things this article does not prove). First, strictness of the last inclusion is *not* established here. No surcomplex Goursat theorem is proved: we do not assert that fine differentiability at every point of a fine-open class implies a power-series representation, and we exhibit no function that is differentiable everywhere and germ-analytic nowhere. The inclusion is recorded in the direction in which it is proved, and no positive theorem about the differentiable class beyond the calculus rules — uniqueness of the derivative, the sum, product, quotient and chain rules, all of which follow from finite inequalities in an ordered field with no limiting sequence — is used anywhere in this article. Second, nothing below asserts that an arbitrary assignment of a germ to each point of a halo arises from a coherent section; Proposition 5.2 says the opposite.

Remark 5.4 (The chart changes the function). Proposition 5.2 must be read beside the scale-embedding theorem of Section 7 (Theorem 7.25), which says that *every* formal series over $\mathbb{K}$, including $\sum n! X^n$, becomes a coherent family on all of $\mathbb{C}$ with polynomial coefficient functions after a single monomial rescaling $z = \rho\zeta$, $\rho = t^\lambda$. There is no contradiction: the obstruction above concerns the original chart at zero, and the positive result concerns a rescaled chart. The rescaling changes which function is being represented. A germ has *some* sufficiently small coherent chart; it does not follow, and it is false, that it has a chart of every larger radius. That is the content of the all-scale rigidity theorem of Section 13.

5.2 The locally constant witness

Example 5.5 (Bounded, nonconstant, with zero derivative). Let $\mathcal{O} = \{z \in \mathbb{K} : v(z) \geq 0\}$ be the class of finite elements and put

$$L(z) = \begin{cases} 1, & z \in \mathcal{O}, \\ 0, & z \notin \mathcal{O}. \end{cases}$$

The class $\mathcal{O}$ is fine-open, since a finite perturbation of a finite number is finite; its complement is fine-open, since $|h| < |z|/2$ leaves an infinite z infinite. Hence L is locally constant. It is therefore germ-analytic at every point of $\mathbb{K}$, being locally the Hahn sum of a constant series, and a fortiori differentiable with $L' \equiv 0$ and every positive-order derivative zero.

Proposition 5.6 (Five classical statements fail in the germ-analytic class). *The function L of Example 5.5 is bounded by 1 on all of $\mathbb{K}$, is nonconstant, vanishes on a nonempty fine-open class, has local maxima of its modulus, and is not an open map. Consequently, for the germ-analytic class on all of $\mathbb{K}$, each of the following is false:*

- (i) *the identity theorem (vanishing on a fine-open class does not force vanishing);*
- (ii) *the strong maximum principle;*
- (iii) *Liouville's theorem (boundedness does not force constancy);*
- (iv) *uniqueness of primitives (adding L to a primitive changes nothing about its derivative);*
- (v) *the open mapping theorem.*

Coherent analogues hold with their stated hypotheses: the identity theorem is Theorem 7.18; maximum and open mapping principles use a connected ordinary base; the coefficientwise Liouville theorem requires $U = \mathbb{C}$ and bounded coefficient functions; and coherent primitive uniqueness is in Section 8. In particular, ordinary-real or surreal boundedness alone does not imply constancy even for coherent sections (Section 11).

Proof. All five assertions about L are immediate from local constancy and the clopenness of $\mathcal{O}$. For (iv), note that L is differentiable on $\mathbb{K}$ with derivative 0, so it may be added to any primitive of any function. For (v), the image of $\mathbb{K}$ is the two-point set $\{0, 1\}$, which contains no fine ball. $\square$

The witness is not confined to the split between finite and infinite elements, and in particular it does not need an infinite scale.

Example 5.7 (The failure already occurs on one halo). On $D^\#$, the halo of the ordinary unit disk, put $q(z) = 0$ for $z \in \mathfrak{m}$ and $q(z) = 1$ otherwise. The cosets of $\mathfrak{m}$ are fine-clopen, so q is locally constant, germ-analytic, bounded, and nonconstant on $D^\#$ — although its ordinary base D is connected. Thus coherence, not connectedness of the base, is what the identity theorem of Section 7 consumes. Moreover no $\mathbb{K}$ -linear functional $\int$ on a class of germ-analytic functions can satisfy $\int_a^b H'(z) dz = H(b) - H(a)$ for all fine-locally analytic H and arbitrary endpoints: apply it to q with a and b in different cosets of $\mathfrak{m}$. This is the finite-endpoint shadow of the no-go theorem below.

5.3 A bounded germ-analytic function with no fine limit

Example 5.5 is locally constant in a rather coarse way. The next witness is sharper: it separates points inside a single monad, at arbitrarily small scales, and it kills the unrestricted Riemann removable-singularity theorem.

Example 5.8 (No fine limit at zero). For $z \neq 0$ define

$$b(z) = \begin{cases} 1, & v(z) \text{ is an ordinal,} \\ 0, & \text{otherwise.} \end{cases}$$

The valuation is locally constant away from zero — if $v(h) > v(z)$ then $v(z+h) = v(z)$ — so b is locally constant on $\mathbb{K}^\times$, hence germ-analytic there, with $b' \equiv 0$, and $|b| \leq 1$. Ordinals are cofinal in $\mathbf{No}$, and for every ordinal α both t^α and $t^{\alpha+1/2}$ lie in the ball $|z| < t^\beta$ once $\alpha > \beta$, while $b(t^\alpha) = 1$ and $b(t^{\alpha+1/2}) = 0$. Hence b has no fine limit at zero, although it is bounded and germ-analytic on a full punctured fine neighbourhood.

Proposition 5.9 (No unrestricted removability theorem). *There is no theorem asserting that a germ-analytic function bounded on a punctured fine neighbourhood of a point extends germ-analytically across that point. The witness is Example 5.8.*

The correct removability statements are coefficientwise and belong to the coherent class: a coherent section extends across an ordinary puncture exactly when each ordinary coefficient function is bounded near it (Section 11), and a scalar surreal bound on $F^\#$ is not sufficient there either. Two further points must not be lost. First, the punctured halo $(U \setminus \{c\})^\#$ deletes the *whole monad* $c + \mathfrak{m}$, not the single point c ; the two puncturing operations are different classes and keeping them apart prevents a false analogy. Second, for a *finite-meromorphic* germ — an element of $\mathbb{K}((X))$, with finitely many negative powers — boundedness on a small punctured ball *does* force a removable singularity, by Proposition 6.28. Example 5.8 is not finite-meromorphic at zero, and it is exactly the gap between “germ-analytic on a punctured neighbourhood” and “finite principal part” that it occupies.

5.4 A structured witness: unit modulus, all derivatives zero

The locally constant examples can be dismissed as pathologies of the indicator function. They cannot: the same phenomenon occurs for a quotient of two everywhere-germ-analytic exponential homomorphisms.

Proposition 5.10 (The twisted-exponential witness). *Let $\text{Exp} = E_0$ be the canonical surcomplex exponential and E_λ ($\lambda \in \mathbb{R}$) the twisted family of Section 12 (Theorem 12.4). For $\lambda \notin 2\pi\mathbb{Z}$ the quotient*

$$\Theta_\lambda := E_\lambda/E_0 : \mathbb{K} \longrightarrow \mathbb{K}^\times$$

is a nonconstant function of unit modulus on all of $\mathbb{K}$, germ-analytic everywhere, with every positive-order derivative identically zero. Indeed $\Theta_\lambda(i\omega) = e^{i\lambda} \neq 1 = \Theta_\lambda(0)$.

Proof. Both E_λ and E_0 are homomorphisms $(\mathbb{K}, +) \rightarrow (\mathbb{K}^\times, \cdot)$ with the same modulus law $|E_\lambda(x + iy)| = \exp_{\mathbf{No}}(x)$, so the quotient is a homomorphism into the unit circle of $\mathbb{K}$. It differs from E_0 only through the real coefficient of the monomial ω in the imaginary part; for an *infinitesimal* increment h that coefficient is zero, so $\Theta_\lambda(a+h) = \Theta_\lambda(a)$ for all $h \in \mathfrak{m}$. Hence Θ_λ is constant on every monad — in particular germ-analytic with constant germ, so all positive-order derivatives vanish — while the displayed values show it is globally nonconstant. $\square$

This is the same obstruction as Example 5.5, but now carried by a function with a complete differential and algebraic description. It shows that $E' = E$, $E(0) = 1$, agreement with $\exp_{\mathbb{C}}$ and with the real surreal exponential, and the correct modulus law do not determine a function on $\mathbb{K}$: local analytic data impose no continuation between parts of $\mathbb{K}$ that are infinitely far apart.

5.5 The contour no-go theorem

Section 8 defines the contour functional coefficientwise, $\int_\Gamma^H F dz := \sum_\gamma t^\gamma \int_\Gamma f_\gamma dz$ along an *ordinary* piecewise- C^1 path Γ , and restricts its fundamental theorem to *coherent* primitives. The following theorem is the reason both decisions are forced. Its input is the global fine-analytic logarithm $\mathcal{L}$ of Section 12 (Theorem 12.11): a function defined on *all* of $\mathbb{K}^\times$, germ-analytic at every point, with $\mathcal{L}'(z) = 1/z$, whose branch is chosen from the fine-locally constant standard part of the unit direction $z/|z|$.

Theorem 5.11 (Contour no-go). *Let $\mathcal{C}$ be any class of functions that are germ-analytic on a fine neighbourhood of the ordinary unit circle $\Gamma_1 = \{|z| = 1\} \subset \mathbb{C}$ and that contains $z \mapsto 1/z$ together with G' for every G germ-analytic on all of $\mathbb{K}^\times$. There is no functional*

$$I : \mathcal{C} \longrightarrow \mathbb{K}$$

satisfying both of the following:

- (i) (classical normalization) $I(1/z) = 2\pi i$;
- (ii) (fundamental theorem for germ-analytic primitives) $I(G') = 0$ for every G that is germ-analytic on all of $\mathbb{K}^\times$.

Proof. Take $G = \mathcal{L}$, the global fine-analytic logarithm of Theorem 12.11. It is germ-analytic at every point of $\mathbb{K}^\times$ and $\mathcal{L}'(z) = 1/z$ identically there. Hypothesis (ii) gives $I(1/z) = I(\mathcal{L}') = 0$; hypothesis (i) gives $I(1/z) = 2\pi i \neq 0$. $\square$

The theorem is a genuine obstruction, not a defect of one construction. Three consequences fix the architecture of the rest of the article.

Corollary 5.12 (The fundamental theorem cannot be widened). *The identity $\int_{\Gamma}^H F' dz = F^\#(\Gamma(1)) - F^\#(\Gamma(0))$ of Section 8 holds for $F \in \mathcal{H}(U)$ and cannot be extended to germ-analytic primitives, even ones that are germ-analytic on all of $\mathbb{K}^\times$. The restriction of the fundamental theorem to the coherent category is therefore forced.*

Corollary 5.13 ($\mathcal{L}$ is not a coherent primitive of $1/z$). *$\mathcal{L}$ is not the evaluation of any element of $\mathcal{H}(\mathbb{C}^\times)$ on $(\mathbb{C}^\times)^\#$.*

Proof. If it were, Corollary 5.12 applied to the closed ordinary unit circle would give $\oint_{\Gamma_1}^H dz/z = 0$, whereas the coefficientwise integral of $1/z$ along Γ_1 is its ordinary value $2\pi i$. $\square$

Corollary 5.14 (Uniqueness of primitives is a coherent statement). *Within $\mathcal{H}(U)$ a primitive is unique up to an additive constant of $\mathbb{K}$. No such uniqueness holds in the germ-analytic or the differentiable class: by Example 5.5 a primitive there may be modified by any locally constant function.*

Remark 5.15 (What the no-go does *not* say). Theorem 5.11 does not say that no useful surreal integration theory exists, and the coefficientwise functional $\int^H$ is not proposed as the unique notion of surreal integration; it addresses a different input class from, for instance, integration programs for surreal extensions of real functions. What it says is that on the ordinary unit circle the classical normalization and an unrestricted fundamental theorem are incompatible, so any functional deserving the name must give up one of them, and this article gives up the second in the explicit and minimal way recorded in Corollary 5.12. Two further points survive with it. The local Laurent residue obstruction of Proposition 6.25 is untouched: $\mathcal{L}$ is germ-analytic at every nonzero point but is *not* a finite-meromorphic germ at zero, so it is not a counterexample to the formal statement that a Laurent differential has a primitive in $\mathbb{K}((X))$ iff its residue vanishes. And the coefficientwise definition of $\int^H$ is independently forced by Section 4: an ordinary parametrized circle is not a fine-continuous path, since every fine-continuous map $[0, 1] \rightarrow \mathbb{K}$ is constant, so no Riemann-sum reading of the symbol is available in the first place.

5.6 Summary of what Section 5 forbids

- (1) No unrestricted identity theorem, strong maximum principle, Liouville theorem, primitive-uniqueness theorem or open mapping theorem for the germ-analytic class (Proposition 5.6); each requires coherence, and each is restated with that hypothesis in Sections 7 and 11.
- (2) No unrestricted removability theorem for germ-analytic functions bounded near a point (Proposition 5.9).
- (3) No determination of a global exponential, or of a global function on $\mathbb{K}$, by local analytic data alone (Proposition 5.10).
- (4) No contour functional on the ordinary unit circle with both the classical value for $1/z$ and an unrestricted fundamental theorem (Theorem 5.11).

6 The local theory: germs over $\mathbb{K}$

We now develop the local power-series class: arbitrary surcomplex coefficients, no growth condition, no ordinary domain. Everything in this part is a statement about germ-analytic functions in the sense of Definition 5.1(a). No result of this part is used for $\mathcal{H}(U)$ or for canonical lifts except through the evaluation theorem of Section 7, which is proved there; conversely, nothing proved for $\mathcal{H}(U)$ is available here, as Section 5 has just shown.

The one surprise of the local theory is how large it is: the germ algebra at any point is exactly $\mathbb{K}[[X]]$, with no radius-of-convergence condition whatsoever. Classical growth restrictions disappear because the sole infinite operation in use is Hahn summability (Section 3) — a support condition, not a limit.

6.1 Separated degree blocks

Everything rests on one piece of interval arithmetic in the surreal value class. The point is to bound *supports*, not merely leading valuations.

Lemma 6.1 (Separated degree blocks). *Let $(a_\nu)_{\nu \in \mathbb{N}^d}$ be a family in $\mathbb{K}$ indexed by the multi-indices of $d < \infty$ variables. Put $S = \bigcup_\nu \text{supp}(a_\nu)$; this is a set, although it need not be well ordered. Choose $\Delta \in \mathbf{No}^{>0}$ with $|s| < \Delta$ for every $s \in S$, which is possible by surreal interpolation, and choose $\lambda > 2\Delta$. Put $\rho = t^\lambda$. Then:*

- (i) *for each total degree $n = |\nu|$ the support of $a_\nu \rho^n$ lies in the open interval $(n\lambda - \Delta, n\lambda + \Delta)$, and these intervals are pairwise disjoint and increasingly ordered in n ;*
- (ii) *the family $(a_\nu \rho^{|\nu|})_\nu$ is Hahn summable, and an exponent occurring in it determines the total degree n ;*
- (iii) *for every $u \in \mathcal{O}^d$ the fully expanded family $(a_\nu \rho^{|\nu|} u^\nu)_\nu$ is Hahn summable.*

Proof. (i) is the choice of Δ and λ : the upper endpoint $n\lambda + \Delta$ of the n th interval is below the lower endpoint $(n+1)\lambda - \Delta$ of the next precisely because $\lambda > 2\Delta$. For (ii), within a fixed total degree there are only finitely many ν , so the union of their shifted well-ordered supports is well ordered and each exponent occurs finitely often there; the ordered union of the disjoint increasing blocks is then well ordered, and by (i) an exponent lies in exactly one block, which fixes n .

For (iii) write $u_j = c_j + \eta_j$ with $c_j \in \mathbb{C}$ and $\eta_j \in \mathfrak{m}$, let B be the union of the supports of the η_j — a well-ordered subset of $\mathbf{No}^{>0}$ — and expand each finite product u^ν by the binomial theorem. All resulting supports lie in $A + \langle B \rangle$, where A is the well-ordered union from (ii). By Neumann [3]’s lemma (Section 3) $A + \langle B \rangle$ is well ordered, and a fixed exponent admits only finitely many decompositions into a member of A and a member of $\langle B \rangle$, only finitely many finite nondecreasing words from B for the second factor, only finitely many ν of the total degree pinned by the first, and only finitely many binomial terms for each such ν . This is joint summability of the full expansion, which is what licenses regrouping. $\square$

Two examples show that the hypothesis really is a bound on supports.

Example 6.2 (A valuation bound is not enough). For $n \geq 1$ put $a_n = t^{\omega - n + 1/n}$. Every a_n is infinitesimal, so every coefficient is small in valuation. Nevertheless at $h = t$ the terms are $a_n h^n = t^{\omega + 1/n}$, whose exponents strictly decrease and have no least element: the family is not Hahn summable. An assertion that “all infinitesimal coefficients may be evaluated at every infinitesimal argument” is therefore false. It is the common bound Δ on the whole union of supports in Lemma 6.1, not a bound on each $v(a_n)$, that does the work.

Example 6.3 (A radius unavailable at rank one). The series $\sum_{n \geq 0} t^{-n^2} X^n$ cannot be evaluated at any nonzero argument of positive *real* valuation λ : the exponents $n\lambda - n^2$ are eventually strictly decreasing. At $X = t^\omega$ the exponents are $n\omega - n^2$, with consecutive differences $\omega - (2n + 1) > 0$, so the support is well ordered and the sum exists. The usable radius can be smaller than t^n for every ordinary integer n ; the value class of **No** has rank far above one and Lemma 6.1 uses all of it.

6.2 Universal realization

Theorem 6.4 (Realization of every formal series). *Every $P \in \mathbb{K}[[X_1, \dots, X_d]]$ has a Hahn-summed evaluation on a fine neighbourhood of zero: with ρ as in Lemma 6.1, the family defining $P(\rho u_1, \dots, \rho u_d)$ is summable for every $u \in \mathcal{O}^d$. On that scaled finite polydisk $\rho \mathcal{O}^d$ the values are bounded in surcomplex modulus by one fixed surreal, namely $|P(\rho u)| < t^{-\Delta-1}$. Formal sums, products, derivatives and compositions with zero constant inner term agree with the corresponding operations on evaluations, after shrinking the neighbourhood if necessary. For each nonzero one-variable series P , its value $P(h)$ is nonzero for every sufficiently small $h \neq 0$, with the scale allowed to depend on P . In any finite number of variables, evaluation as a function germ is injective. There is no injective evaluation of all of $\mathbb{K}[[X]]$ at a fixed point h : the polynomial $X - h$ already precludes that claim, and no one radius realizes every series.*

Proof. Summability on the scaled finite polydisk is Lemma 6.1: every exponent occurring is at least $-\Delta$, so the modulus of the value is below $t^{-\Delta-1}$, a factor of ω dominating every real leading coefficient; cancellation only raises the valuation.

Sums and products are expansions of jointly summable families. For a composition with zero inner constant term, each coefficient of the formal composite is a finite algebraic expression in the input coefficients; include the supports of all those individual products, taken *before* cancellation, in the set used to choose Δ . Then the separated-block argument applies to the fully expanded family, there being only finitely many algebraic terms at each fixed total degree, and grouping by total degree reproduces the formal composite exactly. A further shrinking puts the inner values inside a realization neighbourhood of the outer series.

For one-variable nonvanishing, let m be the least index with $a_m \neq 0$ and write $\alpha_n = v(a_n)$, $\beta = v(h) \geq \lambda$. For $n > m$,

$$(\alpha_n + n\beta) - (\alpha_m + m\beta) = (\alpha_n - \alpha_m) + (n - m)\beta > 0,$$

because $|\alpha_n - \alpha_m| < 2\Delta < \lambda \leq \beta$. So $a_m h^m$ contributes the unique least exponent and the value is nonzero.

For injectivity on multivariable germs, let P_m be the first nonzero homogeneous part of a nonzero formal series. A nonzero polynomial over $\mathbb{K}$ cannot vanish on all of $\mathbb{Q}^d$: induction on the number of variables reduces this to the finite root bound in one variable. Choose $u \in \mathbb{Q}^d$ with $P_m(u) \neq 0$. The one-variable series $P(uX)$ is nonzero, so its evaluations are nonzero at all sufficiently small nonzero X . These points lie in any prescribed fine neighborhood of zero. Thus P cannot represent the zero germ, even when evaluation is restricted to surreal-real coordinate tuples.

Derivative compatibility is proved in Theorem 6.9 using Proposition 6.5. Those proofs use only the summability, algebra and faithfulness clauses established here, not that compatibility as an input. $\square$

Proposition 6.5 (Uniform Taylor remainder). *For a fixed $A = \sum_{n \geq 0} a_n X^n \in \mathbb{K}[[X]]$ there are $r, M \in \mathbf{No}^{>0}$ such that for every $N \geq 0$ and every h with $|h| \leq r$,*

$$\left| A(h) - \sum_{n < N} a_n h^n \right| \leq \frac{M|h|^N}{1 - |h|} \leq 2M|h|^N.$$

Proof. Choose $\Delta > 0$ bounding the absolute values of all exponents in *both* families (a_n) and $(|a_n|)$, and choose $\lambda > 2\Delta$ as in Lemma 6.1. Set $r = t^\lambda$ and $M = t^{-\Delta-1}$. Lemma 1.4 gives $|a_n| < M$ for every n . The separated-block lemma applied to $(|a_n|)$ makes $(|a_n||h|^n)_n$ summable for $|h| \leq r$; the family $(M|h|^n)_n$ is summable by the positive-support geometric series. Their termwise differences are therefore summable and nonnegative. A Hahn sum of a summable family of nonnegative surreals is nonnegative: at the least exponent in the union of nonempty supports, every contributing leading coefficient is positive.

For completeness, the triangle inequality for this tail follows without any limiting argument. Put $T = \sum_{n \geq N} a_n h^n$. If $T \neq 0$, multiply by $\theta = \bar{T}/|T|$ and sum $\operatorname{Re}(\theta a_n h^n) \leq |a_n||h|^n$; multiplication by the fixed θ preserves Hahn summability. The preceding positivity argument gives $|T| \leq \sum_{n \geq N} |a_n||h|^n$; for $T = 0$ this is immediate. Summing the bounds by $M|h|^n$ and using the geometric identity gives the first estimate. Since r is infinitesimal, $1/(1 - |h|) < 2$, giving the second. Here $|h|^0 = 1$, including at $h = 0$. $\square$

The next two results separate infinitesimal realization with ordinary coefficients from the much stronger requirement of one series being summable at every surcomplex argument.

Corollary 6.6 (No growth condition). *For every $\varepsilon \in \mathfrak{m}$ and every complex sequence (c_n) , the family $(c_n \varepsilon^n)_{n \geq 0}$ is Hahn summable, with no growth hypothesis on (c_n) . In particular $\sum_{n \geq 0} (n!)^2 \varepsilon^n$ and $\sum_{n \geq 0} n! \varepsilon^n$ are germ-analytic functions on $\mathfrak{m}$, although the corresponding classical power series have radius of convergence zero.*

Proof. Complex coefficients have support $\{0\}$, so Neumann's lemma applied to $\langle \operatorname{supp} \varepsilon \rangle$ suffices: at each exponent only finitely many finite words from $\operatorname{supp} \varepsilon$ contribute, and the word length fixes n . $\square$

Proposition 6.7 (Everywhere-summable means polynomial). *If $\sum_{n \geq 0} a_n X^n$ is Hahn summable at every $X \in \mathbb{K}$, it is a polynomial.*

Proof. Suppose infinitely many a_n are nonzero, and choose Δ bounding all $|\operatorname{supp} a_n|$ as in Lemma 6.1, and take $\lambda > 2\Delta$. At $X = t^{-\lambda}$ the exponents $v(a_n) - n\lambda$ strictly decrease along the nonzero indices: for $m > n$ their difference is $v(a_m) - v(a_n) - (m - n)\lambda < 2\Delta - \lambda < 0$. They all lie in the union of the term supports, which is therefore not well ordered. $\square$

Proposition 6.7 is a statement about one formal series, not about functions: it emphatically does not say that a globally germ-analytic function on $\mathbb{K}$ is a polynomial, and Section 5 supplies the counterexamples. The classification of the functions that are *coherent at every scale* is the rigidity theorem of Section 13.

6.3 The germ algebra

Definition 6.8 (Germ-analytic germs). For $a \in \mathbb{K}$, two functions germ-analytic at a define the same germ if they agree on some fine ball about a . Write $\mathcal{G}_a$ for the $\mathbb{K}$ -algebra of germs at a .

Theorem 6.9 (The germ algebra is $\mathbb{K}[[X]]$). *Hahn evaluation is an isomorphism of differential $\mathbb{K}$ -algebras*

$$\mathbb{K}[[X]] \xrightarrow{\sim} \mathcal{G}_a, \quad X = z - a.$$

Every germ-analytic function is infinitely differentiable near its centre in the sense of Definition 5.1(d), its derivatives are germ-analytic, differentiation is termwise, and

$$a_n = \frac{F^{(n)}(a)}{n!}.$$

It admits a Hahn-summed Taylor expansion about every sufficiently nearby point:

$$A(h_0 + k) = \sum_{j \geq 0} c_j k^j, \quad c_j = \sum_{n \geq j} \binom{n}{j} a_n h_0^{n-j},$$

all displayed families being summable. Sums, products, quotients by germs with nonzero constant term, and compositions of compatible germs are germ-analytic, with the ordinary sum, product, quotient and chain rules. Every germ has a primitive $C + \sum_{n \geq 0} \frac{a_n}{n+1} X^{n+1}$, unique up to an additive constant as a germ.

Proof. Surjectivity is Definition 6.8 together with Theorem 6.4; injectivity is the series-dependent nonvanishing clause of that theorem, since two series representing the same germ have a difference vanishing at every sufficiently small nonzero h .

For the derivative at the centre write $A(h) = a_0 + a_1 h + h^2 B(h)$; by Proposition 6.5 with $N = 2$ the difference quotient differs from a_1 by at most $2M|h|$, so for a prescribed target tolerance η it suffices to take $|h| < \min\{r, \eta/(2M+1)\}$. Note that this is the full ε - δ derivative of Definition 5.1(d), with both radii ranging over all positive surreals and in particular over infinitesimal tolerances; the quantification is over a proper class of increments, so the discreteness theorem of Section 4 does not make it vacuous.

The recentring formula follows by expanding $a_n(h_0 + k)^n$ with the finite binomial theorem — $n+1$ terms at degree n — and regrouping, which Lemma 6.1(iii) licenses. Applying the first argument at h_0 identifies $F'(h_0) = c_1 = \sum_{n \geq 1} n a_n h_0^{n-1}$; iteration gives all higher derivatives and the coefficient formula. The algebraic operations and the chain rule are the corresponding formal identities, transported by the composition clause of Theorem 6.4. Formal integration divides a_n by $n+1$ and shifts the degree, which changes no support; a formal series with zero derivative has no positive-degree coefficient in characteristic zero, which is the uniqueness clause. $\square$

Remark 6.10 (Where uniqueness of the primitive stops). The uniqueness in Theorem 6.9 is uniqueness of a germ at one point. It does not propagate: by Example 5.5 and Corollary 5.14, a germ-analytic primitive on a disconnected-in-the-fine-sense domain may be modified by any locally constant function without changing its derivative anywhere.

Corollary 6.11 (Isolated zeros of a germ). *A nonzero germ at a has a unique finite order $n \geq 0$, factors as $A(h) = a_n h^n U(h)$ with $U(0) = 1$, and after shrinking the ball $U(h) - 1$ is infinitesimal, so there are no zeros there if $n = 0$, and otherwise the centre is the only zero there. Consequently, if a germ-analytic function has zeros meeting every sufficiently small punctured fine ball about a , its germ at a is zero.*

Proof. Factor formally at the first nonzero coefficient and evaluate; choosing the scale small enough makes every exponent in the evaluated support of $U - 1$ strictly positive, by Lemma 6.1. A nonzero germ then has nonzero values at all sufficiently small nonzero increments. $\square$

The last sentence of Corollary 6.11 permits a proper class of zeros. It is *not* a sequential accumulation statement: by the discreteness theorem of Section 4, no nontrivial set of distinct points accumulates in the fine topology, so no argument in this article may be phrased as “the zeros have an accumulation point”. The global identity theorem of Section 7 instead uses accumulation of *standard parts* in $\mathbb{C}$, which is a statement about the ordinary plane.

6.4 Which derivative? The derivation obstruction

Remark 6.12 (D_z differentiates the variable). The operator D_z used throughout this article differentiates a function of a variable and holds every surcomplex *coefficient* fixed. In particular $D_z(\omega) = 0$, ω being a constant function. It is not the Berarducci [9]–Mantova [11] field derivation

on surreal numbers viewed as transseries, whose normalization has $\partial\omega = 1$. Confusing the two roles of a surreal number would invalidate every derivative in this article. Nothing here imports surreal differential algebra, and nothing here may be combined with it without first checking which operator is meant.

That warning is not a matter of taste. It is a theorem that the two operators cannot be reconciled through the exponential.

Theorem 6.13 (Derivation obstruction). *Let ∂ be any field derivation on $\mathbb{K}$ fixing $\mathbb{C}$ pointwise with $\partial\omega = 1$; the complexification of the normalized Berarducci–Mantova derivation is one. Then the canonical surcomplex exponential Exp of Section 12 cannot satisfy*

$$\partial \text{Exp}(z) = \text{Exp}(z) \partial z \quad \text{for all } z \in \mathbb{K}. \quad (6.1)$$

More generally, if $E : (\mathbb{K}, +) \rightarrow (\mathbb{K}^\times, \cdot)$ is a homomorphism satisfying (6.1) in place of Exp , then every element of $\ker E$ is a differential constant of ∂ . Since $\ker \text{Exp} = 2\pi i \mathbf{Oz} = i(\mathbf{J} + 2\pi\mathbb{Z})$ contains $i\omega$, and $\partial(i\omega) = i \neq 0$, no such ∂ exists for Exp .

Proof. For the general assertion, let $k \in \ker E$, so $E(k) = 1$. Applying (6.1) at $z = k$ gives $0 = \partial(1) = \partial E(k) = E(k) \partial k = \partial k$, so k is a differential constant. For the specific assertion, $\text{Exp}(i\omega) = 1$, so the left side of (6.1) at $z = i\omega$ is $\partial(1) = 0$ while the right side is $\text{Exp}(i\omega) \partial(i\omega) = i \neq 0$. $\square$

Remark 6.14. There is no conflict with $D_z \text{Exp}(z) = \text{Exp}(z)$, which is proved in Section 12. That identity differentiates a variable while holding every surcomplex coefficient fixed; (6.1) differentiates the field’s own elements as transseries. The two are different mathematical tasks, and Theorem 6.13 says precisely that the omnific-period kernel is incompatible with the normalization $\partial\omega = 1$. No composition law compatible with the Berarducci–Mantova derivation is claimed anywhere in this article.

6.5 Cauchy–Riemann equations and harmonic germs

Call a function *real germ-analytic* in the surreal variables (x, y) if it is the Hahn evaluation of a formal series in two variables over $\mathbb{K}$, as in Theorem 6.4 with $d = 2$. Such a function has a Fréchet differential in the fine sense: subtracting its degree-zero and degree-one parts leaves a remainder bounded by $M\|h\|^2$ on a small polydisk, by the support estimates of Lemma 6.1 and a finite sum over degree-two factors. Here “**No**-linear” means linear on the two surreal-real coordinates, not merely $\mathbb{R}$ -linear on **No** viewed as an infinite-dimensional real vector space.

Theorem 6.15 (Cauchy–Riemann criterion for germs). *Let $F = u + iv$ be real germ-analytic near a point of $\mathbf{No}^2$. Then F is germ-analytic in $z = x + iy$ near that point if and only if, on a neighbourhood,*

$$u_x = v_y, \quad u_y = -v_x. \quad (6.2)$$

At a single point, (6.2) is exactly the condition that the real Fréchet differential be $\mathbb{K}$ -linear, i.e. multiplication by one element of $\mathbb{K}$. The real and imaginary parts of a germ-analytic function satisfy $u_{xx} + u_{yy} = v_{xx} + v_{yy} = 0$.

Proof. A real 2×2 matrix commutes with multiplication by i , represented by $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, exactly when it has the form $\begin{pmatrix} p & -q \\ q & p \end{pmatrix}$; that is (6.2) for the differential, and the remainder condition converts it into the complex difference quotient.

For the germ statement introduce the formal coordinates $Z = X + iY$ and $W = X - iY$, invertible over $\mathbb{K}$ because $2i \neq 0$. Equations (6.2) say $\partial_W F = \frac{1}{2}(\partial_X + i\partial_Y)F = 0$. If they hold as functions on a neighbourhood, injectivity of multivariable evaluation as a function germ (Theorem 6.4) makes this a formal identity, and in characteristic zero it forces every term of

positive W -degree to vanish; the remaining formal series depends on Z alone and realizes a germ-analytic function. Conversely ∂_W annihilates a series in Z . Differentiating (6.2) once more, with formal mixed partials commuting, gives the two Laplace equations. $\square$

Theorem 6.16 (Harmonic conjugate germs). *Every real germ-analytic harmonic germ $u(x, y)$ over $\mathbf{No}$ is the real part of a germ-analytic germ. Its harmonic conjugate is unique up to a constant in $\mathbf{No}$ as a germ (the constant need not belong to the ordinary real field $\mathbb{R}$).*

Proof. Translate the centre to $(0, 0)$ and define, with coefficientwise formal primitives,

$$v(X, Y) = - \int_0^X u_Y(s, 0) ds + \int_0^Y u_X(X, \tau) d\tau.$$

Then $v_Y = u_X$, and since $u_{XX} = -u_{YY}$,

$$v_X = -u_Y(X, 0) + \int_0^Y u_{XX}(X, \tau) d\tau = -u_Y(X, Y).$$

So $u + iv$ satisfies (6.2); realize it and apply Theorem 6.15. A difference of two conjugates has both formal partial derivatives zero, hence is a constant formal series. $\square$

Remark 6.17 (Two scope limits). Theorem 6.15 does not assert that the existence of two partial derivatives at a point suffices for complex differentiability, and it does not assert that an arbitrary pointwise differentiable function has a normal-form expansion: the Fréchet or real germ-analytic hypothesis is load-bearing (Remark 5.3). Theorem 6.16 is a differential identity at the level of germs; it asserts nothing about a global Dirichlet problem. The coefficientwise harmonic-conjugate theorem for $\mathcal{H}(U)$ on simply connected ordinary domains, the Poisson formula and the coherent Dirichlet problem are in Section 11 and have different hypotheses — ordinary coefficientwise boundary data on an ordinary disk.

6.6 Inverse, implicit function, and finite ramification

Theorem 6.18 (Local inverse and implicit function theorems for germs). *Let $F = (F_1, \dots, F_d)$ be a germ-analytic germ at $a \in \mathbb{K}^d$ with invertible Jacobian $J_F(a)$. Then F has a germ-analytic inverse between fine-open neighbourhoods of a and $F(a)$.*

If $\Phi(X, Y)$ is a tuple of q germ-analytic germs with $\Phi(0, 0) = 0$ and $\partial\Phi/\partial Y(0, 0)$ an invertible $q \times q$ matrix, there is a unique germ $Y = \varphi(X)$ with $\varphi(0) = 0$ and $\Phi(X, \varphi(X)) = 0$; after shrinking neighbourhoods it gives all sufficiently small solutions.

Proof. Translate centres and write $F(X) = AX + \sum_{n \geq 2} F_n(X)$ with A invertible and F_n homogeneous of degree n . Seek $G(Y) = A^{-1}Y + \sum_{n \geq 2} G_n(Y)$. Once $G_2, \dots, G_{n-1}$ are known, the degree- n part of $F(G(Y)) - Y$ is AG_n plus a known homogeneous polynomial, so exactly one G_n cancels it. The same recursion, or substitution, makes the formal inverse two-sided. Realize the two composition identities by Theorem 6.4. Germ-analytic functions are continuous by Proposition 6.5; choose a fine ball B on which $G \circ F = \text{id}$, then a smaller ball V about $F(a)$ with $F \circ G = \text{id}$ on V and $G(V) \subseteq B$. Then $W = B \cap F^{-1}(V)$ is fine-open and $F : W \rightarrow V$, $G : V \rightarrow W$ are inverse.

For the implicit statement apply the inverse theorem to $(X, Y) \mapsto (X, \Phi(X, Y))$, whose Jacobian is block triangular with invertible diagonal blocks, and restrict the inverse to pairs $(X, 0)$; uniqueness also follows degree by degree from the invertible Y -Jacobian.

Every recursion here is a set-sized list of coefficient equations, solved degree by degree. There is no appeal to completeness, to a contraction principle, or to convergence of a sequence of Newton iterates — none of which is available in the fine topology. $\square$

Theorem 6.19 (Finite ramification normal form). *Let F be a nonconstant germ-analytic germ at a and let $m \geq 1$ be the least index with $F^{(m)}(a) \neq 0$, so that $F(a+X) - F(a) = bX^m + O(X^{m+1})$ with $b \neq 0$. There is a germ-analytic local coordinate ϕ with $\phi(0) = 0$ and $\phi'(0) = 1$ such that*

$$F(a+X) = F(a) + b\phi(X)^m. \quad (6.3)$$

The zero of $F - F(a)$ at a has finite order m and is isolated; F is an open map at a ; on a suitable neighbourhood every target $F(a) + w$ with $w \neq 0$ sufficiently small has exactly m distinct preimages, all simple.

Proof. Factor $F(a+X) - F(a) = bX^m(1+Q(X))$ with $Q(0) = 0$ and put $\phi(X) = X(1+Q(X))^{1/m}$ using the formal binomial series; $\phi'(0) = 1$, so ϕ is an invertible local coordinate by Theorem 6.18, and realization preserves the m th-power identity. In the coordinate $w = \phi(X)$ the equation for the target $F(a) + y$ is $bw^m = y$. Since $\mathbb{K}$ is algebraically closed, for $y \neq 0$ there are exactly m distinct roots, each of modulus $(|y|/|b|)^{1/m}$, so all lie in a prescribed small coordinate ball once $|y|$ is small enough; the derivative $bmw^{m-1}\phi'(X)$ is nonzero at each. For $y = 0$ the only root is $w = 0$, of multiplicity m . The same root argument shows that the image contains a fine ball about $F(a)$. $\square$

Corollary 6.20 (Local openness and no local modulus maximum). *At a point where its germ is nonconstant, a germ-analytic function maps every sufficiently small fine ball onto a class containing a fine ball about its value. Its modulus has no local maximum there, and its real part has no local maximum or minimum.*

Proof. Openness is Theorem 6.19. If $F(a) \neq 0$, every fine ball about $F(a)$ contains $(1+\delta)F(a)$ with $\delta > 0$ infinitesimal, of strictly larger modulus; if $F(a) = 0$, it contains nonzero points. For the real part, a ball about b contains $b \pm \delta$ with $\delta > 0$ infinitesimal real. $\square$

Remark 6.21. The modulus in Corollary 6.20 is $|z| = \sqrt{z\bar{z}}$ in the ordered field $\mathbf{No}$, not an ultrametric absolute value read off the valuation v . The valuation is locally constant on nonzero values, as Example 5.8 exploits, and satisfies no such maximum principle. Note also that “nonconstant germ” cannot be relaxed to “globally nonconstant function”: L of Example 5.5 is globally nonconstant and is not an open map.

Remark 6.22 (Scope of the coherent inverse theorem). Theorem 6.18 is a statement about germs; it produces no section over a common ordinary base and says nothing about supports. The coherent result proved in Section 11 is the *one-variable* local inverse theorem (Theorem 11.6), using positive-support Hahn reversion after a scale change. The multivariable germ theorem above does not by itself produce a coherent section on a common ordinary parameter domain. An earlier version of this remark incorrectly referred to itself as a proved multivariable coherent implicit-function theorem; no such theorem is established in this article.

6.7 Local identity and maximum modulus, as germ statements

Corollary 6.23 (Local identity and maximum-modulus principles). *A germ-analytic germ all of whose derivatives vanish at its centre is the zero germ. A germ-analytic function whose modulus has a local maximum at a point is constant as a germ there.*

Proof. The first is uniqueness of Taylor coefficients (Theorem 6.9). The second is Corollary 6.20: a nonconstant germ is locally open, which contradicts a local maximum. $\square$

These are germ statements and nothing more. Three things must travel with them. They do not propagate to a whole fine-open class: L and q (Examples 5.5, 5.7) have constant germs at every point and are globally nonconstant. They are not sequential accumulation theorems, since by Section 4 no nontrivial set of distinct points accumulates in the fine topology; in particular

no proof in this article may construct a convergent sequence of zeros. And the passage from “constant germ” to “one global constant” is exactly what coherence buys, in Theorem 7.18 and the global maximum principle of Section 11, and it is available for no other reason.

6.8 Formal residues in $\mathbb{K}((X))$

The field of fractions of $\mathbb{K}[[X]]$ is the Laurent field $\mathbb{K}((X))$, whose elements have only finitely many negative powers; multiplication by a large enough power of X makes such an element germ-analytic, and it evaluates on a sufficiently small punctured fine ball. We call these *finite-meromorphic germs* and set

$$\operatorname{Res}_{X=0} \left(\sum_{n \geq n_0} a_n X^n dX \right) = a_{-1} \in \mathbb{K}.$$

Remark 6.24 (Which residue this is). Res is the *formal* residue, the coefficient of X^{-1} in $\mathbb{K}((X))$, taken at any surcomplex centre in an analytic local coordinate X ; the letter X is a local coordinate and never the Hahn monomial parameter t^γ . The actual residue Res_b of Section 10 is this same coefficient operation on the Laurent germ at b , with $X = z - b$; the formal laws apply after that identification. The different operation is the cluster residue Res_a^H at an ordinary centre. For $1/(\zeta - \varepsilon)$ with $\varepsilon \in \mathfrak{m} \setminus \{0\}$ one has $\operatorname{Res}_0^H = 1$ while the actual local residue at the exact point 0 is zero, the pole being at ε . The two bridge theorems of Section 10 relate this cluster operation to actual local residues; no bridge is needed between a Laurent germ’s formal residue and its actual residue.

Proposition 6.25 (The only local primitive obstruction). *A Laurent differential $\alpha \in \mathbb{K}((X)) dX$ has a primitive in $\mathbb{K}((X))$ if and only if $\operatorname{Res}_{X=0} \alpha = 0$. Equivalently*

$$\mathbb{K}((X)) dX / d\mathbb{K}((X)) \simeq \mathbb{K} \frac{dX}{X}.$$

Proof. A formal derivative has no X^{-1} coefficient, since the only possible source is the degree-zero term, which differentiates to zero. Conversely each $a_n X^n dX$ with $n \neq -1$ integrates to $a_n X^{n+1}/(n+1)$, and these coefficients again form a Laurent series with finite negative part. The only omitted term is $a_{-1} dX/X$. $\square$

Related (batch 93, added 4 October 2026). With $\mathbb{R}$ in place of $\mathbb{K}$, this proposition is the case $\Gamma = \mathbb{Z}$, $D = d/dt$ of Corollary 231.3 (pma:bsd:cor:integration) of Part XVII of docs/foundations-and-computation/polish-models-of-omnific-arithmetic (AI-assisted, unrefereed). For every countable $\Gamma \leq \mathbb{R}$ of rational rank one, that corollary shows that every derivation of the field of real series with left-finite support in Γ that is Borel in the coefficients is aE for an Euler derivation E , and that for $a \neq 0$ its image consists of the g for which g/a has constant term zero. For $\Gamma = \mathbb{Z}$ that field is $\mathbb{R}((t))$ and $d/dt = t^{-1}E$, so the condition is the vanishing of the coefficient of t^{-1} . Its Remark 239.5 (pma:bsd:rem:lean) notes that the Lean theorems `exists_derivative_eq_iff` and `range_derivative_eq_ker_residue`, which prove this proposition over every field of characteristic zero, formalize that case; the Borel classification and the other groups Γ are not formalized.

Theorem 6.26 (Change of variable, with the ramification factor). *Let $g(X) = cX^m(1 + O(X))$ with $c \neq 0$ and $m \geq 1$. For every $A(Y) \in \mathbb{K}((Y))$,*

$$\operatorname{Res}_{X=0} A(g(X)) g'(X) dX = m \operatorname{Res}_{Y=0} A(Y) dY. \quad (6.4)$$

In particular residues are invariant under an invertible local coordinate change, the case $m = 1$.

Proof. Only the finitely many negative powers of Y can affect the residue, so it suffices to treat monomials $A(Y) = Y^n$. For $n \neq -1$ the pullback is $d(g^{n+1}/(n+1))$, of residue zero by Proposition 6.25. For $n = -1$ write $g = cX^m u$ with $u(0) = 1$; then $g'/g = m/X + u'/u$ and u'/u is a power series, whose residue is zero, leaving the factor m . Positive powers of Y pull back to series with no negative powers and contribute nothing at exponent -1 . $\square$

Corollary 6.27 (Local argument principle). *For a nonzero finite-meromorphic germ $F = X^m U$ with $U(0) \neq 0$,*

$$\operatorname{Res}_{X=0} \frac{F'}{F} dX = m \in \mathbb{Z},$$

a positive m being a zero order and a negative m minus a pole order.

Proposition 6.28 (Formal residue facts and removability). *For finite-meromorphic germs: the residue of a derivative is zero; a germ bounded on some sufficiently small punctured fine ball by one fixed surreal constant has zero principal part, hence a removable singularity.*

Proof. The first is Proposition 6.25. For the second, a germ with no principal part is germ-analytic and bounded on a small ball by Theorem 6.4. If instead a pole of order $m > 0$ is present, write $A(X) = cX^{-m}(1 + V(X))$ with $V(0) = 0$, so $|1 + V| > 1/2$ on a small ball; given any proposed bound $M > 0$, choose a nonzero X small enough that $|c|/(2|X|^m) > M$. $\square$

Proposition 6.28 is the exact scope of local removability, and Proposition 5.9 is its boundary: the hypothesis “finite-meromorphic” cannot be weakened to “germ-analytic on a punctured fine neighbourhood”.

There is also a formal Cauchy identity. In the *explicit* ring $\mathbb{K}((X))[[Y]]$, interpret

$$\frac{1}{X - Y} = \sum_{n \geq 0} Y^n X^{-n-1};$$

then for $P \in \mathbb{K}[[X]]$, coefficientwise residue in X gives

$$\operatorname{Res}_{X=0} \frac{P(X)}{X - Y} dX = P(Y). \quad (6.5)$$

The ambient ring matters and is part of the statement: an infinite negative Laurent tail in X has *not* been silently declared a member of $\mathbb{K}((X))$.

Theorem 6.29 (Lagrange–Bürmann inversion). *Let $F(X) = a_1 X + a_2 X^2 + \dots$ with $a_1 \neq 0$ and let G be its compositional inverse germ. For $n \geq 1$ and every $H \in \mathbb{K}[[X]]$,*

$$[Y^n] H(G(Y)) = \frac{1}{n} [X^{n-1}] H'(X) \left(\frac{X}{F(X)} \right)^n.$$

In particular,

$$[Y^n] G(Y) = \frac{1}{n} [X^{n-1}] \left(\frac{X}{F(X)} \right)^n.$$

These are identities of realized germ-analytic germs as well as of formal series.

Proof. By Theorem 6.26 with $Y = F(X)$ and $m = 1$,

$$[Y^n] H(G(Y)) = \operatorname{Res}_{X=0} \frac{H(X) F'(X)}{F(X)^{n+1}} dX.$$

The residue of $d(H(X)F(X)^{-n})$ vanishes; expanding that derivative turns the right side into $\frac{1}{n} \operatorname{Res}_{X=0} H'(X) F(X)^{-n} dX$, which is the stated coefficient. Taking $H(X) = X$ gives the second formula. Theorems 6.4 and 6.18 identify the formal inverse with the actual inverse germ and realize both sides on a sufficiently small ball. $\square$

6.9 What the local theory does and does not give

The germ algebra is $\mathbb{K}[[X]]$ exactly: arbitrary coefficients, no growth condition, no radius of ordinary convergence, and the full characteristic-zero formal calculus with actual function values and actual fine derivatives. What it does not give is any global statement whatever. Specifically, and to be read together with Section 5:

- (1) No radius-of-convergence formula is claimed from the ordinary magnitudes of the coefficients; the usable scale is produced by Lemma 6.1 from a bound on the *union of supports*, and Example 6.2 shows a valuation bound will not do.
- (2) A Hahn sum is not a fine limit of its partial sums, and no theorem here may be reproved by exhibiting a convergent sequence.
- (3) Theorem 6.4 does not assert summability at every infinitesimal for arbitrary surcomplex coefficients: the input must be small relative to a field containing the coefficient data. For purely complex coefficients every infinitesimal works (Corollary 6.6).
- (4) Agreement at ordinary complex points does not determine a germ-analytic function; it itself is locally constant on $\mathcal{O}$, agrees with the identity at every ordinary point, and differs at infinitesimal ones.
- (5) The local identity, maximum modulus and open mapping statements are germ statements. Their global forms require coherence and are proved in Sections 7 and 11, for $\mathcal{H}(U)$ only.
- (6) No classification of all functions on a punctured fine neighbourhood is offered: finite-meromorphic germs and coherent quotients are specified subclasses, not an exhaustive list, and no general theory of essential singularities is developed here. The one value-distribution theorem in this article concerns a single specified function and is in Section 12.

7 The coherent theory: $\mathcal{H}(U)$

Section 6 developed the local power-series layer: germ-analytic functions, with arbitrary surcomplex coefficients, no ordinary domain and no growth condition. That layer is too wide to carry a global theorem. The locally constant witness L of Section 5 (the locally constant germ-analytic witness of Section 5) and its germ-analytic refinement b (the germ-analytic witness of Section 5) are germ-analytic on $\mathbb{K}$ and $\mathbb{K}^\times$, respectively, and destroy, at one stroke, the identity theorem, the global maximum principle, Liouville's theorem, uniqueness of primitives and the open mapping theorem.

This part introduces the coherent layer: one *ordinary* connected complex domain carries all coefficient functions at once. This common-domain condition ties different monads to one another; additional hypotheses for each global principle, such as entire coefficientwise boundedness for Liouville, are stated with that theorem, and every global theorem in Sections 8–11 is a theorem about this class and not about the germ class.

7.1 The halo

Definition 7.1 (Halo of an ordinary domain). For an ordinary open $U \subseteq \mathbb{C}$ put

$$U^\# = \text{st}^{-1}(U) = \{a + \varepsilon : a \in U, \varepsilon \in \mathfrak{m}\} \subseteq \mathcal{O}. \quad (7.1)$$

This is an open class in the fine topology, being a union of cosets of $\mathfrak{m}$.

Two warnings belong to the definition and are made once here.

Remark 7.2 (The halo is not a surcomplex ball). $\mathbb{D}^\#$ consists of the *finite* points whose standard part has modulus strictly less than one. It is strictly smaller than $\{z \in \mathbb{K} : |z| < 1\}$: for positive infinitesimal ε the point $1 - \varepsilon$ lies in the surcomplex unit ball but not in $\mathbb{D}^\#$, because its standard part lies on the unit circle. The whole boundary monad is missing. No statement below about $\mathbb{D}^\#$ is a statement about the surcomplex unit ball; in particular the Schwarz–Pick theorem of Section 11 is not, and the failure witness $(1 + t)z$ there turns on exactly the omitted infinitesimal boundary layer.

Remark 7.3 (Puncturing deletes a monad). $(\mathbb{D} \setminus \{0\})^\#$ is the halo of the punctured ordinary disk, and it omits the *entire* monad $\mathfrak{m}$, not the single point 0. It is therefore different from $\mathbb{D}^\# \setminus \{0\}$. The removable-singularity statements of Section 11 concern the former. Keeping the two puncturing operations apart is what prevents a false analogy with the classical punctured disk.

7.2 Coherent Hahn sections

Definition 7.4 (The coherent algebra). Let $U \subseteq \mathbb{C}$ be a nonempty connected ordinary open set. A *coherent Hahn section* on U is a formal expression

$$F = \sum_{\gamma \in S} f_\gamma(Z) t^\gamma, \quad f_\gamma \in \text{Hol}(U), \quad S = \{\gamma : f_\gamma \neq 0\} \text{ a well-ordered set.} \quad (7.2)$$

The class of these is written $\mathcal{H}(U)$; it is a $\mathbb{K}$ -algebra under coefficientwise addition and Hahn multiplication, the latter being well defined because at each exponent only finitely many products contribute (Section 3, 3.2). Put

$$\mathcal{H}_{>0}(U) = \{F \in \mathcal{H}(U) : v_H(F) > 0\} \cup \{0\}, \quad F' = \sum_{\gamma \in S} f'_\gamma t^\gamma,$$

and write D_z for the operator $F \mapsto F'$ when the operator itself is the subject. For $F \neq 0$ define the *family valuation* and the *leading coefficient function*

$$v_H(F) = \min S, \quad f_{v_H(F)} \in \text{Hol}(U).$$

As for the scalar valuation, put $v_H(0) = \infty$, an added symbol.

Remark 7.5 (What “coherent” means, and what it does not). The entire content of the word is: *one* ordinary holomorphic function per exponent, all of them on *one* common connected ordinary domain, with well-ordered set support. Three disclaimers travel with it.

- (i) This is not the technical assertion that some sheaf is coherent as a module in the sense of complex geometry. The word is used in the sense just given and in no other.
- (ii) The hypothesis is not free. An arbitrary family of ordinary holomorphic germs, one at each point, whose radii of convergence shrink to zero, is not an element of $\mathcal{H}(U)$ for any ordinary neighborhood U . It is precisely this hypothesis that links monads with different standard parts, and it is why the identity theorem (7.18), the global maximum principle (Section 11) and the zero-cluster theorem (Section 9) hold at all.
- (iii) Coherence is a genuine restriction on functions, not a reparameterization. The function $z \mapsto \text{st}(z)$ on $U^\#$ is not in $\mathcal{H}(U)$ when U has more than one point: its values at ordinary points would force it to be the canonical lift of the identity, which it is not.

The synonym “holomorphic profile” is recorded once and then dropped.

Remark 7.6 (v_H is not a pointwise valuation). $v_H(F)$ and $f_{v_H(F)}$ are attached to the family. They are not the valuation $v(F^\#(z))$ and the leading complex coefficient of the surcomplex value $F^\#(z)$, and the two disagree exactly where the theory is interesting: at a point z of the halo lying over a zero of $f_{v_H(F)}$, the value has strictly larger valuation than $v_H(F)$. The inequality in one direction is Theorem 7.8.

Because U is connected, $\text{Hol}(U)$ is an integral domain, so $v_H(FG) = v_H(F) + v_H(G)$ and v_H is a valuation on the fraction field of $\mathcal{H}(U)$.

7.3 Faithful evaluation on the halo

Definition 7.7 (Evaluation). For $F \in \mathcal{H}(U)$ and $z = a + \varepsilon \in U^\#$ set

$$F^\#(a + \varepsilon) = \sum_{\gamma \in S} \sum_{n \geq 0} \frac{f_\gamma^{(n)}(a)}{n!} \varepsilon^n t^\gamma. \quad (7.3)$$

Theorem 7.8 (Faithful coherent evaluation). (Class (b).) *Let U be connected and $F \in \mathcal{H}(U)$.*

- (i) *The double family in (7.3) is a single jointly Hahn summable family, so $F^\#$ is a well-defined function $U^\# \rightarrow \mathbb{K}$.*
- (ii) *$F \mapsto F^\#$ is an injective homomorphism of $\mathbb{K}$ -algebras onto its image, and it commutes with differentiation, $(F')^\# = (F^\#)'$, with restriction to a nonempty connected ordinary open subset, and with conjugation: if $U^* = \{\bar{z} : z \in U\}$ and $F^* = \sum_\gamma \overline{f_\gamma(\bar{z})} t^\gamma \in \mathcal{H}(U^*)$, then $(F^*)^\#(z) = \overline{F^\#(\bar{z})}$.*
- (iii) *If $F \neq 0$ and $\lambda = v_H(F)$, then for every $z = a + \varepsilon \in U^\#$*

$$v(F^\#(z)) \geq \lambda, \quad \text{st}(t^{-\lambda} F^\#(a + \varepsilon)) = f_\lambda(a). \quad (7.4)$$

Proof. Let $E = \text{supp}(\varepsilon) \subseteq \mathbf{No}_{>0}$. Every term of (7.3) has support inside $S + \langle E \rangle$. By the Neumann [3] support lemma (3.2) that class is a well-ordered set and each of its elements admits only finitely many decompositions $\gamma + \mu$ with μ a finite nondecreasing word in E ; the word length is exactly the Taylor order n , so only finitely many pairs (γ, n) contribute at a fixed exponent. That is precisely the two-clause summability criterion of Section 3, and it also licenses the regroupings used below.

Sums and products: apply the same argument to two families and use the ordinary Leibniz and Taylor product identities at a ; each Hahn exponent involves a finite convolution only. Constants of $\mathbb{K}$ are represented by constant coefficient functions, so the map is $\mathbb{K}$ -linear and not merely $\mathbb{C}$ -linear.

Injectivity: at an ordinary $a \in U$ the value is $F^\#(a) = \sum_\gamma f_\gamma(a) t^\gamma$, already in normal form. If $F^\# \equiv 0$ then uniqueness of normal forms gives $f_\gamma(a) = 0$ for every γ and every $a \in U$, so every coefficient function vanishes identically.

The identification with the fine derivative is proved below in Theorem 7.11, using the jointly summable Taylor expansion; no derivative identity is needed for the present algebraic arguments. Restriction and conjugation are coefficientwise. For (iii), factor t^λ out of (7.3). Every remaining term except the $n = 0, \gamma = \lambda$ term has strictly positive valuation: either $\gamma > \lambda$, or $n > 0$ and the factor ε^n has positive support. Hence the exponent-zero coefficient of $t^{-\lambda} F^\#(z)$ is $f_\lambda(a)$. $\square$

Corollary 7.9 (Leading coefficients control zero shadows). (Class (b).) *If $F \in \mathcal{H}(U)$ is nonzero and $F^\#(z) = 0$ for some $z \in U^\#$, then $f_{v_H(F)}(\text{st } z) = 0$. Equivalently, no zero of $F^\#$ lies over a non-zero of the leading coefficient function.*

Proof. Immediate from the second identity in (7.4). $\square$

Theorem 7.10 (Exact monad-wide Taylor expansion). (Class (b).) *Let $F \in \mathcal{H}(U)$, let $a \in U$ and let z_0, z lie in the same monad $a + \mathfrak{m}$. Then*

$$F^\#(z) = \sum_{n \geq 0} \frac{(F^{(n)})^\#(z_0)}{n!} (z - z_0)^n, \quad (7.5)$$

the family being Hahn summable. The expansion is exact throughout the whole monad, not merely on some smaller ball around z_0 .

Proof. Write $z_0 = a + \varepsilon_0$ and $z = z_0 + h$ with $\varepsilon_0, h \in \mathfrak{m}$. Expand each coefficient f_γ in its ordinary Taylor series at a and substitute the two infinitesimals. The total support lies in $S + \langle \text{supp } \varepsilon_0 \cup \text{supp } h \rangle$, which by 3.2 is well ordered with finite multiplicity at each exponent. The binomial regrouping $(\varepsilon_0 + h)^n = \sum_j \binom{n}{j} \varepsilon_0^{n-j} h^j$ is therefore legitimate, and collecting powers of h produces exactly the coefficients $(F^{(n)})^\#(z_0)/n!$. No hypothesis on $|h|$ beyond $h \in \mathfrak{m}$ was used, which is the assertion of exactness on the whole monad. $\square$

The next theorem is the bridge without which nothing proved with one derivative transfers to the other. It is used silently throughout Sections 8–11.

Theorem 7.11 (The section derivative is the fine derivative). (Class (b).) *Let $F \in \mathcal{H}(U)$ be nonzero with $\lambda = v_H(F)$, and let $b \in U^\#$. For every finite $N \geq 0$,*

$$F^\#(b + h) - \sum_{n=0}^N \frac{(F^{(n)})^\#(b)}{n!} h^n = h^{N+1} Q_N(b, h), \quad |Q_N(b, h)| < t^{\lambda-1} \quad (h \in \mathfrak{m}). \quad (7.6)$$

Consequently $(F')^\#$ is the derivative of $F^\#$ in the ε - δ sense of the fine topology: for every $\eta \in \mathbf{No}^{>0}$ there is $\delta \in \mathbf{No}^{>0}$ with

$$\left| \frac{F^\#(b + h) - F^\#(b)}{h} - (F')^\#(b) \right| < \eta \quad (0 < |h| < \delta).$$

The quantifiers range over all sufficiently small field elements, not over a chosen sequence.

Proof. Apply Theorem 7.10 at $z_0 = b$ and factor h^{N+1} out of the tail. Every exponent remaining in Q_N is at least λ , so $v(Q_N) \geq \lambda$, and the valuation bound of Section 1 (1.4) gives $|Q_N| < t^{\lambda-1}$; the deliberately generous factor $\omega = t^{-1}$ makes the bound independent of the ordinary real size of any leading coefficient. For the derivative take $N = 1$ and

$$0 < \delta < \min\{\omega^{-1}, \eta t^{1-\lambda}\},$$

so that $|hQ_1(b, h)| < \delta t^{\lambda-1} < \eta$ whenever $0 < |h| < \delta$. $\square$

Remark 7.12 (Which derivative this is). D_z differentiates a function of the variable z and holds every surreal coefficient fixed; in particular $D_z(\omega) = 0$. It is not a field derivation on $\mathbb{K}$ in the sense of surreal differential algebra, whose normalization has $d(\omega) = 1$, and the two are not merely differently normalized but genuinely incompatible — see Section 6, 6.13. A reader importing transseries results under the same symbol would silently change the meaning of every derivative in this article.

7.4 Units and the infinitesimal functional calculus

Proposition 7.13 (Units). (Class (b).) *Let $F \in \mathcal{H}(U)$ be nonzero and put $\lambda = v_H(F)$. Then F is a unit of $\mathcal{H}(U)$ if and only if f_λ is nowhere zero on U , in which case $E = t^{-\lambda}F/f_\lambda - 1$ belongs to $\mathcal{H}_{>0}(U)$ and*

$$F^{-1} = t^{-\lambda} f_\lambda^{-1} \sum_{n \geq 0} (-E)^n \quad (7.7)$$

and $F^\#$ is nowhere zero on $U^\#$.

Proof. If f_λ is zero free, $f_\lambda^{-1} \in \text{Hol}(U)$ and the geometric series is Hahn summable, since $\text{supp}(E^n) \subseteq \langle \text{supp } E \rangle$ with only finitely many words of each exponent (3.2); multiplying by $1 + E$ verifies (7.7). Nonvanishing of $F^\#$ follows from (7.4). Conversely an inverse forces the product of the two leading coefficient functions to be 1, so f_λ cannot vanish anywhere. $\square$

Proposition 7.14 (Infinitesimal functional calculus). (Class (b).) *For $H \in \mathcal{H}_{>0}(U)$ and any $\varphi \in \mathbb{C}[[X]]$, the substitution $\varphi(H) = \sum_n \varphi_n H^n$ lies in $\mathcal{H}(U)$, with support inside $\langle \text{supp } H \rangle$. In particular*

$$(1 + H)^{-1} = \sum_{n \geq 0} (-H)^n, \quad \log(1 + H) = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} H^n, \quad e^H = \sum_{n \geq 0} \frac{H^n}{n!} \quad (7.8)$$

are coherent sections, and $\log$ and $\exp$ are mutually inverse isomorphisms between $(\mathcal{H}_{>0}(U), +)$ and $(1 + \mathcal{H}_{>0}(U), \cdot)$.

Proof. Positivity of $\text{supp } H$ and the word-length clause of 3.2 give summability; the coefficient of each exponent is a finite sum of products of the f_γ , hence holomorphic on U . No growth condition on (φ_n) is needed, and none is imposed. The inverse-isomorphism claim is the formal identity $\exp \log(1 + X) = 1 + X$ evaluated inside a summable family. $\square$

7.5 Support-controlled composition

Theorem 7.15 (Admissible composition). (Class (b).) *Let $V, U \subseteq \mathbb{C}$ be connected ordinary domains, $F \in \mathcal{H}(V)$, and let*

$$G = g_0 + E \in \mathcal{H}(U), \quad g_0 \in \text{Hol}(U) \text{ with } g_0(U) \subseteq V, \quad E \in \mathcal{H}_{>0}(U).$$

Then

$$F \circ G := \sum_{\gamma \in \text{supp } F} \sum_{n \geq 0} \frac{f_\gamma^{(n)} \circ g_0}{n!} E^n t^\gamma \quad (7.9)$$

belongs to $\mathcal{H}(U)$, satisfies $(F \circ G)^\# = F^\# \circ G^\#$ on $U^\#$, and obeys the chain rule $(F \circ G)' = ((F' \circ G) \cdot G')$.

Proof. The support of (7.9) lies in $\text{supp } F + \langle \text{supp } E \rangle$, well ordered with finite multiplicity at each exponent by 3.2; each coefficient is a finite sum of products of ordinary holomorphic functions on U , hence lies in $\text{Hol}(U)$. At $z \in U^\#$, $G^\#(z)$ has standard part $g_0(\text{st } z) \in V$, so $G^\#(z) \in V^\#$ and the right-hand side is defined; expanding both sides at $\text{st } z$ and using the ordinary formal Taylor composition identity inside one summable family gives equality, and injectivity (Theorem 7.8) transports the chain rule from the evaluated identity back to the sections. $\square$

The shape of the hypothesis is the theorem's content. The inner section must be an *ordinary holomorphic map plus a positive-support perturbation*. Composition with an inner section having a negative exponent, or having an exponent-zero part that is not an ordinary holomorphic map into V , is not defined here and is not claimed: the support monoid $\langle \text{supp } E \rangle$, which supplies the finiteness clause, exists only because $\text{supp } E > 0$.

7.6 Affine chart changes

Coherence is a property relative to a chart. The following criterion says exactly which affine changes of chart preserve it.

Proposition 7.16 (Chart-change criterion). (Class (b).) *Let $F \in \mathcal{H}(U)$ and let $z = \alpha + \beta w$ with $\alpha, \beta \in \mathcal{O}$ finite. Write $\alpha = \alpha_0 + \delta_\alpha$ and $\beta = \beta_0 + \delta_\beta$ with $\alpha_0, \beta_0 \in \mathbb{C}$ and $\delta_\alpha, \delta_\beta \in \mathfrak{m}$. If V is a connected ordinary domain with $\alpha_0 + \beta_0 V \subseteq U$, then*

$$F(\alpha + \beta w) = \sum_{s \in \text{supp } F} \sum_{n \geq 0} \frac{f_s^{(n)}(\alpha_0 + \beta_0 w)}{n!} (\delta_\alpha + \delta_\beta w)^n t^s \quad (7.10)$$

defines a section of $\mathcal{H}(V)$ whose evaluation is $w \mapsto F^\#(\alpha + \beta w)$. The degenerate case $\beta_0 = 0$ is allowed, provided $\alpha_0 \in U$.

Proof. All support growth comes from the fixed positive supports of δ_α and δ_β , so the total support lies in $\text{supp } F + \langle \text{supp } \delta_\alpha \cup \text{supp } \delta_\beta \rangle$ and 3.2 applies; the coefficients are finite sums of $f_s^{(n)}(\alpha_0 + \beta_0 w)$ times polynomials in w , hence holomorphic on V . When $\beta_0 = 0$ the inner argument is the constant $\alpha_0 \in U$ and the coefficients are polynomials in w ; the series is still summable for the same reason. $\square$

Remark 7.17 (Where the criterion fails, and why it matters). An *infinite* normalized coefficient β does not satisfy the criterion: the decomposition $\beta = \beta_0 + \delta_\beta$ with $\beta_0 \in \mathbb{C}$ and $\delta_\beta \in \mathfrak{m}$ does not exist, and the powers β^n have decreasing valuations. Thus substitution into a general infinite Taylor series is not justified by this criterion; finite polynomial substitution remains valid. This is the exact mechanism by which a function can be coherent in every finite translated chart and still fail to be coherent at all scales, and it is why the canonical exponential Exp is excluded from the all-scale class of Section 13 (13.12) although it is locally coherent. The rigidity theorem states the exclusion; the failure of (7.10) for infinite β explains it.

7.7 The identity theorem

Theorem 7.18 (Identity theorem for coherent sections). (Class (b).) *Let $U \subseteq \mathbb{C}$ be connected and $F \in \mathcal{H}(U)$. Each of the following four conditions implies $F = 0$.*

- (i) *$F^\#$ vanishes at points of $U^\#$ whose distinct standard parts have an accumulation point in U in the ordinary complex topology.*
- (ii) *All fine derivatives $(F^{(n)})^\#(z_0)$, $n \geq 0$, vanish at one single point $z_0 \in U^\#$.*
- (iii) *$F^\#$ vanishes on a nonempty fine-open subset of $U^\#$.*
- (iv) *$F^\#$ vanishes as a germ at a single point of $U^\#$.*

In particular two coherent sections agreeing under any one of these conditions are equal, and their evaluations agree on all of $U^\#$.

Proof. (i) By Corollary 7.9 the leading coefficient function f_λ , $\lambda = v_H(F)$, vanishes at each of the indicated standard parts. If these accumulate in U , the ordinary identity theorem gives $f_\lambda \equiv 0$, contradicting $\lambda = v_H(F)$ unless $F = 0$.

(ii) Put $a = \text{st } z_0$. Theorem 7.10 applied to every $F^{(j)}$ with increment $-(z_0 - a)$ gives $(F^{(j)})^\#(a) = \sum_{n \geq 0} (F^{(j+n)})^\#(z_0) (- (z_0 - a))^n / n! = 0$ for all j . At the ordinary point a these values are already in normal form, so uniqueness of normal forms yields $f_\gamma^{(j)}(a) = 0$ for all γ and all j . Each coefficient has identically zero Taylor series at a and hence vanishes on the connected U .

(iii) A fine-open set contains a point together with a ball of some positive surreal radius; repeated differentiation of the zero function there gives (ii) at that point.

(iv) The zero-germ hypothesis at z_0 supplies a fine neighborhood on which $F^\#$ vanishes. Thus (iv) is the same hypothesis as (iii) with a specified point, and (iii) already gives $F = 0$. No preparation theorem is needed for this implication. Apply the result to $F - G$ for the last assertion. $\square$

Remark 7.19 (Two things the identity theorem does not say). The accumulation in (i) is accumulation of *standard parts* in $\mathbb{C}$. Nothing here asserts, and nothing in this article may assert, the existence of a nontrivial set-sized convergent sequence in the fine topology: by the discreteness theorem of Section 4 (4.2) every set is closed and discrete and every set-indexed convergent net is eventually equal to its limit.

Second, the theorem is false in the germ class (a). The function L of the locally constant germ-analytic witness of Section 5 is germ-analytic on all of $\mathbb{K}$ and vanishes on a nonempty fine-open class without being zero. Conditions (ii)–(iv) above do not force global vanishing in the germ class. In the coherent class, monad-wide Taylor expansion and the ordinary identity theorem for each coefficient function supply that propagation.

Theorem 7.20 (Gluing over the ordinary base). (Class (b).) *Let a connected ordinary domain U be covered by connected ordinary open sets U_j , and let $F_j \in \mathcal{H}(U_j)$ have $F_j^\# = F_k^\#$ on $(U_j \cap U_k)^\#$ whenever that is nonempty. Then there is a unique $F \in \mathcal{H}(U)$, with a single well-ordered support, restricting to each F_j . Restriction to a nonempty connected ordinary open subset is injective. On a disconnected base one defines sections component by component, allowing different supports on different components; this yields a class-valued sheaf over the ordinary complex base.*

Proof. On a nonempty overlap, evaluation at ordinary points and uniqueness of normal forms force $f_{j,\gamma} = f_{k,\gamma}$ for every γ . A holomorphic function that is nonzero on the connected U_j cannot vanish identically on a nonempty open overlap, so the *exact* supports of F_j and F_k coincide there. The intersection graph of a connected open cover is connected, so one common well-ordered support S propagates through it. At each $\gamma \in S$ the coefficient functions glue to an element of $\text{Hol}(U)$, and coefficientwise uniqueness gives uniqueness of F . Injectivity of restriction is the ordinary identity theorem applied to each coefficient. $\square$

This is a sheaf over the *ordinary* base. It is not an assertion that arbitrary fine-local analytic data glue to a coherent section; that assertion is false, and the locally constant germ-analytic witness of Section 5 is the counterexample. Requiring one well-ordered support across arbitrarily many components of a disconnected base is a strictly stronger and unmotivated condition, and it is not imposed.

7.8 Canonical classical lifts

Definition 7.21 (Canonical lift). For $f \in \text{Hol}(U)$, its *canonical lift* is the evaluation of the coherent section supported at exponent 0 alone:

$$f^\#(a + \varepsilon) = \sum_{n \geq 0} \frac{f^{(n)}(a)}{n!} \varepsilon^n, \quad a \in U, \varepsilon \in \mathfrak{m}. \quad (7.11)$$

The same superscript is used for lifts and for evaluations of coherent sections because a lift *is* a section, namely one supported at exponent zero. Class (c) denotes the lifts, class (b) the coherent sections.

Theorem 7.22 (Functoriality of the lift). (Class (c).) $f \mapsto f^\#$ is an injective $\mathbb{C}$ -algebra homomorphism $\text{Hol}(U) \rightarrow \{U^\# \rightarrow \mathbb{K}\}$, and

$$\text{st}(f^\#(z)) = f(\text{st } z), \quad (f^\#)' = (f')^\#,$$

and

$$(g \circ f)^\# = g^\# \circ f^\# \quad \text{if } f(U) \subseteq V, \ g \in \text{Hol}(V),$$

together with the reflection identity $(f^*)^\#(z) = \overline{f^\#(\bar{z})}$ for $f^*(Z) = \overline{f(\bar{Z})} \in \text{Hol}(U^*)$. Lifting commutes with restriction and with ordinary analytic continuation, because one continues the ordinary coefficient function first and then applies (7.11).

Proof. This is Theorem 7.8 for the support $\{0\}$, together with one extra point: at $z = c + \varepsilon$ the difference $f^\#(z) - f(c)$ is infinitesimal, so substituting its Hahn-summed Taylor series into the Taylor series of g is licensed by the positive-support clause of 3.2; collection by total degree is exactly the ordinary formal chain rule. Taking the exponent-zero coefficient of (7.11) gives $\text{st}(f^\#(z)) = f(\text{st } z)$. Conjugation is coefficientwise. $\square$

Proposition 7.23 (A lift creates no displaced zeros). (Class (c).) *Let $f \in \text{Hol}(U)$ be nonzero and $a \in U$ a zero of f of order m . Then the zeros of $f^\#$ in the monad $a + \mathfrak{m}$ are exactly the single point a , with multiplicity m ; and $f^\#$ has no zeros over a non-zero of f .*

Proof. Write $f(Z) = (Z - a)^m u(Z)$ with u ordinary holomorphic and $u(a) \neq 0$ on a small disk about a . Then $f^\#(a + \varepsilon) = \varepsilon^m u^\#(a + \varepsilon)$, and $\text{st}(u^\#(a + \varepsilon)) = u(a) \neq 0$, so the second factor is a unit; hence $\varepsilon = 0$ is the only solution in the monad. The second assertion is Corollary 7.9 for the support $\{0\}$. $\square$

Remark 7.24 (The point of the previous proposition). A canonical lift cannot split an ordinary multiple zero into infinitesimally separated roots. The splitting phenomenon of Section 9 therefore requires genuinely surcomplex coefficients at positive exponents: class (b) strictly contains class (c), and the containment is visible already in the zero geometry. The three classes satisfy

$$\begin{aligned} \text{(c) lifts} &\subsetneq \text{(b) } \mathcal{H}(U) \\ &\subsetneq \{ \text{(a) germ-analytic functions on } U^\# \} \\ &\subsetneq \{ \text{bare differentiable functions on } U^\# \}. \end{aligned}$$

The first two inclusions are proper; the factorial series gives the second obstruction (Proposition 7.26), while a locally constant nonconstant function on the halo gives a direct global witness. Strictness of the last inclusion is not proved here (Remark 5.3). In particular L cannot witness it, since L is itself germ-analytic. Beyond the calculus rules, no regularity theorem for the bare differentiable class is proved here.

7.9 The scale-embedding bridge, and its limit

The germ layer and the coherent layer meet in the following theorem. Coherence is *not* restricted to classically convergent germs.

Theorem 7.25 (Scale embedding of arbitrary formal series). (From class (a) to class (b).) *For every $P(X) = \sum_{n \geq 0} a_n X^n \in \mathbb{K}[[X]]$ — including series whose coefficients are infinite and whose ordinary radius of convergence is zero — there are a positive monomial $\rho = t^\lambda$ and a section $\mathcal{P} \in \mathcal{H}(\mathbb{C})$ with*

$$\mathcal{P}^\#(\zeta) = P(\rho\zeta) \quad (\zeta \in \mathcal{O}). \quad (7.12)$$

Every coefficient function of $\mathcal{P}$ may be taken to be a polynomial; in the separated construction each nonzero coefficient is a complex constant times a single power Z^n .

Proof. Choose λ by the separated-block lemma of Section 6 (6.1): a single Δ bounds the — possibly not well ordered — union $\bigcup_n \text{supp}(a_n)$, by the surreal cut property, and $\lambda > 2\Delta$ places the support of $a_n \rho^n$ inside the band $(n\lambda - \Delta, n\lambda + \Delta)$. These bands are pairwise disjoint and increasingly ordered, so writing $a_n \rho^n = \sum_{\gamma \in S_n} c_{n,\gamma} t^\gamma$ the union $S = \bigcup_n S_n$ is well ordered and every $\gamma \in S$ lies in exactly one block, determining $n = n(\gamma)$ uniquely. Set the coefficient function at γ to be $c_{n,\gamma} Z^n$, an entire polynomial. Evaluating at finite ζ reproduces the Hahn summable family $a_n \rho^n \zeta^n$, including its finite binomial expansion about $\text{st } \zeta$, which is (7.12). $\square$

A germ centred at an arbitrary $a \in \mathbb{K}$ is obtained by substituting $z = a + \rho\zeta$. The theorem must be read next to the following obstruction, which concerns a *different chart*.

Proposition 7.26 (An obstruction in the original chart). (Classes (a) and (b), contrasted.) *The series $\sum_{n \geq 0} n! \varepsilon^n$ is a legitimate germ-analytic function on $\mathfrak{m}$, by the substitution corollary of Section 3 (3.3). It is not the restriction to a monad of any $F \in \mathcal{H}(U)$, for any ordinary neighborhood U of 0, having those same derivatives at 0.*

Proof. Such an F would have $(F^{(n)})^\#(0) = n! \cdot n!$ for all n . Evaluating at the ordinary point 0 and using uniqueness of normal forms, the exponent-zero coefficient function $f_0 \in \text{Hol}(U)$ would have $f_0^{(n)}(0)/n! = n!$, i.e. Taylor coefficients $n!$ at 0. No ordinary holomorphic function on a neighborhood of 0 has Taylor coefficients of that size, since a positive complex radius of convergence bounds them geometrically. $\square$

Remark 7.27 (The two statements are compatible). Theorem 7.25 and Proposition 7.26 look like a contradiction and are not. The obstruction concerns the *original* chart: a section on an ordinary neighborhood of zero with the prescribed derivatives at zero. The bridge concerns a *rescaled* chart: after $z = \rho\zeta$ the degree- n term lands in its own Hahn exponent band, each Hahn coefficient becomes a finite sum and hence a polynomial, and what is being represented is a different function of a different variable. The rescaling changes the chart; it does not manufacture a holomorphic function with factorial Taylor coefficients.

The positive result has an exact limit, which must travel with it: every germ has *some* sufficiently small coherent chart, but there is no implication that it has a chart of every larger radius. That distinction is the content of the all-scale rigidity theorem of Section 13 (13.12), and the mechanism of failure is Remark 7.17.

8 Contours, and the order-valued estimate

Section 4 showed that an ordinary circle is not a path in $\mathbb{K}$: every fine-continuous map from a real interval into $\mathbb{K}$ is constant (4.3). A contour integral over $\mathbb{K}$ therefore cannot be a limit of Riemann sums along a fine-continuous parameterization. It must be defined some other way, and Section 5 showed that there is essentially no choice: the contour no-go theorem (5.11) proves that no functional on the ordinary unit circle can simultaneously give the classical value $2\pi i$ for $1/z$ and satisfy the fundamental theorem of calculus for every globally fine-analytic primitive on $\mathbb{K}^\times$. The definition below, and the restriction of the fundamental theorem to *coherent* primitives in §8.6, are forced by that theorem and are not conservatism.

8.1 The coefficientwise contour functional

Definition 8.1 (Contour and coefficientwise integral). A *contour* in U is an ordinary piecewise- C^1 path $\gamma : [a, b] \rightarrow U$ in the *ordinary* complex domain. For $F = \sum_{s \in S} f_s t^s \in \mathcal{H}(U)$ set

$$\int_{\gamma}^H F(\zeta) d\zeta := \sum_{s \in S} \left(\int_{\gamma} f_s(\zeta) d\zeta \right) t^s. \quad (8.1)$$

The integrals on the right are ordinary complex contour integrals; the outer sum is a Hahn sum whose support can only shrink, so the result is a well-defined surcomplex number. The superscript H is permanent.

Proposition 8.2 (Elementary calculus of the functional). (Class (b).) *The map $F \mapsto \int_{\gamma}^H F d\zeta$ is $\mathbb{K}$ -linear — not merely $\mathbb{C}$ -linear — additive under concatenation of contours, sign-reversing under reversal of orientation, and invariant under ordinary piecewise- C^1 reparameterization. The same definition applies verbatim when the coefficient functions are merely continuous along γ , and when they are meromorphic on a common ordinary domain with no pole on γ . It commutes with any Hahn summable family of sections whose coefficientwise integrals are defined.*

Proof. Every clause except $\mathbb{K}$ -linearity is the corresponding ordinary property applied at each exponent. $\mathbb{K}$ -linearity holds because multiplication by a fixed Hahn series has finite convolution at each exponent (3.2), so $\int_{\gamma}^H (cF) = c \int_{\gamma}^H F$ for $c \in \mathbb{K}$ by comparing coefficients. The last clause holds because each exponent of the sum involves only finitely many members of the family. $\square$

Definition 8.3 (Affine transport). Fix $b \in \mathbb{K}$ and $r \in \mathbb{K}^{\times}$, and let $F \in \mathcal{H}(U)$ define a function on $b + rU^{\#}$ by $f(b + rw) = F^{\#}(w)$. For an ordinary contour γ in U set

$$\int_{b+r\gamma}^H f(z) \, dz := r \int_{\gamma}^H F(w) \, dw. \quad (8.2)$$

The centre b may be infinite and $|r|$ may be infinitesimal or infinite.

Differentiation in the transported chart introduces r^{-n} and the differential introduces r ; the two cancel on a simple-pole residue, so Cauchy's formula, root counts and the residue theorem transfer to the chart with the usual affine change-of-variable factors. Two admissible descriptions giving the same pointwise pulled-back integrand have equal integrals, by uniqueness of normal forms applied to their coefficient functions.

Remark 8.4 (What the symbol is not). $\int^H$ is a coefficientwise functional. It is not a fine-topological Riemann integral, and $b + r\gamma$ is a chart-defined algebraic contour, not a claim that its parameterization is fine-continuous into $\mathbb{K}$. The superscript is kept everywhere for exactly this reason. Nothing below proposes $\int^H$ as *the* unique notion of surreal integration; what is proved is that a functional with the two properties in the no-go theorem does not exist, so some restriction of this kind is unavoidable.

8.2 Positivity and the surcomplex ML inequality

A coefficientwise definition does not by itself yield an inequality about the surreal *modulus*. The next lemma supplies one, and it is the load-bearing step of this part.

Lemma 8.5 (Positivity of the coefficientwise integral). *Let $A(s) = \sum_{\lambda \in S} a_{\lambda}(s)t^{\lambda}$ be a family with one common well-ordered support S and continuous real coefficient functions a_{λ} on a compact interval $[\alpha, \beta]$. If $A(s) \geq 0$ in **No** for every ordinary $s \in [\alpha, \beta]$, then*

$$\int_{\alpha}^{H^{\beta}} A(s) \, ds \geq 0.$$

Proof. If $A \equiv 0$ or the interval is degenerate there is nothing to prove. Otherwise discard identically zero coefficient functions and let λ_0 be the least remaining exponent. For each fixed s , the sign of the surreal number $A(s)$ is the sign of its leading nonzero coefficient; since $A(s) \geq 0$ for all s , we get $a_{\lambda_0}(s) \geq 0$ for every s — were it negative at some s , the value $A(s)$ would be negative there. Now a_{λ_0} is continuous, nonnegative and not identically zero, so $\int_{\alpha}^{\beta} a_{\lambda_0} > 0$. Hence the coefficient of t^{λ_0} in $\int^H A$ is positive and all smaller exponents are absent, so the integral is positive. Piecewise versions follow by splitting into finitely many intervals and adding. $\square$

Theorem 8.6 (Surcomplex ML inequality). (Class (b), and more generally any Hahn family with continuous coefficient functions along the path.) *Let γ be a piecewise- C^1 ordinary path of ordinary length L , let F have continuous complex coefficient functions with common well-ordered support along γ , and suppose*

$$|F^{\#}(\gamma(s))| \leq M \quad \text{for every ordinary } s,$$

for one fixed $M \in \mathbf{No}^{\geq 0}$. Then

$$\left| \int_{\gamma}^H F(z) dz \right| \leq LM. \quad (8.3)$$

Proof. Write $I = \int_{\gamma}^H F dz$. If $I = 0$ the bound is trivial. Otherwise put $\alpha = \bar{I}/|I|$, so that $|\alpha| = 1$ and $\operatorname{Re}(\alpha I) = |I|$. Pointwise along the path,

$$M|\gamma'(s)| - \operatorname{Re}(\alpha F^{\#}(\gamma(s))\gamma'(s)) \geq 0,$$

because $|\operatorname{Re}(\alpha F\gamma')| \leq |F||\gamma'| \leq M|\gamma'|$ in the ordered field $\mathbf{No}$. The left-hand side is a Hahn family of continuous real coefficient functions on the parameter interval, with one common support (the union of $\operatorname{supp} M$ and $\operatorname{supp} \alpha + \operatorname{supp} F$). Apply Lemma 8.5 and then $\mathbb{K}$ -linearity of $\int^H$ (Proposition 8.2) to obtain $LM - |I| \geq 0$. $\square$

The separating scalar α is what converts a coefficientwise statement into an ordered-field positivity statement; without it the theorem is not available.

Remark 8.7 (Two Cauchy-type statements that are not in conflict). Theorem 8.6 and the coefficientwise estimates of §8.4 look like a clash and are not. They bound different quantities.

- It is *false* that a single coarse surreal bound on $F^{\#}$ controls every coefficient function uniformly. Witness: $F(w) = w(1 + it)$ on the ordinary unit circle has $|F^{\#}| = \sqrt{1 + t^2} < 1 + t^2$ pointwise, so $M = 1 + t^2$ is a legitimate pointwise bound, whereas the coefficientwise majorant $B_1(F) = 1 + t$ is strictly larger than M . No estimate below asserts the contrary.
- It is *true* that a single surreal bound controls the surreal modulus of the integral, and hence of the derivatives; that is (8.3) and Corollary 8.9. The conclusion is weaker than the false one and is proved above.

The two are consistent on the standard test case: for $F = \varepsilon Z$ with $\varepsilon \in \mathbf{m} \cap \mathbf{No}_{>0}$, the bound on $|z| = R$ is $M = \varepsilon R$, and Corollary 8.9 gives $|(F')^{\#}(0)| \leq \varepsilon$, which is exact. Reading the coefficientwise caution as a prohibition would delete the strongest estimate available, and that reading is rejected here.

8.3 Cauchy's theorem and formula at surcomplex points

Theorem 8.8 (Cauchy's theorem and integral formula). (Class (b).) *Let U be a connected ordinary domain and $F \in \mathcal{H}(U)$.*

- (i) *If the ordinary closed path γ is null-homologous in U , then $\oint_{\gamma}^H F(\zeta) d\zeta = 0$.*
- (ii) *Let Ω be a bounded ordinary Jordan domain with piecewise- C^1 positively oriented boundary $\Gamma = \partial\Omega$ and $\bar{\Omega} \subset U$. Then for every genuinely surcomplex point $z = c + \varepsilon \in \Omega^{\#}$ and every ordinary finite $k \geq 0$,*

$$\frac{(F^{(k)})^{\#}(z)}{k!} = \frac{1}{2\pi i} \oint_{\Gamma}^H \frac{F(\zeta)}{(\zeta - z)^{k+1}} d\zeta, \quad (8.4)$$

the kernel being understood through its uniform Hahn expansion on an ordinary neighborhood of Γ .

Proof. (i) is the ordinary Cauchy theorem applied at each exponent; this is exactly what is inherited coefficientwise from complex analysis, and nothing more.

(ii) Since $c \in \Omega$ is an ordinary interior point, $\zeta - c$ is bounded away from zero on Γ , and on an ordinary neighborhood of Γ

$$\frac{1}{(\zeta - c - \varepsilon)^{k+1}} = \sum_{n \geq 0} \binom{n+k}{k} \frac{\varepsilon^n}{(\zeta - c)^{n+k+1}}.$$

The support of ε is positive, so this is a Hahn summable family with coefficient functions holomorphic near Γ ; its product with F is jointly summable inside $\text{supp } F + \langle \text{supp } \varepsilon \rangle$. Integrating coefficientwise and applying the ordinary Cauchy derivative formula to each f_γ ,

$$\frac{k!}{2\pi i} \oint_{\Gamma} \frac{F(\zeta)}{(\zeta - z)^{k+1}} d\zeta = \sum_{\gamma, n} t^\gamma \varepsilon^n k! \binom{n+k}{k} \frac{f_\gamma^{(n+k)}(c)}{(n+k)!} = \sum_{\gamma, n} t^\gamma \frac{f_\gamma^{(n+k)}(c)}{n!} \varepsilon^n,$$

which is $(F^{(k)})^\#(z)$ by Theorem 7.10. Only finitely many terms are combined at any fixed Hahn exponent, so every interchange is justified by summability and not by an unproved convergence theorem; no real limit is exchanged with a surreal limit. $\square$

Corollary 8.9 (Cauchy estimates from a pointwise surreal bound). (Class (b).) *Let $c \in \mathbb{C}$, let $R > 0$ be ordinary with $\overline{B(c, R)} \subset U$, and suppose $|F^\#(\zeta)| \leq M$ for one fixed $M \in \mathbf{No}^{\geq 0}$ at every point ζ of the ordinary circle $|\zeta - c| = R$. Then for every ordinary finite $n \geq 0$*

$$|(F^{(n)})^\#(c)| \leq \frac{n! M}{R^n}, \quad (8.5)$$

and for $z = c + \varepsilon$ with $\varepsilon \in \mathbf{m}$,

$$|(F^{(n)})^\#(z)| \leq \frac{n! R M}{(R - |\varepsilon|)^{n+1}}. \quad (8.6)$$

Proof. Apply Theorem 8.6 to (8.4) with $L = 2\pi R$ and, for (8.6), with the lower bound $|\zeta - z| \geq R - |\varepsilon|$ on the kernel. The bound M is allowed to be an arbitrary nonnegative surreal, infinite or infinitesimal. $\square$

Corollary 8.10 (Cauchy estimates in an affine chart). (Class (b), in a transported chart.) *Suppose $f(a + rw) = F^\#(w)$ with $a \in \mathbb{K}$, $r \in \mathbf{No}^{>0}$ and $F \in \mathcal{H}(B(0, 2))$. If $|f(a + re^{i\theta})| \leq M$ for every ordinary real θ , then*

$$|f^{(n)}(a)| \leq \frac{n! M}{r^n} \quad (n \geq 0). \quad (8.7)$$

Cauchy's formula, root counts and the residue theorem transfer to this chart with the usual affine factors.

Proof. The chain rule gives $(F^{(n)})^\#(0) = r^n f^{(n)}(a)$. Apply (8.5) on the unit circle, where the bound M may be surreal. The contour assertions follow by substitution in (8.2), the differential $dz = r dw$ cancelling the affine scaling of a simple-pole residue. $\square$

The qualification “coherent chart” is essential and is not decorative: a germ-analytic function has *some* sufficiently small coherent chart (Theorem 7.25), but there is no implication that it has a chart of every larger radius. See Section 13.

8.4 The coefficientwise majorant

The second estimate has a different shape. It bounds each coefficient separately and assembles the bounds into a surreal number, and it is retained as a distinct tool.

Lemma 8.11 (Cauchy–Schwarz majorant lemma). *Let $Z = \sum_s c_s t^s \in \mathbb{K}$ with $c_s \in \mathbb{C}$ and put $B = \sum_s |c_s| t^s \in \mathbf{No}^{\geq 0}$. Then $|Z| \leq B$.*

Proof.

$$B^2 - |Z|^2 = \sum_{s,u} (|c_s||c_u| - \operatorname{Re}(c_s \overline{c_u})) t^{s+u} \geq 0,$$

since every coefficient of this Hahn summable family is a nonnegative real number and the family is summable by 3.2. Both B and $|Z|$ are nonnegative, so $|Z| \leq B$. $\square$

Proposition 8.12 (Coefficientwise Cauchy estimate). (Class (b).) *Let $\overline{B(a, r)} \subset U$ with $a \in \mathbb{C}$ and $r > 0$ ordinary real. For $F = \sum_s f_s t^s \in \mathcal{H}(U)$ put*

$$M_s = \max_{|\zeta - a| = r} |f_s(\zeta)|, \quad B_r(F) = \sum_s M_s t^s \in \mathbf{No}^{\geq 0},$$

ordinary compact maxima followed by a Hahn sum. Then for every ordinary finite $n \geq 0$

$$|(F^{(n)})^\#(a)| \leq \frac{n!}{r^n} B_r(F). \quad (8.8)$$

Proof. The ordinary Cauchy estimates give $|f_s^{(n)}(a)| \leq n! M_s / r^n$ for every s . By Lemma 8.11, the surreal modulus of $(F^{(n)})^\#(a) = \sum_s f_s^{(n)}(a) t^s$ is at most $\sum_s |f_s^{(n)}(a)| t^s$, and increasing nonnegative coefficients preserves the surreal order on a common support. $\square$

Remark 8.13 (The scope of the majorant). $B_r(F)$ is not in general the least upper bound of $|F^\#|$ on a surreal circle; its coefficientwise definition is part of the statement of (8.8). It is not claimed to be a norm. The coefficientwise order used here is much stronger than the surreal order, and the two estimates (8.5) and (8.8) use different input data. They are not logically incomparable as modulus bounds: the majorant lemma gives $|F^\#(\zeta)| \leq B_r(F)$ at every ordinary point of the circle, so the ML estimate with $M = B_r(F)$ implies (8.8). For $F(w) = w(1 + it)$ on $r = 1$ we have $B_1(F) = 1 + t$ while the pointwise bound is $M = 1 + t^2 < B_1(F)$, so the ML estimate is the sharper one there. The coefficientwise bounds still retain information that a single surreal inequality loses: the Liouville and growth theorem 11.9 applies ordinary estimates to each coefficient separately. A pointwise surreal bound always exists on a fixed circle and, by Proposition 11.7, even on the finite halo for macroscopically entire sections; such a bound alone does not recover those coefficientwise growth hypotheses.

8.5 Morera's criterion

Theorem 8.14 (Coefficientwise Morera). *Let S be a well-ordered set and let $A = \sum_{s \in S} a_s t^s$ have ordinary continuous coefficient functions a_s on an ordinary domain U , with contour integrals defined coefficientwise by (8.1). The following are equivalent.*

(i) *Every a_s is holomorphic on U , i.e. $A \in \mathcal{H}(U)$.*

(ii) *$\oint_{\partial T}^H A(\zeta) d\zeta = 0$ for every ordinary triangle T whose closed interior lies in U .*

If the a_s are C^1 , both are further equivalent to the coefficientwise Cauchy–Riemann equations $\partial_{\bar{z}} a_s = 0$ for every s . When they hold, A has a unique coherent evaluation on $U^\#$ by (7.3).

Proof. Uniqueness of normal forms makes the vanishing of the Hahn-valued triangle integral equivalent to the vanishing of every ordinary coefficient integral $\oint_{\partial T} a_s$. Ordinary Morera's theorem applied to each continuous a_s gives (ii) $\Rightarrow$ (i), and ordinary Cauchy (i) $\Rightarrow$ (ii). For C^1 coefficients use the ordinary Cauchy–Riemann criterion at each exponent. Existence and uniqueness of the evaluation are Theorem 7.8. $\square$

The regularity hypothesis is part of the theorem. It is continuity of the *ordinary coefficient functions* on the ordinary domain, and the theorem asserts nothing whatever about a function that is merely continuous, or even differentiable, for the fine topology. Before the conclusion is reached, the object A is a Hahn-valued function on an ordinary domain; no Taylor extension of a merely continuous coefficient to the halo is presumed at that stage.

8.6 Coherent primitives, periods and endpoint corrections

Theorem 8.15 (Coherent primitives). (Class (b).) *Let U be a connected ordinary domain and $F \in \mathcal{H}(U)$.*

- (i) *If U is simply connected and $a_0 \in U$, there is a unique $P \in \mathcal{H}(U)$ with $P' = F$ and $P^\#(a_0) = 0$, and $\text{supp } P \subseteq \text{supp } F$. Every other coherent primitive differs from P by a constant of $\mathbb{K}$, and for an ordinary path γ in U from a_1 to a_2 ,*

$$\int_{\gamma}^H F(\zeta) d\zeta = P^\#(a_2) - P^\#(a_1).$$

- (ii) *For general connected U , a coherent primitive exists if and only if all ordinary closed-path coefficientwise integrals vanish, i.e. $\oint_{\sigma}^H F d\zeta = 0$ for every ordinary closed path σ in U .*

Proof. (i) On a simply connected U each f_s has a unique ordinary primitive p_s with $p_s(a_0) = 0$, by path independence; set $P = \sum_s p_s t^s$. Its support is contained in $\text{supp } F$ — a primitive can only lose exponents, when f_s has an identically zero primitive, which happens only for $f_s \equiv 0$ — so $P \in \mathcal{H}(U)$ and $P' = F$. If $H' = 0$ then every $h'_s = 0$, and connectedness of U makes each h_s a complex constant, so H is a constant of $\mathbb{K}$; this is uniqueness. The endpoint identity holds at each exponent.

(ii) Vanishing of the Hahn period $\oint_{\sigma}^H F$ is, by uniqueness of normal forms, equivalent to vanishing of every ordinary period $\oint_{\sigma} f_s$, which is the classical criterion for each f_s to have an ordinary primitive on U . Run the construction of (i) with those primitives. $\square$

For endpoints that are genuinely surcomplex, $z_j = a_j + \varepsilon_j \in U^\#$ with $a_j \in U$ and $\varepsilon_j \in \mathfrak{m}$, define the microscopic endpoint correction

$$T_F(a, \varepsilon) = \sum_{n \geq 0} \frac{(F^{(n)})^\#(a)}{(n+1)!} \varepsilon^{n+1}, \quad (8.9)$$

Hahn summable by the support calculation of Theorem 7.8, and set, for an ordinary path γ from a_1 to a_2 ,

$$\int_{z_1}^{H z_2} F dz := -T_F(a_1, \varepsilon_1) + \int_{\gamma}^H F dz + T_F(a_2, \varepsilon_2). \quad (8.10)$$

For a coherent primitive P this equals $P^\#(z_2) - P^\#(z_1)$, by Theorem 7.10; on a simply connected base it is independent of the ordinary path chosen. This gives a concrete endpoint integral between surcomplex endpoints without pretending that the connecting ordinary path is fine-continuous.

Remark 8.16 (Uniqueness belongs to the coherent category only). The uniqueness clause in Theorem 8.15(i) is a statement about class (b) and is false outside it. An arbitrary differentially holomorphic primitive may still be modified by a locally constant function: $P + L$ has the same derivative as P everywhere, with L as in the locally constant germ-analytic witness of Section 5. The restriction cannot be lifted, and the theorem that shows it cannot is the contour no-go theorem (5.11).

Remark 8.17 (The unit circle, and the global logarithm). For $F(z) = 1/z$ on $\mathbb{C}^\times$ the coefficient-wise unit-circle period is $\oint_{|z|=1} dz/z = 2\pi i \neq 0$. By Theorem 8.15(ii) there is therefore *no coherent primitive of $1/z$ on $(\mathbb{C}^\times)^\#$* — no coherent global logarithm there.

This is consistent with, and is the exact complement of, the global fine-analytic logarithm $\mathcal{L}$ constructed in Section 12 (12.11) on all of $\mathbb{K}^\times$, which satisfies $\mathcal{L}'(z) = 1/z$ everywhere. That function is germ-analytic, class (a); its restriction to $(\mathbb{C}^\times)^\#$ is not the evaluation of an element of $\mathcal{H}(\mathbb{C}^\times)$, as the computation above proves. On an ordinary domain avoiding the argument cut, it does agree with the canonical lift of an ordinary logarithm branch. If it were coherent, the fundamental theorem (Theorem 8.15(i)) would force the unit-circle integral to be 0 rather than $2\pi i$. The monodromy obstruction survives in the coherent category and disappears in the more permissive local one; that is the whole content of the no-go theorem of Section 5, and it is why the fundamental theorem above is stated only for coherent primitives.

9 Preparation, division, and infinitesimal zero geometry

Everything in this part is a statement about the *coherent* class $\mathcal{H}(U)$ of Section 7: one common connected ordinary domain U , one ordinary holomorphic coefficient function per exponent, well-ordered set support. Nothing here is claimed for the germ-analytic class of Section 6, and nothing here is a statement about canonical lifts. The reason is structural rather than cautious: the whole mechanism below is ordinary Hermite division applied to the coefficient functions *simultaneously on one domain*, and it is exactly that common domain which ties the monads over different standard points together. A family of germs whose radii shrink to zero admits no such division, and the zero-cluster theorem is false for it.

Throughout, $U \subseteq \mathbb{C}$ is a connected ordinary domain, $\Omega \subset \mathbb{C}$ is a bounded Jordan domain with piecewise C^1 boundary $\Gamma = \partial\Omega$ and $\overline{\Omega} \subset U$, and V denotes an ordinary open neighbourhood of $\overline{\Omega}$ with $\overline{\Omega} \subset V \subseteq U$. We write a nonzero $F \in \mathcal{H}(U)$ in the normalized form

$$F = t^\beta(f_0 + E), \quad \beta = v_H(F), \quad f_0 = f_{v_H(F)} \in \text{Hol}(U) \setminus \{0\}, \quad \text{supp } E \subset \mathbf{No}_{>0}, \quad (9.1)$$

so that f_0 is the leading coefficient *function* of F — not the leading coefficient of any one of its values — and $E \in \mathcal{H}_{>0}(U)$.

9.1 Ordinary Hermite division data

Definition 9.1 (The division data attached to (f_0, Ω)). Assume f_0 has no zero on Γ . Then f_0 has finitely many zeros $c_1, \dots, c_r$ in Ω , of multiplicities $m_1, \dots, m_r$, because f_0 is holomorphic near the compact set $\overline{\Omega}$ and its zeros are isolated. Put

$$d = \sum_{j=1}^r m_j, \quad P_0(Z) = \prod_{j=1}^r (Z - c_j)^{m_j} \in \mathbb{C}[Z],$$

choose V containing no further zeros of f_0 , and set

$$u_0 = f_0/P_0 \in \text{Hol}(V)^\times.$$

If f_0 has no zero in Ω , take $d = 0$, $P_0 = 1$, $u_0 = f_0$.

The compactness used here is compactness of an ordinary complex set. No compactness of a surcomplex ball is invoked anywhere in this part, and none is available.

Lemma 9.2 (Ordinary division, extended coefficientwise). *There are $\mathbb{C}$ -linear maps $T: \text{Hol}(V) \rightarrow \mathbb{C}[Z]_{<d}$ and $Q_0: \text{Hol}(V) \rightarrow \text{Hol}(V)$ with*

$$h = Th + P_0 Q_0 h \quad (h \in \text{Hol}(V)), \quad (9.2)$$

and the decomposition (9.2) is unique among pairs (polynomial of degree $< d$, element of $\text{Hol}(V)$). Applied exponent by exponent, T and Q_0 extend to $\mathbb{K}$ -linear maps on $\mathcal{H}(V)$ that never enlarge a support: $\text{supp } Th$ and $\text{supp } Q_0 h$ are contained in $\text{supp } h$. For $d = 0$ one takes $T = 0$ and $Q_0 = \text{id}$.

Proof. Th is the Hermite interpolation polynomial: the unique polynomial of degree $< d$ whose derivatives through order $m_j - 1$ at c_j agree with those of h , for every j . Then $h - Th$ vanishes to order m_j at each c_j , so $(h - Th)/P_0$ has removable singularities and defines $Q_0 h \in \text{Hol}(V)$. If $Th + P_0 Q_0 h = \tilde{T}h + P_0 \tilde{Q}_0 h$ then the difference of the polynomial parts has degree $< d$ and is divisible by P_0 , hence is zero. The coefficientwise extension is immediate, and it cannot create exponents because it acts inside each exponent separately. That last clause is not bookkeeping: it is what makes the recursion of Theorem 9.3 live inside the Neumann [3] monoid of a fixed positive set. $\square$

9.2 Hahn–Weierstrass preparation: the global form

The theorem is stated on the whole Jordan domain at once. The local single-zero form is recovered as Corollary 9.6. The order matters: the exact weighted argument principle (Theorem 10.14) and the contour-moment reconstruction (Theorem 9.17) both require one monic polynomial valid over the entire domain, and neither is derivable from local factorizations at the separate zeros, since assembling those would require a common factorization that the local theorem does not supply.

Theorem 9.3 (Hahn–Weierstrass preparation, global form). *Let $F \in \mathcal{H}(U)$ be nonzero, normalized as in (9.1), and suppose f_0 has no zero on Γ . Let P_0, u_0, d, V be the division data of Definition 9.1, and put*

$$H = E/u_0 \in \mathcal{H}_{>0}(V), \quad S = \text{supp } H \subset \mathbf{No}_{>0}.$$

Then there exist unique

$$p \in \mathbb{K}[Z], \quad \deg p < d, \quad q \in \mathcal{H}_{>0}(V)$$

such that

$$F = t^\beta u_0 (1 + q) W, \quad W = P_0 + p. \quad (9.3)$$

Every coefficient of p lies in $\mathfrak{m}$, the factor $u_0(1 + q)$ is a unit of $\mathcal{H}(V)$, and

$$\text{supp } p \cup \text{supp } q \subset \langle S \rangle \setminus \{0\},$$

where $\langle S \rangle$ is the Neumann monoid generated by S . The degree d is the total ordinary zero count of f_0 in Ω . When $d = 0$ one has $W = 1$ and $1 + q = 1 + H$.

Proof. Dividing (9.1) by $t^\beta u_0$ turns the assertion into the equation

$$H = p + P_0 q + pq \quad (9.4)$$

for p of degree $< d$ and q of positive support. Introduce a formal bookkeeping parameter s and look for homogeneous corrections $p = \sum_{n \geq 1} p_n$, $q = \sum_{n \geq 1} q_n$, where p_n, q_n carry formal degree n in H . Define

$$p_1 = TH, \quad q_1 = Q_0H, \quad C_n = \sum_{j=1}^{n-1} q_j p_{n-j}, \quad p_n = -TC_n, \quad q_n = -Q_0C_n \quad (n \geq 2). \quad (9.5)$$

All operations take place in $\mathcal{H}(V)$, and each p_n is a polynomial of degree $< d$ by construction. By induction, using Lemma 9.2 (the operators do not enlarge supports) and the fact that a product adds supports,

$$\text{supp } p_n \cup \text{supp } q_n \subset S^{(n)} := \{\sigma_1 + \cdots + \sigma_n : \sigma_i \in S\}.$$

Since S is a well-ordered set of positive surreals, Neumann's lemma in its finite-word form (Sections 3–4) says that $\langle S \rangle = \bigcup_{n \geq 0} S^{(n)}$ is well ordered and that each of its elements is $S^{(n)}$ -representable for only finitely many n , with finitely many words in each case. Hence $(p_n)_{n \geq 1}$ and $(q_n)_{n \geq 1}$ are Hahn summable, and

$$p = \sum_{n \geq 1} p_n, \quad q = \sum_{n \geq 1} q_n$$

exist, with supports in $\langle S \rangle \setminus \{0\}$. Because the polynomial degrees are uniformly bounded by d , the coefficients of p are single elements of $\mathfrak{m}$ and p is one polynomial over $\mathbb{K}$.

Equation (9.2) makes the degree-one part of (9.4) read $H = p_1 + P_0q_1$, and for $n \geq 2$ the degree- n part reads $0 = p_n + P_0q_n + C_n$, which is (9.5). Summing the identities — a licensed Hahn sum of a summable family, evaluated at the bookkeeping value $s = 1$, and not a claim that the partial sums converge for any topology — gives (9.4), hence (9.3). The leading coefficient function of $u_0(1 + q)$ is u_0 , nowhere zero on V , so that factor is a unit of $\mathcal{H}(V)$ by the unit criterion of Section 7.

Uniqueness. Suppose (p, q) and $(\tilde{p}, \tilde{q})$ both solve (9.4) with the stated support conditions, and let α be the least exponent occurring in $\text{supp}(p - \tilde{p}) \cup \text{supp}(q - \tilde{q})$, assuming this set is nonempty. The difference of the two copies of the nonlinear term pq is $(p - \tilde{p})q + \tilde{p}(q - \tilde{q})$; each summand is a difference of valuation $\geq \alpha$ times a factor of strictly positive support, so it has valuation $> \alpha$ and contributes nothing at exponent α . Comparing coefficients of t^α in (9.4) therefore gives

$$[t^\alpha](p - \tilde{p}) + P_0[t^\alpha](q - \tilde{q}) = 0$$

in $\text{Hol}(V)$, with the first summand a polynomial of degree $< d$. Uniqueness in (9.2) forces both to vanish, contradicting the choice of α . $\square$

Remark 9.4 (A second proof, by recursion on the exponent). The same factorization may be built by transfinite recursion along the well-ordered set $M = \langle S \rangle_{>0}$ instead of along the formal degree. Writing $H = \sum_{\delta} t^\delta h_\delta$, $p = \sum_{\delta} t^\delta p_\delta$, $q = \sum_{\delta} t^\delta q_\delta$, equation (9.4) at exponent δ reads

$$r_\delta = h_\delta - \sum_{\substack{\alpha, \beta \in M \\ \alpha + \beta = \delta}} p_\alpha q_\beta, \quad p_\delta = T(r_\delta), \quad q_\delta = Q_0(r_\delta),$$

the sum being finite with all indices strictly below δ . This is ordinary transfinite recursion on a well-ordered set; limit stages require no analytic limit. The two constructions produce the same (p, q) by the uniqueness clause.

Remark 9.5 (What the proof does not use). No norm on $\mathcal{H}(V)$ is introduced, no contraction estimate is made, no completeness is invoked, and it is *not* assumed that $n \min S$ is cofinal in the surreal value class. The last point is the one that would be needed for a metric fixed-point

argument and is exactly what a high-rank Hahn field does not supply: corrections of valuation $n\delta$ need not eventually be smaller than every prescribed surreal tolerance. The construction avoids the issue because coefficients are assigned on a well-ordered support with finitely many earlier contributions at each exponent. The evaluation at $s = 1$ in the first proof is a licensed normal sum, not convergence of a sequence of partial sums.

Corollary 9.6 (Local preparation at a single zero). *Let $F \in \mathcal{H}(U)$ be nonzero with leading coefficient function f_0 , and let $a \in U$ satisfy $\text{ord}_a(f_0) = m \geq 1$. On a sufficiently small ordinary disk $a + D_\varrho \subset U$ on which f_0 has no other zero,*

$$F(a + w) = t^\beta \left(w^m + \sum_{j=0}^{m-1} p_j w^j \right) V(w), \quad p_j \in \mathfrak{m}, \quad (9.6)$$

where $V \in \mathcal{H}(D_\varrho)$ has leading coefficient function $u_0(w) = f_0(a + w)/w^m$, nowhere zero on D_ϱ ; hence V is a unit and $V^\#$ vanishes nowhere on $D_\varrho^\#$.

Proof. This is Theorem 9.3 with $\Omega = a + D_\varrho$, for which $P_0(w) = w^m$ and $r = 1$. Equivalently, one may run the recursion (9.5) with the single-zero division operators

$$T_m g = \sum_{j=0}^{m-1} \frac{g^{(j)}(0)}{j!} w^j, \quad R_m g = \frac{g - T_m g}{w^m},$$

which are the specialization of T, Q_0 to $P_0 = w^m$; the apparent singularity of $R_m g$ at 0 is removable. The most restrictive input form — leading part exactly w^m , i.e. $A = w^m + H$ with $H \in \mathcal{H}_{>0}$ — is obtained by first dividing by u_0 , and the general statement follows by multiplying the unit back in. $\square$

No uniform complex bound on the infinitely many perturbation coefficients is required anywhere above. The common holomorphic domain together with the positive well-ordered support replaces every hypothesis of that kind; in particular the coefficient functions of E may grow arbitrarily fast with the exponent.

9.3 Division by the prepared polynomial

Theorem 9.7 (Hahn–Weierstrass division). *Let $W = P_0 + p$ be the monic polynomial of Theorem 9.3. Every $A \in \mathcal{H}(V)$ has a unique decomposition*

$$A = WQ + R, \quad Q \in \mathcal{H}(V), \quad R \in \mathbb{K}[Z], \quad \deg R < d. \quad (9.7)$$

Proof. Let $\mathcal{L}(Q) = Q_0(pQ)$, a $\mathbb{K}$ -linear operator on $\mathcal{H}(V)$. Each application of $\mathcal{L}$ adds an element of the positive set $\text{supp } p$ to the support, while Q_0 alters only coefficient functions. Hence the family $((-\mathcal{L})^n Q_0 A)_{n \geq 0}$ is supported in $\text{supp } A + \langle \text{supp } p \rangle$, with finitely many contributions at each exponent by Neumann’s lemma, and

$$Q = \sum_{n \geq 0} (-\mathcal{L})^n Q_0 A \in \mathcal{H}(V)$$

is a Hahn sum satisfying $Q = Q_0 A - Q_0(pQ) = Q_0(A - pQ)$. Put $R = T(A - pQ)$; since the degrees are uniformly bounded by d , R is a single polynomial over $\mathbb{K}$. Applying (9.2) to $A - pQ$ gives $A - pQ = P_0 Q + R$, i.e. (9.7).

If $WQ + R = 0$ with $(Q, R) \neq (0, 0)$, let α be the least exponent occurring in Q or R . The term pQ has support in $\text{supp } p + \text{supp } Q$, all of whose elements exceed α , so at exponent α the identity reduces to $P_0 q + r = 0$ with $q \in \text{Hol}(V)$ and $\deg r < d$. Uniqueness in (9.2) forces $q = r = 0$, a contradiction. For $d = 0$ the statement is the identity $A = 1 \cdot A + 0$. $\square$

9.4 Roots of a monic polynomial with infinitesimal corrections

Lemma 9.8 (Finite roots and infinitesimal corrections). *Let $W(Z) = Z^d + \sum_{j < d} b_j Z^j$ with every $b_j \in \mathcal{O}$ and $d \geq 1$. Then W has exactly d roots in $\mathbb{K}$ counted with algebraic multiplicity, and every root is finite. If all the b_j are infinitesimal, every root lies in $\mathfrak{m}$. Moreover, if all the b_j lie in a set-sized subfield $\mathbb{K}_G = \mathbb{C}((t^G))$ for a divisible set-sized ordered subgroup $G \subset \mathbf{No}$, then all the roots lie in $\mathbb{K}_G$.*

Proof. $\mathbb{K}$ is algebraically closed, which gives the d roots. Let ξ be a root. If $v(\xi) < 0$ then $v(b_j \xi^j) - v(\xi^d) = v(b_j) + (j - d)v(\xi) > 0$ for $j < d$, so the monic term ξ^d is the unique term of least valuation and cannot be cancelled — this is the exact place where a root of negative valuation is excluded, and it uses only the ordered-field valuation, no analysis. Thus every root is finite. If all the b_j are infinitesimal and $v(\xi) = 0$, reducing $W(\xi) = 0$ modulo $\mathfrak{m}$ gives $\text{st}(\xi)^d = 0$, impossible for $\text{st}(\xi) \neq 0$. In this special case every root is infinitesimal (including the possible root 0). For the last clause, $\mathbb{K}_G$ is itself algebraically closed (Poonen), so it already splits W ; passing to the full class field adds no further roots. $\square$

9.5 The zero-cluster theorem

Theorem 9.9 (Exact zero clustering; specialization of the zero divisor). *Let $F \in \mathcal{H}(U)$ be nonzero, normalized as in (9.1), with f_0 having no zero on Γ , and let d, W be as in Theorem 9.3. Then:*

- (i) $F^\#$ has exactly d zeros in the halo $\Omega^\#$, counted with analytic multiplicity, and they are precisely the roots in $\mathbb{K}$ of W ;
- (ii) for every $a \in \Omega$, the monad $a + \mathfrak{m}$ contains exactly $\text{ord}_a(f_0)$ of them; in particular there are no zeros over points where f_0 does not vanish, a simple zero of f_0 lifts to exactly one simple surcomplex zero, and a zero of order m splits into exactly m roots (repetitions allowed);
- (iii) equivalently, with zero divisors carrying multiplicities,

$$\text{st}_*(\text{div}_0 F) = \text{div}_0(f_0); \tag{9.8}$$

- (iv) the zeros lie in a set-sized subfield $\mathbb{K}_G$, and the full zero set of $F^\#$ in $U^\#$ is at most countable.

Proof. (i) By (9.3), $F^\# = t^\beta u_0^\# (1 + q)^\# W^\#$ on $V^\#$. The first three factors are units: u_0 is nowhere zero on V , and $1 + q$ has inverse $\sum_{n \geq 0} (-q)^n$, a Hahn sum by Neumann's lemma, so $(1 + q)^\#$ takes values in $1 + \mathfrak{m}$ at every halo point. Hence the zeros of $F^\#$, with their local analytic multiplicities, are exactly the roots of W with their polynomial multiplicities: multiplying by a germ-invertible factor changes neither the location nor the least nonzero Taylor degree at a root.

The lower coefficients of $W = P_0 + p$ are finite, so by Lemma 9.8 every root of W is finite, so it has a standard part. Applying the standard-part ring homomorphism to the factorization $W = \prod_{k=1}^d (Z - \xi_k)$ gives

$$P_0(Z) = \prod_{k=1}^d (Z - \text{st } \xi_k) \quad \text{in } \mathbb{C}[Z],$$

so the $\text{st } \xi_k$ are exactly the zeros $c_1, \dots, c_r$ of f_0 in Ω , each occurring m_j times. This proves (ii) and (9.8), and also proves that all d roots lie in $\Omega^\#$. Over a point where $f_0(a) \neq 0$ the faithful-evaluation estimate of Section 7, $\text{st}(t^{-\beta} F^\#(a + \epsilon)) = f_0(a)$, already excludes a zero; so does the standard-part identity just derived.

(iv) The coefficients of W form a set, so they lie in some $\mathbb{K}_G = \mathbb{C}((t^G))$ with G a divisible set-sized ordered subgroup of $\mathbf{No}$; by Lemma 9.8 the roots lie in $\mathbb{K}_G$ too. For countability, apply Corollary 9.6 at each zero of f_0 in U : the zeros of the nonzero ordinary holomorphic function f_0 on the planar domain U form a discrete, hence at most countable, set, and each standard-part fibre contains at most $\text{ord}_a(f_0)$ distinct roots. The union of countably many finite sets is at most countable. $\square$

Remark 9.10 (Three things the theorem does not say). First, the count is exact but it is *not* a reduction modulo infinitesimals: the root positions may carry surreal terms at arbitrarily deep scales, indeed at scales below every finite power of a chosen infinitesimal, while the integer d is exact. Second, $\Omega^\# = \text{st}^{-1}(\Omega)$ is not the surcomplex ball over Ω : points infinitesimally inside a standard boundary point are absent from the halo, so a root sitting in the monad of a boundary point is counted by neither side. Third, neither this theorem nor the Rouché statements below count zeros whose standard-scale coordinate is infinite; those live outside every halo of a bounded ordinary domain and require a different affine scale chart (Section 13).

Corollary 9.11 (Simple-root lifting and its displacement). *Suppose $f_0(c) = 0$ and $a := f'_0(c) \neq 0$. Then $F^\#$ has exactly one zero $\xi = c + \delta$ in $c + \mathfrak{m}$, it is simple, and δ has support inside $\langle \text{supp } E \rangle$. Writing $H = E$ and $\sigma = \min \text{supp } H > 0$ when $H \neq 0$,*

$$\delta = -\frac{H^\#(c)}{a} + \frac{H^\#(c)(H')^\#(c)}{a^2} - \frac{f''_0(c)H^\#(c)^2}{2a^3} + R, \quad v(R) \geq 3\sigma. \quad (9.9)$$

In particular, if $E = t^\rho g + (\text{exponents} > \rho)$ with $\rho > 0$, the leading correction is

$$\xi = c - t^\rho \frac{g(c)}{f'_0(c)} + o(t^\rho), \quad (9.10)$$

and if $g(c) = 0$ the displayed correction vanishes and the actual displacement is smaller.

Proof. Existence, uniqueness and simplicity are Theorem 9.9(ii) with $m = 1$; there $W = Z - c + p_0$ and $\xi = c - p_0$, whose support lies in $\langle \text{supp } E \rangle$ by Theorem 9.3. For (9.9), substitute $\delta = \delta_1 + \delta_2 + \dots$ into the Taylor identity

$$0 = a\delta + \frac{1}{2}f''_0(c)\delta^2 + \dots + H^\#(c) + (H')^\#(c)\delta + \dots$$

assigning formal degree one to H and to each of its derivatives. The degree-one and degree-two equations give the two displayed corrections; at every higher degree, division by a solves the next term uniquely. The degree- n term has support inside $(\text{supp } H)^{(n)}$, so Neumann's lemma sums the expansion and bounds the remaining valuation below by 3σ . Formal substitution identifies the resulting element with the root already produced by preparation. This is a homogeneous support argument; it is not an appeal to convergence of Newton iterates. $\square$

9.6 Three Rouché theorems, and the inequality that is not enough

The three statements below differ in where the strict inequality is placed, and — a distinction which must not be flattened — in whether they preserve the count monad by monad or only in total.

Theorem 9.12 (Valuation Rouché: fibrewise stability). *Let $F \in \mathcal{H}(U)$ be normalized as in (9.1) with f_0 nonzero on Γ , and let $G \in \mathcal{H}(U)$ satisfy $v_H(G) > v_H(F)$. Then $F^\#$ and $(F + G)^\#$ have the same number of zeros in $\Omega^\#$ counted with multiplicity, and, more precisely, the same number in every individual monad $a + \mathfrak{m}$ with $a \in \Omega$.*

Proof. $F + G$ has the same leading exponent and the same leading coefficient function f_0 as F , and its perturbation $E + t^{-\beta}G$ still has strictly positive support. Apply Theorem 9.9(ii) to each of F and $F + G$: both counts in $a + \mathfrak{m}$ equal $\text{ord}_a(f_0)$. $\square$

Theorem 9.13 (Ordinary-margin Rouché: stability of the total). *Let $F, G \in \mathcal{H}(U)$ and $\beta \in \mathbf{No}$ be such that*

$$t^{-\beta}F = f + E_F, \quad t^{-\beta}G = g + E_G, \quad E_F, E_G \in \mathcal{H}_{>0}(U), \quad f, g \in \text{Hol}(U),$$

and suppose the ordinary strict inequality

$$|g(z)| < |f(z)| \quad (z \in \Gamma)$$

holds on the boundary. Then $F^\#$ and $(F + G)^\#$ have the same total number of zeros in $\Omega^\#$, counted with multiplicity.

Proof. The boundary inequality makes f and $f + g$ nonzero on Γ , and ordinary Rouché gives them equal total zero counts in Ω . Theorem 9.9(i) transfers each of those ordinary counts to the corresponding surcomplex count in $\Omega^\#$. $\square$

Remark 9.14 (Total, not fibrewise). Theorem 9.13 generally preserves only the *total*: f and $f + g$ may have zeros at different ordinary points, so the distribution among monads can change completely. Theorem 9.12 is the form that preserves the count in each fibre, because it does not move the leading coefficient function at all.

Corollary 9.15 (Real-margin Rouché: a surcomplex-valued hypothesis). *Write $t^{-\beta}F = f_0 + E_F$ and $t^{-\beta}G = g_0 + E_G$ with $f_0, g_0 \in \text{Hol}(U)$ and $E_F, E_G \in \mathcal{H}_{>0}(U)$. Thus f_0, g_0 are the exponent-zero coefficient functions (a zero coefficient is allowed). Suppose f_0 has no zero on Γ . If for one ordinary real q with $0 \leq q < 1$*

$$|G^\#(z) - F^\#(z)| \leq q |F^\#(z)| \quad (z \in \Gamma), \quad (9.11)$$

then $F^\#$ and $G^\#$ have the same number of zeros in $\Omega^\#$, counted with multiplicity.

Proof. Divide (9.11) by the positive surreal t^β and take standard parts. Because q is an ordinary real strictly below 1, the reduced inequality is $|g_0 - f_0| \leq q|f_0| < |f_0|$ on Γ , which is strict. Ordinary Rouché now applies and also shows g_0 has no zero on Γ ; Theorem 9.9(i) lifts both counts. $\square$

Example 9.16 (Why a bare strict surreal inequality is not enough). Replace (9.11) by the bare strict surreal inequality $|G^\# - F^\#| < |F^\#|$ on Γ . Strictness can disappear under the standard part, and then nothing is left to apply. Take $\Omega = D$ the unit disk, Γ the unit circle,

$$F(z) = z - 2, \quad G^\# - F^\# \equiv 1 - t.$$

On Γ one has $|F^\#| = |z - 2| \geq 1$ with equality exactly at $z = 1$, while $|G^\# - F^\#| = 1 - t < 1$, so the strict surreal inequality holds at every boundary point. Its standard-part reduction is $1 \leq |z - 2|$, which is not strict at $z = 1$; and indeed $G = z - 1 - t$ has leading coefficient function $z - 1$, which vanishes at the boundary point 1. So G does not satisfy the hypothesis of the zero-cluster theorem at all, and its unique zero $1 + t$ sits in the monad of a boundary point, outside $\Omega^\#$. Since the halo is fine-clopen, this point does lie in the fine interior of its complement. This example exhibits loss of the boundary-reduction hypothesis, but both halo zero counts happen to be zero.

For an actual failure of the zero-count conclusion, instead take $F(Z) = Z$ and $G(Z) = Z - 1 + t$ on the same unit circle. Then $|G^\# - F^\#| = 1 - t < 1 = |F^\#|$ everywhere on the circle. Yet F has one zero in $\Omega^\#$ and G has none: its unique zero $1 - t$ has standard part 1 and is excluded from that halo. No manuscript in the collection asserts the version of Rouché with a bare surreal inequality, and this merged document does not either: the ordinary real margin $q < 1$ of Corollary 9.15, or the ordinary boundary inequality of Theorem 9.13, is exactly the hypothesis the proofs use.

9.7 Contour moments and Newton's identities

The prepared polynomial W has a second, completely independent constructive description: it is determined by finitely many Hahn-valued contour integrals. The proof consumes the exact weighted argument principle, Theorem 10.14 of Section 10, which is proved there from Theorem 9.3 and is independent of this subsection.

Theorem 9.17 (Contour moments of a zero cluster). *In the situation of Theorem 9.3, let $\xi_1, \dots, \xi_d \in \Omega^\#$ be the zeros of $F^\#$ listed with multiplicity, and define the moments*

$$s_k = \frac{1}{2\pi i} \oint_{\Gamma}^H w^k \frac{F'(w)}{F(w)} dw \quad (k \geq 0). \quad (9.12)$$

Then $s_0 = d$ and $s_k = \sum_{j=1}^d \xi_j^k$ for all $k \geq 0$. The moments $s_1, \dots, s_d$ determine W exactly, by Newton's identities:

$$e_0 = 1, \quad k e_k = \sum_{j=1}^k (-1)^{j-1} e_{k-j} s_j \quad (1 \leq k \leq d), \quad (9.13)$$

$$W(Z) = Z^d - e_1 Z^{d-1} + e_2 Z^{d-2} - \dots + (-1)^d e_d.$$

Proof. The moment identity is Theorem 10.14 applied to the weight $A(Z) = Z^k$, whose evaluation at ξ_j is ξ_j^k . Newton's identities (9.13) are an identity of symmetric functions over any field of characteristic zero — obtained by comparing coefficients in the logarithmic derivative of $\prod_j (1 + \xi_j T)$ — and $\mathbb{K}$ has characteristic zero, so the divisions by k are legitimate. The recursion determines $e_1, \dots, e_d$, hence the monic polynomial with those elementary symmetric functions, which by Theorem 9.9(i) is W . $\square$

Remark 9.18. The moments s_k are genuinely surcomplex numbers, and neither description of W — the recursion (9.5) by holomorphic division, or the moments (9.12) by symmetric functions — discards infinitesimal corrections to the roots. Both require the *global* preparation: the moment integral is taken around the whole of Γ and is matched against one polynomial with all d roots.

10 Residues: three notions, and the theorems that join them

Three different objects in this subject are called “the residue”, they are attached to different classes of function, and two of them take *different numerical values at the same point*. They are kept apart here by symbol:

$$\begin{aligned} \text{Res} & \text{ formal, in } \mathbb{K}((X)), \\ \text{Res}_a^H & \text{ cluster, at an ordinary centre } a, \\ \text{Res}_b & \text{ actual, at a surcomplex point } b. \end{aligned}$$

Using one symbol for the first two turns a correct statement into a false one; the standing example of §10.4 shows exactly how. Two bridge theorems say when and how they agree, and no manuscript in the collection contains both.

10.1 R1: the formal residue in $\mathbb{K}((X))$

Definition 10.1 (Formal residue). For $A = \sum_{n \geq -N} a_n X^n \in \mathbb{K}((X))$ — a Laurent series with coefficients in $\mathbb{K}$ and only *finitely many* negative powers — set

$$\text{Res}_{X=0}(A(X) dX) = a_{-1}.$$

For a meromorphic germ at a point $b \in \mathbb{K}$ one takes $X = z - b$ and writes Res_b for the resulting coefficient; this is the notion R3 of §10.3.

This is a purely algebraic notion, valid at *any* surcomplex centre, and it belongs to the germ layer of Section 6, not to the coherent class. Its two basic laws are proved there and are used freely below: the change-of-variable law $\text{Res}_{Y=0} A(\psi(Y))\psi'(Y)dY = m \text{Res}_{X=0} A(X)dX$ for ψ of order m (with $m = 1$ the invertible case), and the local argument principle $\text{Res}_{X=0} (A'/A)dX = m$ for $A = X^m U$ with $U(0) \neq 0$. The primitive obstruction is likewise a statement about $\mathbb{K}((X))$: a formal meromorphic differential has a formal meromorphic primitive if and only if its residue vanishes, every monomial except X^{-1} being integrable.

10.2 R2: the cluster residue at an ordinary centre

Definition 10.2 (The class $\mathcal{M}(U)$ and the cluster residue). Let $\mathcal{M}(U)$ consist of the Hahn series $A = \sum_{s \in S} a_s t^s$ with $S \subset \mathbf{No}$ a well-ordered *set* and every a_s meromorphic on the connected ordinary domain U . The poles are free to vary with the exponent s , and their orders are not assumed bounded in s . For $a \in U$ define the *cluster residue* at the *ordinary* centre a by

$$\text{Res}_a^H(A(\zeta)d\zeta) = \sum_{s \in S} \left(\text{Res}_{\zeta=a} a_s(\zeta) d\zeta \right) t^s. \quad (10.1)$$

The support of the right-hand side is contained in S , so it is a surcomplex number.

The coefficient ring of $\mathcal{M}(U)$ is a field, so a nonzero element of $\mathcal{M}(U)$ is invertible by leading-term factorization together with the infinitesimal functional calculus of Section 7. The adjective “cluster” is the whole point: Res_a^H is attached to an ordinary centre and records a whole infinitesimal cluster sitting above it. It is *not* the residue of a meromorphic germ at the exact point a .

10.3 R3: the actual residue at a surcomplex point

Definition 10.3 (Actual local residue). Let $A, B \in \mathcal{H}(U)$ with $B \neq 0$, and let $b \in U^\#$ be a point at which the germ of A/B is a formal Laurent series with finite principal part,

$$\frac{A(b+h)}{B(b+h)} = \sum_{n \geq -N} C_n h^n \in \mathbb{K}((h)),$$

which holds at every $b \in U^\#$: the coherent identity theorem excludes the zero germ of B , so its order there is finite. Its *actual local residue* is $\text{Res}_b((A/B)(z)dz) = C_{-1}$. It is centred at b , not at $\text{st } b$.

R3 is defined on quotients of coherent families. It is *not* available on all of $\mathcal{M}(U)$: an arbitrary element of $\mathcal{M}(U)$ need not be such a quotient, and need not have any actual surcomplex pole in a given monad at all.

10.4 The standing example: $1/(\zeta - \epsilon)$

Example 10.4 (The three residues disagree, and both disagreements are correct). Let $\epsilon \in \mathbf{m}$, $\epsilon \neq 0$, and $R(\zeta) = 1/(\zeta - \epsilon)$. On an ordinary annulus about 0,

$$\frac{1}{\zeta - \epsilon} = \sum_{n \geq 0} \epsilon^n \zeta^{-n-1}, \quad (10.2)$$

which, after expanding each ϵ^n in the monomial scale, is an element of $\mathcal{M}(\mathbb{C})$ whose coefficient functions all have their poles at the single ordinary point 0. Then

$$\text{Res}_0^H(R(\zeta)d\zeta) = 1, \quad \text{Res}_\epsilon(R(\zeta)d\zeta) = 1, \quad \text{Res}_0(R(\zeta)d\zeta) = 0.$$

The first value is read off (10.2): only the $n = 0$ coefficient ζ^{-1} contributes a residue at 0, with value 1. The second is immediate, the actual pole being the simple pole at ϵ . The third is also immediate and is not a competing computation: on the fine neighborhood ϵm , equivalently when ζ/ϵ is infinitesimal, the function expands as $-\epsilon^{-1} \sum_{n \geq 0} (\zeta/\epsilon)^n$, a power series with no negative powers, so the meromorphic germ at the exact point 0 is regular there. The weaker inequality $|\zeta| < |\epsilon|$ does not ensure Hahn summability: at $\zeta = \epsilon/2$ all nonzero geometric terms have valuation zero before multiplication by $-\epsilon^{-1}$. The cluster residue at the ordinary centre 0 correctly accounts for the actual pole at ϵ ; the local residue at 0 correctly reports that there is no pole at 0.

Two scope statements travel with the example. The expansion (10.2) is *not* asserted to evaluate throughout the monad containing the pole; it is a coefficientwise identity valid away from that monad. And the term $\epsilon^n \zeta^{-n-1}$ has a pole of order $n + 1$ in its ordinary coefficient. To check that these orders survive Hahn regrouping, put $\alpha = v(\epsilon) > 0$ and $c = \text{lc}(\epsilon)$. At exponent $n\alpha$, the n th term contributes $c^n \zeta^{-n-1}$; terms with larger index have valuation $> n\alpha$, and smaller indices have pole order at most n . Hence the coefficient at $n\alpha$ has pole order exactly $n + 1$. These unbounded orders are excluded by the hypothesis of Theorem 10.11.

10.5 The coefficientwise residue theorem, in maximal generality

Theorem 10.5 (Cluster residue theorem on $\mathcal{M}(U)$). *Let $A \in \mathcal{M}(U)$, let Ω be a bounded Jordan domain with piecewise C^1 boundary Γ and $\overline{\Omega} \subset U$, and suppose no coefficient a_s has a pole on Γ . Then*

$$\frac{1}{2\pi i} \oint_{\Gamma}^H A(\zeta) d\zeta = \sum_{a \in \Omega} \text{Res}_a^H(A(\zeta) d\zeta). \quad (10.3)$$

Only a set of ordinary centres contributes. The family on the right is Hahn summable even when infinitely many centres contribute in total.

Proof. Fix an exponent $s \in S$. The coefficient a_s is meromorphic on U and $\overline{\Omega} \subset U$ is compact, so a_s has finitely many poles in $\overline{\Omega}$, none on Γ . Hence only finitely many centres contribute at the fixed exponent s , and the ordinary residue theorem gives $\frac{1}{2\pi i} \int_{\Gamma} a_s = \sum_{a \in \Omega} \text{Res}_a a_s$, a finite sum. The union of the supports of the family $(\text{Res}_a^H A)_{a \in \Omega}$ is contained in the single fixed well-ordered set S , and each exponent s occurs for only finitely many a by the previous sentence: these are exactly the two clauses of Hahn summability. Summing the ordinary identity coefficientwise, which is the definition (10.1) of Res_a^H and the definition of $\oint^H$, gives (10.3). $\square$

Remark 10.6 (The hypotheses that are not needed, and the conclusion that does not transfer). Two stronger hypotheses appear in the sources and are unnecessary: a *common* closed discrete pole set for all coefficients, and a single *fixed finite* pole set. The proof above uses neither; meromorphy of each coefficient on the connected domain U is enough, because compactness of $\overline{\Omega}$ localizes the count at each fixed exponent.

The price of that generality must be stated. On $\mathcal{M}(U)$ there need be no factorization into a quotient of coherent families, so the conclusion of Theorem 10.8 — finitely many actual poles, and a *finite* sum of actual residues — is *not* available and must not be imported. For a concrete separation, take $U = D$, $\Omega = \{|z| < \frac{1}{2}\}$, distinct points $b_n = 1/(2n + 4) \in \Omega$, and

$$A = \sum_{n \geq 1} \frac{t^n}{\zeta - b_n} \in \mathcal{M}(D).$$

Its support $\{1, 2, 3, \dots\}$ is a well-ordered set and each coefficient function is meromorphic on D with one pole, so $A \in \mathcal{M}(D)$; but infinitely many ordinary centres contribute, and $\sum_{a \in \Omega} \text{Res}_a^H A = \sum_{n \geq 1} t^n = t/(1 - t)$, a Hahn sum with infinitely many nonzero terms. The poles accumulate at $0 \in \Omega$, so the coefficient family is not that of any quotient of coherent

sections, whose coefficient poles lie in the finite zero set of one leading coefficient function inside $\bar{\Omega}$. Summability, not finiteness, is what Theorem 10.5 provides.

Proposition 10.7 (Quotients of coherent families lie in $\mathcal{M}(U)$). *Let $A, B \in \mathcal{H}(U)$ with $B \neq 0$, and write $B = t^\beta f_0(1 + E_B)$ with f_0 the leading coefficient function of B and E_B of strictly positive support (with meromorphic coefficients, poles only at the zeros of f_0). Then $A/B \in \mathcal{M}(U)$: each of its coefficient functions is a finite sum of products of meromorphic functions on U , hence meromorphic, and its poles lie over the zeros of f_0 . The inclusion is proper.*

Proof. $1/B = t^{-\beta} f_0^{-1} \sum_{n \geq 0} (-E_B)^n$, and the family $((-E_B)^n)_{n \geq 0}$ is Hahn summable by Neumann's lemma since $\text{supp } E_B$ is a well-ordered set of positive surreals. Hence at each exponent only finitely many terms contribute, and each coefficient of $1/B$ is a finite sum of finite products of the meromorphic functions f_0^{-1} and the coefficients of E_B . Multiplying by A preserves this. Properness is the example of Remark 10.6. $\square$

This containment is what makes the two residue theorems compatible rather than competing: on the quotient class both apply and agree, which is the content of Bridge 1. What it does not give is the converse, and the converse is false.

10.6 Bridge 1: the cluster residue is a finite sum of actual residues

Theorem 10.8 (Residue-cluster theorem). *Let $A, B \in \mathcal{H}(U)$ with $B \neq 0$, let $a \in U$, and suppose the leading coefficient function f_0 of B has a zero of order $m \geq 1$ at a and no other zero on a closed ordinary disk $\bar{D}_\rho(a) \subset U$. Let $\gamma_a = \partial D_\rho(a)$, positively oriented. Then A/B has at most m poles in the monad $a + \mathfrak{m}$, all of them actual points of $\mathbb{K}$, and*

$$\text{Res}_a^H\left(\frac{A}{B}(z) dz\right) = \frac{1}{2\pi i} \oint_{\gamma_a}^H \frac{A(z)}{B(z)} dz = \sum_{\substack{b \in a + \mathfrak{m} \\ b \text{ a pole of } A/B}} \text{Res}_b\left(\frac{A}{B}(z) dz\right), \quad (10.4)$$

a finite sum of actual surcomplex residues. Cancelled zeros of the denominator contribute no pole.

Proof. The first equality is Theorem 10.5 applied on $D_\rho(a)$, legitimate by Proposition 10.7: the only ordinary centre inside is a .

For the second, translate a to 0. Prepare $B = t^\beta u_0(1 + q)W$ by Theorem 9.3 on a neighbourhood of $\bar{D}_\rho$, with W monic of degree m and infinitesimal lower coefficients. The section $C = t^{-\beta} A/(u_0(1 + q))$ lies in $\mathcal{H}(V)$ because $u_0(1 + q)$ is a unit. Theorem 9.7 gives $C = WQ + R$ with $Q \in \mathcal{H}(V)$ and $R \in \mathbb{K}[Z]$, $\deg R < m$, so

$$\frac{A}{B} = Q + \frac{R}{W}. \quad (10.5)$$

The first summand is coherent holomorphic on V , so its coefficientwise contour integral vanishes by Cauchy's theorem (Section 8). Over the algebraically closed field $\mathbb{K}$, the proper fraction R/W has a finite partial-fraction expansion

$$\frac{R(Z)}{W(Z)} = \sum_j \sum_{k=1}^{m_j} \frac{c_{j,k}}{(Z - b_j)^k},$$

where $b_1, \dots$ are the distinct roots of W , all infinitesimal by Lemma 9.8, and absent or cancelled principal parts are recorded by zero coefficients. At b_j the other partial fractions and Q are analytic on a small fine disk, so the actual local residue of A/B at b_j is exactly $c_{j,1}$; this identifies

the right-hand side of (10.4) with $\sum_j c_{j,1}$, and shows that the poles are finite in number and lie in $\mathfrak{m}$.

Finally, on the translated ordinary circle γ_a , where $|z|$ is a fixed positive real and b_j is infinitesimal,

$$\frac{1}{(z - b_j)^k} = \sum_{n \geq 0} \binom{k+n-1}{n} \frac{b_j^n}{z^{k+n}}, \quad (10.6)$$

a Hahn summable family by the all-lengths clause of Neumann's lemma applied to the positive support of b_j . Increasing leading valuations alone would not justify the finiteness clause in higher rank. The case $b_j = 0$ leaves only the $n = 0$ term. Its coefficientwise circle integral divided by $2\pi i$ is the ordinary residue of each displayed monomial at 0: only the term with $k+n=1$, that is $k=1$ and $n=0$, survives, giving 1; all terms with $k \geq 2$ give 0. Multiplying by the fixed constants $c_{j,k}$ and adding finitely many terms gives $\sum_j c_{j,1}$. This is the second equality.

Note that the argument is finite partial fractions plus one summable kernel expansion. It is not a limit, and at no point are distinct nearby surcomplex poles replaced by their common standard part: standard parts organize the cluster, the residues themselves stay in $\mathbb{K}$. $\square$

Theorem 10.9 (Residue theorem for coherent quotients, global form). *Let $A, B \in \mathcal{H}(U)$ with $B \neq 0$ and the leading coefficient function of B nonzero on $\Gamma = \partial\Omega$. Then $M = A/B$ has finitely many poles in $\Omega^\#$, all of them actual points of $\mathbb{K}$, and*

$$\oint_{\Gamma}^H M(w) dw = 2\pi i \sum_{b \in \Omega^\#} \text{Res}_b(M(z)dz). \quad (10.7)$$

The sum is finite; its individual terms may be infinite surcomplex numbers.

Proof. By Theorem 9.9 the leading coefficient function of B has finitely many zeros in Ω and every pole of M lies in one of their monads. By Proposition 10.7, $M \in \mathcal{M}(U)$ with coefficient poles only at those ordinary points, so Theorem 10.5 replaces Γ by finitely many small disjoint ordinary circles around them. On each circle apply Theorem 10.8. Alternatively, prepare B once on the whole of V and use (10.5) globally: the coherent term integrates to zero and the proper rational term is handled by (10.6) exactly as above, with all roots of W lying in $\Omega^\#$. $\square$

Remark 10.10. The decomposition (10.5) is considerably more than a coefficientwise residue computation at fixed ordinary poles: it separates the *actual* poles of $\mathbb{K}$, including infinitesimally split ones, from a coherent holomorphic remainder. That separation is supplied by preparation and division, and by nothing else in this article.

10.7 Bridge 2: when the cluster residue is the residue at the exact point

Theorem 10.11 (Uniformly bounded pole order). *Let $F = \sum_{s \in S} f_s t^s$ have common well-ordered support S , with each f_s holomorphic on $D_\varrho(a) \setminus \{a\}$, and suppose there is one finite integer $d \geq 1$ such that*

$$(z - a)^d f_s(z) \text{ extends holomorphically across } a \text{ for every } s \in S.$$

Then $G := (z - a)^d F$ is a coherent section in $\mathcal{H}(D_\varrho(a))$ with the same support. The quotient $G^\#/(z - a)^d$ extends the original evaluation from $(D_\varrho(a) \setminus \{a\})^\#$ to $D_\varrho(a)^\# \setminus \{a\}$ and has at most a pole of order d at the exact point a , and

$$\text{Res}_a^H(F(z)dz) = \text{Res}_a(F^\#(z)dz) = \frac{G^{(d-1)}(a)}{(d-1)!}. \quad (10.8)$$

Proof. The extensions have the same index set S , which is well ordered, so $G \in \mathcal{H}(D_\varrho(a))$. At the exact point a , writing $h = z - a$, the germ of $F^\#$ is $\sum_{n \geq 0} \frac{G^{(n)}(a)}{n!} h^{n-d} \in \mathbb{K}((h))$, a Laurent series with finite principal part; its coefficient of h^{-1} is $G^{(d-1)}(a)/(d-1)!$, which is the middle and right members of (10.8) by Definition 10.1. For the left member, the ordinary residue of f_s at a is $g_s^{(d-1)}(a)/(d-1)!$ with $g_s = (z - a)^d f_s$, so by (10.1)

$$\text{Res}_a^H(F dz) = \sum_{s \in S} \frac{g_s^{(d-1)}(a)}{(d-1)!} t^s = \frac{G^{(d-1)}(a)}{(d-1)!},$$

the last step because differentiation of a coherent section and evaluation at an ordinary point are both coefficientwise (Section 7). $\square$

Remark 10.12 (Exactly what the hypothesis excludes, and what is not claimed). The uniform bound is what excludes the displaced-pole phenomenon. In the standing Example 10.4 the coefficient of ϵ^n has a pole of order $n + 1$ at 0, so no single finite d works, and indeed $\text{Res}_0^H = 1$ while $\text{Res}_0 = 0$. Conversely, under the hypothesis of Theorem 10.11 the whole array is one meromorphic object at the exact point and the two residues coincide. What is *not* asserted is that uniform boundedness is necessary for numerical equality in every individual instance: the theorem is a sufficient criterion, and the example shows it cannot simply be dropped. Note also that R2 and R3 may both be defined and still differ in location: Theorem 10.8 equates Res_a^H with a sum over the whole cluster $a + \mathfrak{m}$, while (10.8) equates it with the single value at the centre. The uniform pole-order hypothesis makes all possible poles lie at that centre and thus ensures agreement; numerical equality by cancellation can also occur without that hypothesis.

Remark 10.13 (Two meanings of a Laurent expansion). The same distinction appears one level up. For $F \in \mathcal{H}(\mathcal{A})$ on an ordinary annulus $\mathcal{A} = \{r < |z - a| < R\}$, each coefficient has its ordinary Laurent expansion $f_s(z) = \sum_{n \in \mathbb{Z}} c_{s,n}(z - a)^n$, and the resulting object is a *coefficientwise* Laurent expansion: the inner series is summed in the ordinary complex sense first and the results are then lifted. There is no general justification for reading the whole array (s, n) as one Hahn sum at a nonzero ordinary argument, since infinitely many ordinary scalar terms would land at the single exponent s . By contrast a *meromorphic germ* at an arbitrary point $b \in \mathbb{K}$ is an element of $\mathbb{K}((z - b))$ with finitely many negative powers. R2 belongs to the first picture and R1/R3 to the second.

10.8 The exact argument principle

Theorem 10.14 (Exact weighted argument principle). *Let $F \in \mathcal{H}(U)$ be nonzero with leading coefficient function f_0 having no zero on Γ , and let $\xi_1, \dots, \xi_d \in \Omega^\#$ be the zeros of $F^\#$ repeated with multiplicity, as supplied by Theorem 9.9. Then for every weight $A \in \mathcal{H}(U)$,*

$$\frac{1}{2\pi i} \oint_{\Gamma}^H A(w) \frac{F'(w)}{F(w)} dw = \sum_{j=1}^d A^\#(\xi_j). \quad (10.9)$$

In particular, with $A = 1$,

$$\frac{1}{2\pi i} \oint_{\Gamma}^H \frac{F'(w)}{F(w)} dw = \sum_{z \in \Omega^\#} \text{ord}_z F^\# = \sum_{a \in \Omega} \text{ord}_a f_0 = d \in \mathbb{N}, \quad (10.10)$$

an ordinary integer with no infinitesimal error term, and locally $\text{Res}_a^H((F'/F)d\zeta) = \text{ord}_a f_0$.

Proof. Global preparation (Theorem 9.3) writes $F = \mathcal{U}W$ on V with $\mathcal{U} = t^\beta u_0(1 + q)$ a unit of $\mathcal{H}(V)$ and W the single monic cluster polynomial of degree d . Then

$$\frac{F'}{F} = \frac{\mathcal{U}'}{\mathcal{U}} + \frac{W'}{W}.$$

The first summand lies in $\mathcal{H}(V)$, so AU'/U is coherent holomorphic and its coefficientwise integral around the closed contour Γ vanishes by Cauchy's theorem (Section 8). Algebraically $W'/W = \sum_{j=1}^d (w - \xi_j)^{-1}$ in $\mathbb{K}(w)$. Cauchy's integral formula at the genuinely surcomplex point $\xi_j \in \Omega^\#$ (Section 8) evaluates $\frac{1}{2\pi i} \oint_{\Gamma}^H A(w)(w - \xi_j)^{-1} dw = A^\#(\xi_j)$, the kernel being handled there by its uniform Hahn expansion near the ordinary contour. Adding the finitely many terms gives (10.9). The identification of d with both zero counts in (10.10) is Theorem 9.9. $\square$

Second proof of the unweighted identity (10.10). Near Γ write $F = t^\beta f_0(1+E)$ with $\text{supp } E > 0$, which is legitimate in $\mathcal{M}$ because f_0 has no zero there. Then

$$\frac{F'}{F} = \frac{f_0'}{f_0} + (\log(1+E))',$$

where $\log(1+E) = \sum_{n \geq 1} (-1)^{n-1} E^n/n$ is a Hahn sum by Neumann's lemma and is a *single-valued* coherent section along the whole contour. Each coefficient of the second summand is the derivative of a function holomorphic near Γ , so it has zero residue everywhere and zero integral around the closed contour. The remaining term contributes the ordinary winding integer $\sum_{a \in \Omega} \text{ord}_a f_0$ by the classical argument principle. This proof isolates where the integrality comes from; preparation is what identifies that integer with the actual surcomplex roots, split multiplicities included. $\square$

Corollary 10.15 (Zeros minus poles). *Let $A, B \in \mathcal{H}(U)$ be nonzero with both leading coefficient functions nonzero on Γ , and let $M = A/B$. Then*

$$\frac{1}{2\pi i} \oint_{\Gamma}^H \frac{M'(w)}{M(w)} dw = \sum_{b \in \Omega^\#} \text{ord}_b M^\# = N_A(\Omega^\#) - N_B(\Omega^\#) \in \mathbb{Z},$$

where N counts zeros with multiplicity. Only finitely many terms in the middle are nonzero, and cancellations are counted with their signs.

Proof. $M'/M = A'/A - B'/B$ and the integral is $\mathbb{K}$ -linear; apply (10.10) to each. Equivalently, M'/M is a coherent quotient whose only poles in $\Omega^\#$ are at the zeros of A and B , with actual local residue $\text{ord}_b M^\#$ there by the local argument principle of §10.1; then apply Theorem 10.9. $\square$

Remark 10.16. The normalized integral is an exact ordinary integer even though the zeros, the derivatives and the residues used to compute it live in $\mathbb{K}$. It is an analytic index for the specified contour pairing. It is *not* identified with the winding number of a continuous loop in the fine topology: every continuous map $[0, 1] \rightarrow \mathbb{K}$ is constant (Sections 3–4), so no such nonconstant loop exists.

10.9 Deformation of the contour

Theorem 10.17 (Deformation invariance under admissible infinitesimal deformations). *Let γ_0 be a closed ordinary piecewise- C^1 path in U and let*

$$\tilde{\gamma}(\tau) = \gamma_0(\tau) + \eta(\tau), \quad \eta(\tau) = \sum_{\lambda \in S} \eta_\lambda(\tau) t^\lambda, \quad S \subset \mathbf{No}_{>0} \text{ a well-ordered set},$$

with every η_λ piecewise C^1 and with matching endpoints, the integral over $\tilde{\gamma}$ being defined by Taylor substitution in the integrand followed by coefficientwise integration of $R(\tilde{\gamma}(\tau))\tilde{\gamma}'(\tau)$. Let $R = A/B$ be a quotient of coherent families whose denominator's leading coefficient function has no zero on γ_0 . Then

$$\oint_{\tilde{\gamma}}^H R(z) dz = \oint_{\gamma_0}^H R(z) dz.$$

Proof. Use the ordinary real homotopy $\gamma_u = \gamma_0 + u\eta$, $0 \leq u \leq 1$. The support calculus makes every substitution admissible: η has positive well-ordered support, so the Taylor expansion of $R(\gamma_u)$ about γ_0 is Hahn summable, and at each exponent the expression is a *finite polynomial* in u with piecewise continuous coefficient functions of τ . The chain rule gives

$$\frac{\partial}{\partial u}(R(\gamma_u)\gamma'_u) = \frac{d}{d\tau}(R(\gamma_u)\eta),$$

and integrating coefficientwise in τ kills the right-hand side, the endpoint terms cancelling because the parametrizations are closed. Hence each coefficient is independent of u ; evaluate at $u = 0$ and $u = 1$. $\square$

This is invariance under a specified algebraic-analytic deformation of a classical contour. It is *not* an unrestricted homotopy theorem for maps into the fine-topological class field, and none is proved anywhere in this article.

10.10 The global rational residue theorem on $\mathbb{P}^1(\mathbb{K})$

The last theorem of this part uses no contour at all. It is a theorem about rational functions over $\mathbb{K}$, hence about the formal/actual residue R1–R3 and not about Res^H , and it holds over the entire proper-class field.

Definition 10.18. For $R \in \mathbb{K}(z)$ put $\text{Res}_\infty(R(z)dz) = \text{Res}_{w=0}(-w^{-2}R(1/w))dw$, using $w = 1/z$.

Theorem 10.19 (Global rational residue theorem). *A rational differential $R(z)dz$ over $\mathbb{K}$ has finitely many poles in $\mathbb{P}^1(\mathbb{K}) = \mathbb{K} \cup \{\infty\}$ and*

$$\sum_{b \in \mathbb{P}^1(\mathbb{K})} \text{Res}_b(R(z)dz) = 0. \quad (10.11)$$

Moreover $R(z)dz$ has a rational primitive if and only if all its affine residues vanish.

Proof. Algebraic closedness of $\mathbb{K}$ gives the finite partial-fraction decomposition

$$R(z) = P(z) + \sum_b \sum_{j=1}^{m_b} \frac{c_{b,j}}{(z-b)^j}, \quad P \in \mathbb{K}[z],$$

with finitely many b . The affine residues are the $c_{b,1}$. In the expansion at infinity the coefficient of z^{-1} is $\sum_b c_{b,1}$, so the residue at infinity is its negative, proving (10.11): a simple-pole term contributes its coefficient at its pole and the negative at infinity, while polynomial terms and terms with $j \geq 2$ contribute zero in total. For the second assertion, every summand with $j \geq 2$ and every polynomial term has a rational primitive (characteristic zero), while a derivative of a rational function has zero residue at every point, so the $j = 1$ terms are the whole obstruction. $\square$

No contour, no completeness, no topological connectedness, and no compactness enters the proof. In the same spirit, Definition 10.1 gives the local statement behind it: modulo derivatives, $\mathbb{K}((X))dX/d\mathbb{K}((X)) \cong \mathbb{K}dX/X$.

10.11 What this part does not assert

- (a) Res , Res_a^H and Res_b are three symbols for three notions, and at $1/(\zeta - \epsilon)$ the first and second take the values 0 and 1 at the same ordinary point. Nothing in this article licenses writing one symbol for two of them.

- (b) Theorem 10.5 produces a Hahn-summable sum of cluster residues. It does not produce actual surcomplex poles; on $\mathcal{M}(U)$ there need be none, and the finite-sum conclusion of Theorem 10.8 is licensed only for quotients of coherent families.
- (c) Theorem 10.11 is a sufficient criterion. Uniform boundedness of the pole orders is not claimed to be necessary for numerical coincidence in an individual case.
- (d) The expansions (10.2) and (10.6) are coefficientwise identities valid away from the monad containing the relevant pole. They are not asserted to evaluate inside it.
- (e) Theorem 10.17 moves a classical contour by an admissible infinitesimal deformation. No unrestricted homotopy invariance is claimed, and the coefficientwise contour functional is not offered as the unique possible notion of surcomplex integration.

11 Global principles on the halo

This part collects the classical global theorems that survive on a halo $U^\#$, each with the exact hypothesis that makes it true and, where one exists, with the counterexample that shows the hypothesis cannot be weakened. Every statement below is a statement about the *coherent* class $\mathcal{H}(U)$ of Section 7, with the sole exception of §11.5, whose sharp inequalities hold only for canonical classical lifts $f^\#$ and are *false* for general coherent self-maps of the halo. Nothing in this part is asserted for the germ-analytic class of Section 6: the identity theorem, the global maximum principle, Liouville, uniqueness of primitives and the open mapping theorem all fail there, as the locally constant witnesses of Section 5 show.

11.1 The normalization lemma and local finite degree

The engine of the local mapping theorems is a rescaling whose radius is an explicitly computed *monomial*. Its motivating case is a point b at which $F^{\#'}(b)$ is nonzero but itself infinitesimal; such a point is invisible to any argument that looks only at the reduction to $\mathbb{C}$.

Lemma 11.1 (Infinitesimal rescaling; monomial radius). *Let $U \subseteq \mathbb{C}$ be a connected ordinary domain, let $F \in \mathcal{H}(U)$ be nonconstant, and let $b \in U^\#$. There is a least integer $m \geq 1$ with*

$$a_m := \frac{(D_z^m F)^\#(b)}{m!} \neq 0.$$

Write $b = a + \varepsilon$ with $a = \text{st}(b) \in U$ and $\varepsilon \in \mathfrak{m}$, and let σ be any lower bound for the exponents occurring in the common support $\text{supp}(F) + \langle \text{supp}(\varepsilon) \rangle$ of the Taylor coefficients $a_n = (D_z^n F)^\#(b)/n!$; one may take $\sigma = v_H(F)$. Choose a surreal

$$\rho > \max\{0, v(a_m) - \sigma\}, \quad r := t^\rho > 0. \quad (11.1)$$

Then the normalized function

$$G(w) = \frac{F^\#(b + rw) - F^\#(b)}{a_m r^m} \quad (11.2)$$

is the evaluation of a section of $\mathcal{H}(\mathbb{C})$ of the form $w^m + H(w)$ with $v_H(H) > 0$. Increasing ρ makes r smaller than any prescribed positive surreal radius, so $b + r\mathfrak{m}$ may be placed inside any prescribed fine-open neighbourhood of b .

Proof. That some m exists is the identity theorem of Section 7 applied to $F - F^\#(b)$: were every positive-order derivative to vanish at b , that section would vanish identically. Concretely, its

first nonzero leading coefficient function has finite vanishing order at a , so some positive-order derivative is nonzero at b .

The exact Taylor expansion of Section 7, valid throughout the monad and here applied with the infinitesimal increment rw for finite w , gives

$$G(w) = w^m + \sum_{n>m} \frac{a_n}{a_m} r^{n-m} w^n.$$

The support of the tail lies in $(\text{supp}(F) + \langle \text{supp}(\varepsilon) \rangle + \text{supp}(a_m^{-1})) + \{\rho, 2\rho, 3\rho, \dots\}$. The reciprocal support must be included in full; its minimum is $-v(a_m)$, but it need not be a singleton. The displayed set is a sum of a fixed well-ordered set with the positive multiples of ρ , so it is well ordered, and each exponent receives a contribution from only finitely many n ; hence the coefficient of each exponent is an ordinary *polynomial* in w , in particular an entire function of w . By (11.1) the least exponent occurring in the tail is at least $\sigma - v(a_m) + \rho > 0$, so $v_H(H) > 0$. The last assertion is immediate, since $\rho \mapsto t^\rho$ is strictly decreasing. $\square$

The choice of a *monomial* radius rather than a real one is what makes the lemma usable at arbitrary surcomplex points: ρ is allowed to absorb the valuation $v(a_m)$ of an infinitesimal derivative, which no real radius can do.

Theorem 11.2 (Local finite degree and the open mapping theorem). *With the notation of Lemma 11.1,*

$$F^\#(b + r\mathfrak{m}) = F^\#(b) + a_m r^m \mathfrak{m}, \quad (11.3)$$

and every value on the right has exactly m preimages in $b + r\mathfrak{m}$, counted with multiplicity. Consequently every nonconstant $F \in \mathcal{H}(U)$ evaluates to a map $F^\# : U^\# \rightarrow \mathbb{K}$ that carries fine-open subclasses to fine-open subclasses.

Proof. For $\eta \in \mathfrak{m}$ the section $G - \eta$ of $\mathcal{H}(\mathbb{C})$ has leading coefficient function w^m , whose only zero in $\mathbb{C}$ is a zero of order m at the origin. The zero-cluster theorem of Section 9 therefore produces exactly m solutions of $G(w) = \eta$ in $\mathfrak{m}$, counted with multiplicity, and no solutions in any other monad. Conversely $G(w) \in \mathfrak{m}$ for $w \in \mathfrak{m}$, because $G - w^m$ has positive support and $w^m \in \mathfrak{m}$. Undoing the two affine substitutions of (11.2) gives (11.3) and the count. For openness, given a fine-open neighbourhood of b , choose ρ by the last clause of Lemma 11.1 so that $b + r\mathfrak{m}$ lies inside it; its image contains the fine-open neighbourhood $F^\#(b) + a_m r^m \mathfrak{m}$ of $F^\#(b)$. $\square$

This is an assertion about actual open classes and actual surcomplex preimages, not merely about the reduction to $\mathbb{C}$.

Corollary 11.3 (Strong local maximum modulus). *Let U be connected and $F \in \mathcal{H}(U)$ nonconstant. Then $|F^\#|$ has no local maximum at any point of $U^\#$ in the fine topology. Equivalently: if $|F^\#|$ has a fine-local maximum at some $b \in U^\#$, then F is a constant element of $\mathbb{K}$ and $F^\#$ is that constant on all of $U^\#$.*

Proof. By Theorem 11.2 every fine-neighbourhood of $F^\#(b)$ contains values of $F^\#$; and every fine-neighbourhood of a point of $\mathbb{K}$ contains points of strictly larger modulus — take $(1 + \eta)c$ with $\eta > 0$ a sufficiently small surreal if $c = F^\#(b) \neq 0$, and any sufficiently small nonzero value if $c = 0$. This contradicts a local maximum. For the second formulation, a local maximum forces the germ of $F^\#$ at b to be constant; subtracting that constant, which is itself an element of $\mathcal{H}(U)$ with constant coefficient functions, and applying the identity theorem of Section 7 in its vanishing-germ form, gives F constant on the whole halo over the connected ordinary base. $\square$

Remark 11.4 (Exactly what the maximum principle does and does not say). Two scope statements must travel with Corollary 11.3. First, this is a genuine *neighbourhood* maximum principle in the fine topology, not merely a statement about standard parts; coherence enters only in the last step, the passage from a constant germ to one global constant. Second, nothing here asserts that $|F^\#|$ attains a maximum on a closed surcomplex disk. No closed disk in $\mathbb{K}$ is compact, and it is precisely this missing compactness that prevents the classical route from the maximum principle to a boundary-maximum or Liouville statement. The Liouville question is settled instead by §11.3, and by hypotheses of an entirely different shape.

11.2 Hahn reversion and the coherent local inverse

Lemma 11.5 (Hahn reversion). *Let $G(w) = w + H(w)$ with $H \in \mathcal{H}_{>0}(\mathbb{C})$. There is a unique $K \in \mathcal{H}_{>0}(\mathbb{C})$ such that $J(y) = y + K(y)$ is a two-sided inverse of G under infinitesimal Hahn composition, and $J^\#$ is a bijection $\mathcal{O} \rightarrow \mathcal{O}$ inverse to $G^\#$. Explicitly,*

$$K(y) = \sum_{n \geq 1} \frac{(-1)^n}{n!} D_y^{n-1} (H(y)^n), \quad (11.4)$$

a Hahn sum whose n -th term has support inside $n \cdot \text{supp}(H)$.

Proof. Expanding $H(y + K)$ as the formal Taylor series in the positive-support section K , the equation $K = -H(y + K)$ determines the part of K of degree n in occurrences of H from the parts of smaller degree. Derivatives of entire coefficient functions are entire, so every term lies in $\mathcal{H}_{>0}(\mathbb{C})$, and the support of a degree- n term is a sum of n elements of $\text{supp}(H)$. Neumann [3]’s lemma in its finite-word form (Sections 3–4) makes the family summable, since each exponent of $\langle \text{supp}(H) \rangle$ admits only finitely many representations as a word. Uniqueness follows by comparing least exponents: altering K on the right multiplies the difference by a section of strictly positive valuation. For (11.4), introduce an auxiliary ordinary formal parameter q and solve $u = -qH(y + u)$; formal Lagrange inversion gives $[q^n]u = \frac{(-1)^n}{n!} D_y^{n-1} H(y)^n$, and summability licenses the graded evaluation at $q = 1$ — a normal sum, not the convergence of partial sums. Finally, for a finite target $y = c + \eta$ the zero-cluster theorem applied to $G - y$ gives exactly one solution in $c + \mathfrak{m}$ and none elsewhere, so $G^\#$ is a bijection of $\mathcal{O}$; faithful evaluation identifies its inverse with $J^\#$. $\square$

Theorem 11.6 (Coherent local inverse). *Let $F \in \mathcal{H}(U)$ and $b \in U^\#$ with $F^{\#'}(b) \neq 0$. Then $F^\#$ has a local inverse Φ between fine-open neighbourhoods of b and of $F^\#(b)$; after the affine rescaling of Lemma 11.1 with $m = 1$ the inverse is itself coherent, and*

$$\Phi'(F^\#(z)) = \frac{1}{F^{\#'}(z)}$$

on a sufficiently small such neighbourhood.

Proof. Apply Lemma 11.1 with $m = 1$, invert by Lemma 11.5, and undo the affine normalizations; the derivative formula is the chain rule. Shrinking once more makes $F^{\#'}$ nonvanishing throughout, by the normalized leading-derivative estimate. The hypothesis $F^{\#'}(b) \neq 0$ is the right one: it does *not* require $F^{\#'}(b)$ to have nonzero standard part, and the monomial radius of (11.1) is exactly what accommodates an infinitesimal derivative. $\square$

11.3 The Liouville hierarchy: three inequivalent hypotheses

Call $F \in \mathcal{H}(\mathbb{C})$ *macroscopically entire*; its evaluation lives on $\mathbb{C}^\# = \mathcal{O}$, the finite elements, not on all of $\mathbb{K}$. Three different boundedness hypotheses are in circulation for such F , and they give three genuinely different theorems. They are stated here together, with their separate conclusions. The ordinary-real and coefficientwise bounds are incomparable; neither should be substituted for the other.

(a) A single surreal bound is vacuous.

Proposition 11.7 (Automatic surreal boundedness). *Every nonzero $F \in \mathcal{H}(\mathbb{C})$ satisfies*

$$|F^\#(z)| < t^{v_H(F)-1} \quad (z \in \mathbb{C}^\# = \mathcal{O}), \quad (11.5)$$

with a bound independent of z . There are even nonconstant $F \in \mathcal{H}(\mathbb{C})$ bounded on $\mathcal{O}$ by the ordinary constant 1.

Proof. Evaluation at a finite point adds only positive exponents to the coefficient supports, so $v(F^\#(z)) \geq v_H(F)$ whenever the value is nonzero. Since $v(|w|) = v(w)$ and $t^{\gamma-1}$ has valuation $\gamma-1 < \gamma$, any w with $v(w) \geq \gamma$ satisfies $|w| < t^{\gamma-1}$; take $\gamma = v_H(F)$. For the second assertion take $F(Z) = tZ$: its value at every finite point is infinitesimal, hence of modulus less than 1, yet $F' = t \neq 0$. $\square$

Thus “bounded by some surreal constant on the finite plane” is a property of *every* macroscopically entire function and cannot be the hypothesis of any Liouville theorem. Note also why the order-valued Cauchy estimate of Section 8 does not rescue the argument: for $F = tZ$ and the real bound $M = 1$ on the ordinary circle $|z| = R$, that estimate gives $t \leq 1/R$ for every ordinary real $R > 0$, which is true for every infinitesimal t . The classical step is an archimedean limit in R , and there is no such limit through radii cofinal in **No**.

(b) A single ordinary real bound: an exact characterization, not constancy.

Theorem 11.8 (Real-bounded macroscopically entire functions). *For $F \in \mathcal{H}(\mathbb{C})$ the following are equivalent.*

- (i) *There is an ordinary real $M > 0$ with $|F^\#(z)| \leq M$ for every $z \in \mathcal{O}$.*
- (ii) *There are $c \in \mathbb{C}$ and $R \in \mathcal{H}_{>0}(\mathbb{C})$ with $F = c + R$.*

Hence the algebra of real-bounded macroscopically entire functions is exactly

$$\mathcal{B} = \mathbb{C} \oplus \mathcal{H}_{>0}(\mathbb{C}), \quad \mathcal{B}/\mathcal{H}_{>0}(\mathbb{C}) \cong \mathbb{C}. \quad (11.6)$$

More generally, a bound $|F^\#(z)| \leq Mt^\beta$ with M ordinary real holds if and only if $F = t^\beta(c + R)$ with $c \in \mathbb{C}$ and R of positive support.

Proof. Assume (i). If $v_H(F) < 0$, pick an ordinary point c at which the leading coefficient function $f_{v_H(F)}$ is nonzero; the value $F^\#(c)$ then has negative valuation and is infinite, contradicting a real bound. So $F \in \mathcal{H}_{\geq 0}(\mathbb{C})$. Let f_0 be its exponent-zero coefficient function (possibly zero). Taking standard parts of the bound at ordinary points gives $|f_0(c)| \leq M$ for all $c \in \mathbb{C}$, so the ordinary Liouville theorem makes f_0 a constant $c \in \mathbb{C}$. All remaining support is strictly positive, which is (ii).

Conversely, if $F = c + R$ with $\text{supp}(R) > 0$ then $R^\#(z) \in \mathfrak{m}$ for every finite z , so $|F^\#(z)| < |c| + 1$: a single ordinary real bound, despite the fact that the individual coefficient functions of R may grow arbitrarily fast. The quotient statement is immediate, and multiplying by $t^{-\beta}$ gives the graded version. $\square$

This is the only sharp statement in the collection of what a scalar bound buys, and it is what reconciles the other two hypotheses: $tZ = 0 + tZ$ lies in $\mathbb{C} \oplus \mathcal{H}_{>0}(\mathbb{C})$, exactly as (11.6) predicts.

(c) **Coefficientwise growth: the classical hypothesis.**

Theorem 11.9 (Coefficientwise growth and polynomial degree). *Let $d \geq 0$ be an ordinary integer and $F = \sum_{\gamma \in S} f_\gamma t^\gamma \in \mathcal{H}(\mathbb{C})$. The following are equivalent.*

- (i) *For every $\gamma \in S$ there is an ordinary real $M_\gamma > 0$ with $|f_\gamma(c)| \leq M_\gamma(1 + |c|)^d$ for all $c \in \mathbb{C}$.*
- (ii) *There are $A_0, \dots, A_d \in \mathbb{K}$ with $F^\#(z) = \sum_{j=0}^d A_j z^j$ for all $z \in \mathcal{O}$.*

The case $d = 0$ is the coefficientwise Liouville theorem: if every f_γ is bounded on $\mathbb{C}$ by its own real constant, then F is a constant element of $\mathbb{K}$.

Proof. Assume (i) and apply the ordinary Cauchy estimates to f_γ on the circle of ordinary radius R about 0: for $n > d$,

$$|f_\gamma^{(n)}(0)| \leq n! M_\gamma \frac{(1 + R)^d}{R^n}.$$

Letting the *ordinary real* R tend to infinity — a limit inside $\mathbb{C}$, taken separately for each γ , not a limit through surreal radii — makes this derivative vanish. So each f_γ is a polynomial of degree at most d . Put $A_j = \sum_{\gamma \in S} \frac{f_\gamma^{(j)}(0)}{j!} t^\gamma$; these sums exist with support inside S , and regrouping the finitely many powers of z gives (ii). Conversely, expand the finitely many A_j into normal form: each ordinary coefficient function is a polynomial of degree at most d and obeys a bound of the required shape, and injectivity of coherent evaluation identifies those coefficients with the original f_γ . $\square$

Remark 11.10 (The hierarchy, and the two counterexamples). For the boundedness case $d = 0$, hypotheses (b) and (c) both imply (a), but are incomparable. The constant section $F = \omega$ has bounded ordinary coefficient functions and no ordinary-real bound on its values, so (c) does not imply (b). Conversely $F = tZ$ satisfies (b) and violates (c). The earlier assertion of a linear hierarchy (c) $\Rightarrow$ (b) $\Rightarrow$ (a) was therefore false. Importing the conclusion of Theorem 11.9 under hypothesis (b) is false: $F = tZ$ is real-bounded by 1 on the whole finite plane and is not constant. Importing it under hypothesis (a) is worse, since (a) holds for every macroscopically entire F by Proposition 11.7. The two standing counterexamples are recorded once here: tZ , and εZ for a positive real infinitesimal ε , whose nonzero coefficient functions are nonzero scalar multiples of Z and are therefore unbounded, so that (c) is genuinely violated.

These are a *different* obstruction from the locally constant function L of Section 5. The functions tZ and εZ are coherent, and their domain is $\mathbb{C}^\#$, the halo of the ordinary plane; L is defined on all of $\mathbb{K}$ and is germ-analytic. Its restriction to every finite halo is the coherent constant 1; it is its global behavior on $\mathbb{K}$ that gives the different counterexample. Different analytic class, different domain; the two must not be presented as instances of one phenomenon.

11.4 Removable singularities

Theorem 11.11 (Coefficientwise removal is necessary and sufficient). *Let $U \subseteq \mathbb{C}$ be a domain, $c \in U$, and $F = \sum_{\gamma \in S} f_\gamma t^\gamma \in \mathcal{H}(U \setminus \{c\})$. Then F extends to an element of $\mathcal{H}(U)$ if and only if every f_γ has a removable singularity at c . In particular the extension exists whenever each f_γ is bounded on some ordinary punctured neighbourhood of c , with a bound allowed to depend on γ . The extension is unique.*

Proof. Necessity: any coherent extension restricts coefficientwise to the given functions, by uniqueness of normal forms at ordinary points, so each f_γ extends holomorphically. Sufficiency: the ordinary removable singularity theorem extends each f_γ ; the extended family has the same well-ordered support S , hence is an element of $\mathcal{H}(U)$. Uniqueness is the ordinary identity theorem applied coefficient by coefficient. $\square$

Remark 11.12 (A scalar bound is not enough, and the puncture is a monad). A bound on the surcomplex value $F^\#$ does not suffice. On the halo of the ordinary punctured disk $0 < |Z| < 1$ the family t/Z is infinitesimal at every point, hence bounded by the ordinary constant 1; but its coefficient at exponent 1 is $1/Z$, which has a pole, so by Theorem 11.11 it has no coherent extension across the origin. The family $t e^{1/Z}$ is bounded by 1 in the same way and its coefficient has an essential singularity. This is the same phenomenon as in Remark 11.10: a bound on the leading scale of the value says nothing at all about the smaller scales.

Note also which domain is in play. The halo $(\mathbb{D} \setminus \{0\})^\#$ excludes the *whole monad* of the origin; it is not the class obtained by deleting the single point 0 from $\mathbb{D}^\#$. Keeping these two puncturing operations distinct prevents a false analogy with the ordinary punctured disk.

11.5 Schwarz–Pick for canonical classical lifts

This subsection, alone in Section 11, is about the class of canonical classical lifts $f^\#$ of ordinary holomorphic f . The sharp inequalities below are *false* for general coherent self-maps of $\mathbb{D}^\#$; see Example 11.16. Let $\mathbb{D} = \{Z \in \mathbb{C} : |Z| < 1\}$, let $f : \mathbb{D} \rightarrow \mathbb{D}$ be ordinary holomorphic, and put, for $z, w \in \mathbb{D}^\#$,

$$d(z, w) = \left| \frac{z - w}{1 - \bar{w}z} \right|.$$

The denominator is invertible because its standard part $1 - \bar{w}_0 z_0$ is a nonzero complex number, z_0, w_0 being the standard parts. Since standard parts commute with evaluation, $f^\#$ maps $\mathbb{D}^\#$ into $\mathbb{D}^\#$.

Theorem 11.13 (Schwarz–Pick for canonical lifts). *For all $z, w \in \mathbb{D}^\#$,*

$$d(f^\#(z), f^\#(w)) \leq d(z, w), \quad \frac{|(f^\#)'(z)|}{1 - |f^\#(z)|^2} \leq \frac{1}{1 - |z|^2}. \quad (11.7)$$

For $z \neq w$, equality in the first inequality holds if and only if f is an ordinary disk automorphism; equality in the second at any point has the same characterization.

Proof. Suppose first that f is not an ordinary disk automorphism. Then the ordinary Schwarz–Pick inequality is strict at distinct ordinary points, and its infinitesimal form is strict at every ordinary point.

If $\text{st } z \neq \text{st } w$, the quotient of the two sides of the first inequality in (11.7) has standard part strictly less than one, so the surreal inequality is strict.

If $\text{st } z = \text{st } w = c$ and $z \neq w$, this argument is unavailable: both pseudohyperbolic distances have standard part zero, and taking standard parts of the classical inequality yields only $0 \leq 0$. One must divide out $z - w$ *before* reducing. Consider the divided difference

$$G(z, w) = \frac{f^\#(z) - f^\#(w)}{z - w}.$$

The coherent Taylor expansion at c gives a strongly summable extension of G across the diagonal, with $\text{st } G(z, w) = f'(c)$. Hence

$$\text{st} \left(\frac{d(f^\#(z), f^\#(w))}{d(z, w)} \right) = |f'(c)| \frac{1 - |c|^2}{1 - |f(c)|^2} < 1,$$

by the strict ordinary infinitesimal Schwarz–Pick inequality at c . So the strict inequality holds even when $z - w$ is arbitrarily small. The derivative inequality follows from the same strict ordinary inequality by taking standard parts.

If f is an automorphism, write $f(Z) = e^{i\theta}(Z - a)/(1 - \bar{a}Z)$ with $a \in \mathbb{D}$ and θ ordinary real. The usual algebraic identities for this expression are polynomial identities valid over the real closed field **No** and its complexification $\mathbb{K}$, so both inequalities in (11.7) become equalities. This proves the equality characterizations. $\square$

Corollary 11.14 (Schwarz lemma for canonical lifts). *If in addition $f(0) = 0$ then*

$$|f^\#(z)| \leq |z| \quad (z \in \mathbb{D}^\#), \quad |(f^\#)'(0)| \leq 1,$$

with equality at a nonzero z , or in the derivative bound, if and only if $f(Z) = uZ$ for an ordinary complex u with $|u| = 1$.

Proof. Put $w = 0$ in Theorem 11.13; an automorphism fixing the origin is an ordinary rotation. $\square$

Corollary 11.15 (Stability away from the equality case). *Let $F = f_0 + E \in \mathcal{H}(\mathbb{D})$ with $E = 0$ or $v_H(E) > 0$, and suppose $f_0 : \mathbb{D} \rightarrow \mathbb{D}$ is ordinary holomorphic and is not a disk automorphism. Then $F^\#$ satisfies both inequalities of (11.7), strictly, the first one at distinct points.*

Proof. The standard parts of $F^\#$ and of $F^{\# \prime}$ are f_0 and f_0' . At distinct standard parts use the strict ordinary inequality for f_0 ; in a single monad use the coherent Taylor expansion, whose divided difference has standard part $f_0'(c)$. The proof of Theorem 11.13 then applies verbatim. $\square$

Corollary 11.15 is a statement about the coherent class, but it is strictly weaker than Theorem 11.13: it asserts only the strict inequalities, and it requires the ordinary part f_0 to be non-extremal. The sharp inequality, with its equality case, is not available in $\mathcal{H}(\mathbb{D})$ at all.

Example 11.16 (Why the restriction to lifts is substantive). For the positive infinitesimal $t = \omega^{-1}$ the coherent function $F(Z) = (1 + t)Z$ maps $\mathbb{D}^\#$ bijectively onto itself, since it does not change a point's standard part, and it fixes the origin. Nevertheless

$$F^{\# \prime}(0) = 1 + t > 1, \quad |F^\#(z)| > |z| \quad (z \neq 0).$$

So Schwarz's lemma is false for arbitrary coherent self-maps of the halo. Two features of the setting are exposed by this example. First, the ordinary part of F here *is* an automorphism, so Corollary 11.15 does not apply — the equality case of the ordinary theorem is exactly what is unstable under infinitesimal perturbation. Second, $\mathbb{D}^\#$ is not the surcomplex unit ball $\{z \in \mathbb{K} : |z| < 1\}$: the latter contains points such as $1 - \varepsilon$ with ε a positive infinitesimal, whose standard part lies on the ordinary unit circle, and no point of that boundary monad belongs to $\mathbb{D}^\#$. It is precisely this omitted infinitesimal boundary layer that leaves room for $(1 + t)Z$.

11.6 Conformal equivalence of halos

Theorem 11.17 (Lifting of infinitesimally deformed biholomorphisms). *Let $f_0 : U \rightarrow V$ be an ordinary biholomorphism of ordinary complex domains and let $F = f_0 + H \in \mathcal{H}(U)$ with $H = 0$ or $v_H(H) > 0$. Then there is $G = g_0 + J \in \mathcal{H}(V)$ with $g_0 = f_0^{-1}$ and $J = 0$ or $v_H(J) > 0$, such that*

$$F \circ G = \text{id}_V, \quad G \circ F = \text{id}_U$$

as coherent sections. Consequently $F^\# : U^\# \rightarrow V^\#$ is a bijection with germ-analytic inverse $G^\#$, and $F^{\# \prime}$ vanishes nowhere.

Proof. Let $M = \langle \text{supp}(H) \rangle$ and seek $J = \sum_{\gamma \in M, \gamma > 0} j_\gamma t^\gamma$. In the coefficient equation at exponent γ of $F(g_0 + J) = \text{id}$, the only term containing the unknown j_γ is $f_0'(g_0) j_\gamma$; every other term involves strictly smaller exponents of J or known coefficients of H , and by Neumann's lemma each such equation has finitely many terms. Since $f_0'(g_0)$ is nowhere zero on V , the equation determines a single ordinary holomorphic j_γ on V . Well-ordered recursion over $M_{>0}$ constructs J , and the support-controlled composition law of Section 7 (inner ordinary holomorphic map plus positive-support perturbation) justifies the composition identities.

For the evaluated statement: $\text{st}(F^\#(a + \eta)) = f_0(a)$, so the image lies in $V^\#$, and the evaluated identity $F^\# \circ G^\# = \text{id}$ gives surjectivity. For injectivity take $w = b + \varepsilon \in V^\#$; the section $F - w$ has leading coefficient function $f_0 - b$, which has a unique and simple zero at $a = g_0(b)$, so the zero-cluster theorem of Section 9 gives exactly one solution of $F^\#(z) = w$ in $a + \mathfrak{m}$ and none in any other monad. Faithfulness of evaluation gives the remaining composition identity, and $\text{st}(F^{\#'}(a + \eta)) = f'_0(a) \neq 0$ gives the last assertion. $\square$

Corollary 11.18 (Lifted conformal equivalence). *The case $H = 0$: if $f : U \rightarrow V$ is an ordinary biholomorphism, then $f^\# : U^\# \rightarrow V^\#$ is a bijection, both it and its inverse $(f^{-1})^\#$ are germ-analytic, and $(f^\#)'$ is nowhere zero.*

Corollary 11.19 (Standard-halo Riemann mapping theorem). *If $U \subsetneq \mathbb{C}$ is nonempty, open and simply connected, then $U^\#$ is analytically equivalent to $\mathbb{D}^\#$ by a canonical classical lift.*

Proof. Choose an ordinary Riemann map and apply Corollary 11.18. $\square$

The existence input in Corollary 11.19 is precisely the *ordinary* Riemann mapping theorem. No uniformization of arbitrary fine-open subclasses of $\mathbb{K}$ is claimed, and none is proved anywhere in this article. Theorem 11.17 is strictly stronger than Corollary 11.18, and the extra strength is real: for example $G(Z) = Z + tZ^2$ is a non-identity biholomorphism of $\mathbb{D}^\#$ fixing the origin with $G^{\#'}(0) = 1$, so even the normalization that would characterize an ordinary disk automorphism leaves infinitesimal coherent freedom. There is no conflict with Corollary 11.14, whose hypothesis is a canonical lift.

11.7 A coefficientwise Montel theorem

Definition 11.20 (Coefficientwise compact-open convergence). Fix a well-ordered support set S and let $F_n = \sum_{\gamma \in S} f_{n,\gamma} t^\gamma$ and $F = \sum_{\gamma \in S} f_\gamma t^\gamma$ lie in $\mathcal{H}(U)$. Say $F_n \rightarrow F$ *coefficientwise compact-open* if, for each fixed $\gamma \in S$, $f_{n,\gamma} \rightarrow f_\gamma$ uniformly on every ordinary compact subset of U .

Theorem 11.21 (Countable-support Montel theorem). *Let $(F_n)_{n \geq 1}$ in $\mathcal{H}(U)$ have a common countable well-ordered support S , and suppose that for each $\gamma \in S$ and each ordinary compact $K \Subset U$,*

$$\sup_n \sup_{z \in K} |f_{n,\gamma}(z)| < \infty.$$

Then some subsequence converges coefficientwise compact-open to an element of $\mathcal{H}(U)$, and every fixed coefficientwise derivative $D_z^k F_n$ converges in the same sense along that subsequence.

Proof. Enumerate S as $\gamma_1, \gamma_2, \dots$. The ordinary Montel theorem applied to $(f_{n,\gamma_1})_n$ gives a locally uniformly convergent subsequence; refine for γ_2 , and so on. The diagonal subsequence converges for every $\gamma \in S$. Each limit function is ordinary holomorphic and the limits are supported in the same well-ordered set S , so they assemble to an element of $\mathcal{H}(U)$. Local uniform convergence of the ordinary derivatives, obtained from the ordinary Cauchy formula, gives the statement about D_z^k . $\square$

Remark 11.22 (What this theorem is a theorem about). Definition 11.20 describes a topology on *coefficient data*. It is not, and must not be read as, pointwise convergence of evaluated values in the fine topology: by the discreteness theorem of Sections 3–4 every set-indexed convergent net in $\mathbb{K}$ is eventually equal to its limit, so a sequential Montel theorem in the fine topology would be the wrong statement. Countability of S is also not cosmetic. The diagonal argument extracts one subsequence controlling countably many independent coefficient sequences; it does not automatically produce a subsequence controlling uncountably many. Product-compactness arguments for suitable coefficient spaces and nets can be substituted, but they too concern the coefficient topology and not the topology of values.

11.8 Harmonic components, Poisson's formula, and reflection

If $F \in \mathcal{H}(U)$ and $F^\# = u + iv$ with $z = x + iy$, then the Cauchy–Riemann equations of Section 6, together with the equality of mixed second derivatives, give

$$u_{xx} + u_{yy} = 0, \quad v_{xx} + v_{yy} = 0,$$

where the derivatives are genuine fine partial derivatives, by the bridge theorem of Section 7 identifying the coefficientwise derivative with the fine difference-quotient derivative. These are differential identities; no global Dirichlet problem is asserted by them. Theorem 11.24 is what fills that gap, and only on ordinary disks with coefficientwise boundary data.

A *coherent harmonic datum* on an ordinary domain U is a family $u = \sum_{\gamma \in S} u_\gamma t^\gamma$ with common well-ordered support and ordinary real-valued harmonic coefficient functions u_γ . Its evaluation at $z = c + \varepsilon$ is defined by the two-real-variable Taylor expansion of each u_γ at c ; harmonic functions are real analytic, and the support lemma applies to $\operatorname{Re} \varepsilon$ and $\operatorname{Im} \varepsilon$, so $u^\#$ is well defined and real-surreal valued.

Theorem 11.23 (Coefficientwise harmonic conjugates). *On a simply connected ordinary domain U , every coherent harmonic datum is the real part of an element of $\mathcal{H}(U)$. A harmonic conjugate is unique modulo one real surreal constant, $u^\#$ satisfies the Laplace equation in the fine differentiable sense, and a nonconstant coherent harmonic datum on a connected ordinary base has no fine-local maximum or minimum.*

Proof. Fix $a \in U$ and let v_γ be the ordinary harmonic conjugate of u_γ normalized by $v_\gamma(a) = 0$. The functions $u_\gamma + iv_\gamma$ are ordinary holomorphic, and their Hahn sum with weights t^γ is an element of $\mathcal{H}(U)$ whose real part evaluates to $u^\#$, by the ordinary Cauchy–Riemann equations and uniqueness of Taylor expansions. Two conjugates differ coefficientwise by real constants, hence by one real surreal constant. The Laplace equation holds coefficientwise and is preserved by Taylor evaluation. For the extremum assertion, work on a small simply connected ordinary disk about the standard part of a proposed extremum; the local holomorphic conjugate is nonconstant unless the datum is constant on that disk, in which case ordinary unique continuation propagates constancy through the connected base. In the nonconstant case Theorem 11.2 supplies values with both larger and smaller real part in every fine-neighbourhood, excluding an extremum. $\square$

Theorem 11.24 (Poisson's formula and a coherent Dirichlet problem). *Let the coefficients of a coherent harmonic datum u be harmonic on a neighbourhood of the closed ordinary disk $\overline{B}(c, R)$, with $c \in \mathbb{C}$ and $R > 0$ ordinary. Then for every $z \in B(c, R)^\#$,*

$$u^\#(z) = \frac{1}{2\pi} \int_{[0, 2\pi]}^H \frac{R^2 - |z - c|^2}{|c + Re^{i\theta} - z|^2} u(c + Re^{i\theta}) d\theta. \quad (11.8)$$

More generally, a well-ordered Hahn family of ordinary continuous real boundary data on the ordinary circle has a unique coherent harmonic extension whose coefficient functions extend continuously to the closed disk with the prescribed boundary values.

Proof. At an ordinary interior point apply the ordinary Poisson formula to each coefficient. For $z = a + \varepsilon$ with a ordinary and ε infinitesimal, the kernel's denominator has a positive ordinary lower bound on the circle before perturbation; expand its inverse in the infinitesimal coordinates of ε . The resulting support lies in the sum of the original support with a positive-support Neumann monoid, so the family is Hahn summable and each coefficient is an ordinary continuous function of θ , integrable in the coefficientwise sense of Section 8. Ordinary differentiation under the compact integral identifies each Taylor coefficient of the right-hand side with the corresponding Taylor coefficient of u at a , proving (11.8). For general continuous boundary coefficients take their ordinary Poisson extensions and reassemble with the same support; uniqueness is the uniqueness of the ordinary Dirichlet problem, coefficient by coefficient. $\square$

The boundary condition in Theorem 11.24 is an ordinary coefficientwise trace. It is not a claim about compactness, nor about convergence to the boundary in the fine topology, and it says nothing about boundary data that are not presented as a common-support Hahn family of ordinary continuous functions. Boundary data given as coefficientwise measures or distributions, and the Herglotz positivity problem, are treated in the companion report held in docs/surcomplex/hahn-herglotz-positivity/ (herg:thm:hierarchy).

Theorem 11.25 (Coherent Schwarz reflection). *Let $F = \sum_{\gamma \in S} f_{\gamma} t^{\gamma}$ be a coherent datum on an upper half-disk, and suppose each f_{γ} extends continuously to the real diameter and is real valued there. Then the coefficientwise reflection $\tilde{f}_{\gamma}(Z) = \overline{f_{\gamma}(\overline{Z})}$ on the lower half-disk joins each f_{γ} to a function holomorphic on the full disk, the unchanged support gives a coherent extension $\tilde{F}$ of F to the full disk, and $\tilde{F}^{\#}(\bar{z}) = \tilde{F}^{\#}(z)$. The extension is unique.*

Proof. Ordinary Schwarz reflection applies to each f_{γ} . The support is unchanged, so the reflected family is coherent, and evaluation commutes with conjugation of coefficients and of the variable. Uniqueness is the coherent identity theorem of Section 7. $\square$

The reality hypothesis in Theorem 11.25 concerns *all* Hahn coefficients on the ordinary diameter. It cannot be replaced by a hypothesis about an arbitrarily selected fine-open component of the halo, since the fine topology supplies no connection between such components.

12 Exponentials and logarithms: local rigidity, global freedom

Every function constructed in this part is germ-analytic in the sense of Section 6 — it has a Hahn-summable power-series expansion on a fine neighbourhood of every point of its domain. Their coherent restrictions and their class-wide domains must be distinguished: the exponential and its twists are coherent on finite translated charts, while the global logarithm is not coherent on the halo of the whole punctured ordinary plane. That specific obstruction, proved in §12.4, is what the contour no-go theorem consumes.

12.1 The infinitesimal and finite exponentials, and polar form

Proposition 12.1 (Infinitesimal exponential and logarithm). *For $h \in \mathfrak{m}$ and $\eta \in \mathfrak{m}$ the families*

$$\exp_{\mathfrak{m}}(h) = \sum_{n \geq 0} \frac{h^n}{n!}, \quad \log_{\mathfrak{m}}(1 + \eta) = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \eta^n$$

are Hahn summable, and they are mutually inverse isomorphisms

$$\exp_{\mathfrak{m}} : (\mathfrak{m}, +) \xrightarrow{\sim} (1 + \mathfrak{m}, \cdot), \quad \log_{\mathfrak{m}} : (1 + \mathfrak{m}, \cdot) \xrightarrow{\sim} (\mathfrak{m}, +).$$

Both commute with conjugation. In particular, if $v \in 1 + \mathfrak{m}$ has $|v| = 1$, then $\log_{\mathfrak{m}}(v)$ is purely imaginary.

Proof. Summability is Neumann’s lemma applied to the positive well-ordered set $\text{supp}(h)$, or $\text{supp}(\eta)$: complex scalar coefficients add no exponents and each exponent of $\langle \text{supp}(h) \rangle$ has finitely many word representations. The identities $\exp_{\mathfrak{m}}(h+k) = \exp_{\mathfrak{m}}(h) \exp_{\mathfrak{m}}(k)$, $\log_{\mathfrak{m}}((1+\eta)(1+\xi)) = \log_{\mathfrak{m}}(1+\eta) + \log_{\mathfrak{m}}(1+\xi)$, $\log_{\mathfrak{m}}(\exp_{\mathfrak{m}}(h)) = h$ and $\exp_{\mathfrak{m}}(\log_{\mathfrak{m}}(1+\eta)) = 1+\eta$ are formal identities in finitely many infinitesimal variables and hold after evaluation by the same support argument. Conjugation acts on the complex coefficients only. For the last assertion, $|v| = 1$ means $v\bar{v} = 1$, so $\log_{\mathfrak{m}}(v) + \log_{\mathfrak{m}}(\bar{v}) = \log_{\mathfrak{m}}(v\bar{v}) = 0$. $\square$

Proposition 12.2 (The finite exponential and the polar form). *For $z = c + h \in \mathcal{O}$ with $c = \text{st}(z) \in \mathbb{C}$ and $h \in \mathfrak{m}$, set*

$$E_{\text{fin}}(z) = e^c \exp_{\mathfrak{m}}(h).$$

Then E_{fin} is a surjective group homomorphism from $(\mathcal{O}, +)$ onto the group of finite units of $\mathbb{K}$ with nonzero standard part, and $\ker E_{\text{fin}} = 2\pi i\mathbb{Z}$. Moreover every $z \in \mathbb{K}^\times$ has a polar representation

$$z = |z| E_{\text{fin}}(i\theta), \quad \theta \in \mathbf{No} \text{ finite}, \quad (12.1)$$

and θ is unique modulo $2\pi\mathbb{Z}$ among finite angles.

Proof. The homomorphism property combines $e^{c+c'} = e^c e^{c'}$ with Proposition 12.1. A finite unit is $c(1+h)$ with $c \neq 0$ complex and $h \in \mathfrak{m}$; choosing an ordinary logarithm of c and applying $\log_{\mathfrak{m}}$ to $1+h$ gives surjectivity. For the kernel, take standard parts first, obtaining $e^c = 1$ and hence $c \in 2\pi i\mathbb{Z}$, then use the injectivity of $\exp_{\mathfrak{m}}$ on $\mathfrak{m}$.

For (12.1), put $u = z/|z|$, so $|u| = 1$ and u is finite. Its standard part $u_0 \in \mathbb{C}$ has $|u_0| = 1$; write $u = u_0(1+\delta)$ with $\delta \in \mathfrak{m}$. Since $|1+\delta| = 1$, Proposition 12.1 makes $\log_{\mathfrak{m}}(1+\delta) = i\eta$ with η a real infinitesimal. Choose an ordinary real θ_0 with $u_0 = e^{i\theta_0}$ and set $\theta = \theta_0 + \eta$, a finite real surreal. Uniqueness modulo $2\pi\mathbb{Z}$ is the kernel computation. Shifting by an ordinary integer multiple of 2π normalizes θ to lie in $(-\pi, \pi]$; for a negative real z this normalized angle is π . $\square$

Note what (12.1) does *not* require: no value of a sine or cosine at an infinite real argument is used, because every angle occurring is finite.

12.2 The canonical surcomplex exponential

For a real surreal y split the normal form uniquely as

$$y = y_{\text{PI}} + y_0 + y_{\text{inf}}, \quad y_{\text{PI}} \in \mathbf{J}, \quad y_0 \in \mathbb{R}, \quad y_{\text{inf}} \in \mathfrak{m} \cap \mathbf{No}, \quad (12.2)$$

where y_{PI} collects the monomials t^γ with $\gamma < 0$, and put $\text{Fin}(y) = y_0 + y_{\text{inf}}$. Here $\mathbf{J}$ is the additive group of purely infinite real surreals; it is a real *vector space*, a fact used in Remark 12.5. All three projections in (12.2) are additive, and Fin retains all terms at nonnegative exponents, not merely the standard part.

Definition 12.3 (Canonical exponential). For $x, y \in \mathbf{No}$ put

$$\text{Exp}(x + iy) = \exp_{\mathbf{No}}(x) E_{\text{fin}}(i \text{Fin}(y)), \quad (12.3)$$

where $\exp_{\mathbf{No}}$ is the Gonshor [1] real surreal exponential.

Theorem 12.4 (Properties of the canonical exponential). *The map $\text{Exp} : (\mathbb{K}, +) \rightarrow (\mathbb{K}^\times, \cdot)$ is a surjective group homomorphism extending both $\exp_{\mathbf{No}}$ and the ordinary complex exponential, with*

$$|\text{Exp}(x + iy)| = \exp_{\mathbf{No}}(x), \quad \ker \text{Exp} = 2\pi i \mathbf{Oz}, \quad \mathbf{Oz} = \mathbf{J} + \mathbb{Z}. \quad (12.4)$$

It is germ-analytic at every point of $\mathbb{K}$: for every $b \in \mathbb{K}$ and every $h \in \mathfrak{m}$,

$$\text{Exp}(b + h) = \text{Exp}(b) \sum_{n \geq 0} \frac{h^n}{n!} = \text{Exp}(b) \exp_{\mathfrak{m}}(h), \quad (12.5)$$

and consequently $\text{Exp}' = \text{Exp}$ in the fine sense.

Proof. Additivity of Fin together with the group laws for $\exp_{\mathbf{No}}$ and E_{fin} gives the homomorphism property. The modulus identity holds because E_{fin} of a purely imaginary finite argument has modulus one, by Proposition 12.1.

Kernel: the modulus identity forces $x = 0$; then $E_{\text{fin}}(i \text{Fin}(y)) = 1$, so $\text{Fin}(y) \in 2\pi\mathbb{Z}$ by Proposition 12.2, which says exactly $y \in \mathbf{J} + 2\pi\mathbb{Z}$. Hence $\ker \text{Exp} = i(\mathbf{J} + 2\pi\mathbb{Z})$; see Remark 12.5 for the identification with $2\pi i \mathbf{Oz}$.

Surjectivity is (12.1): for $z \neq 0$ write $z = |z| E_{\text{fin}}(i\theta)$ with θ finite, and then $z = \text{Exp}(\log_{\mathbf{No}}|z| + i\theta)$, since $\text{Fin}(\theta) = \theta$ for finite θ .

For (12.5), let $h = p + iq \in \mathbf{m}$. Then q is infinitesimal, so $y_{\text{PI}}(q) = 0$ and $\text{Fin}(y+q) = \text{Fin}(y)+q$; the real infinitesimal Taylor identity for $\exp_{\mathbf{No}}$ and Proposition 12.1 give $\text{Exp}(h) = \exp_{\mathbf{m}}(h)$, and the group law gives (12.5). Factoring h out of the linear remainder bounds the difference-quotient error by $|\text{Exp}(b)|\omega|h|$, which is fine-small, so $\text{Exp}' = \text{Exp}$. $\square$

The existence of a canonical surcomplex exponential with kernel $2\pi i \mathbf{Oz}$ is established in Ehrlich and Kaplan [20, Proposition 11.2 and Theorem 11.1, pp. 33–34 of arXiv:2002.07739v3]; what is proved here is its local analytic calculus. Note also that Exp does *not* have a Hahn-summable Maclaurin series at every argument: at an infinite positive real argument it is defined by $\exp_{\mathbf{No}}$, not by declaring $\sum z^n/n!$ summable.

Remark 12.5 (The kernel written two ways). The two standard presentations of the kernel,

$$\ker \text{Exp} = 2\pi i \mathbf{Oz} \quad \text{with } \mathbf{Oz} = \mathbf{J} + \mathbb{Z}, \quad \text{and} \quad \ker \text{Exp} = i(\mathbf{J} + 2\pi\mathbb{Z}),$$

are equal, and the identity that makes them equal must be stated, because the first form invites a false reading. Indeed $2\pi(\mathbf{J} + \mathbb{Z}) = 2\pi\mathbf{J} + 2\pi\mathbb{Z}$, and $\mathbf{J}$ is a real vector space, so $2\pi\mathbf{J} = \mathbf{J}$. Written as $2\pi i \mathbf{Oz}$ the kernel looks like a uniform scaling of a discrete group, suggesting that multiplication by 2π changes the purely infinite subgroup. It changes individual elements but preserves that subgroup as a whole.

Related (batch 33). On the additive domain $\overline{\mathbb{Q}} + \mathbf{J}$, whose imaginary parts are finite algebraic numbers, Exp restricts to $a + p \mapsto e^a \exp_{\mathbf{No}}(p)$, which is the exponential `exr:def:domain` of the companion report held in `docs/foundations-and-computation/exponential-relations-over-omnific-integers/`. There the values at finitely many distinct points of the domain are linearly independent over $\overline{\mathbb{Q}}$ (`exr:thm:independence`, from Gonshor’s monomial theorem and the Lindemann–Weierstrass theorem), and the Laurent-polynomial relations among finitely many such values are generated by the binomials $X^\ell - 1$, with ℓ in the lattice of integer relations of the arguments (`exr:thm:toric`). No such independence holds on a domain containing $i\mathbf{J}$, which lies in the kernel: $\text{Exp}(i\omega) = \text{Exp}(0)$ (Remark 12.6).

Remark 12.6 ($\text{Exp}(i\omega) = 1$, and how it is *not* obtained). Since $\text{Fin}(\omega) = 0$, formula (12.3) gives $\text{Exp}(i\omega) = 1$. This value is *not* inferred by summing $\sum_{n \geq 0} (i\omega)^n/n!$: that family is not Hahn summable, its term valuations $-n$ forming a strictly decreasing sequence with no well-ordered union of supports. The value records the chosen treatment of infinite imaginary phases in (12.3), and §12.3 shows that the choice is not forced by any local analytic axiom.

12.3 The twisted family: local axioms do not fix the infinite phase

For a real surreal y with normal form $\sum_\gamma y_\gamma t^\gamma$, let $c_\omega(y) = y_{-1} \in \mathbb{R}$ be the real coefficient of the monomial $\omega = t^{-1}$. This is an additive real-linear functional vanishing on every finite y .

Theorem 12.7 (The twisted global exponentials). *For each ordinary real λ define*

$$E_\lambda(x + iy) = \exp_{\mathbf{No}}(x) E_{\text{fin}}(i(\text{Fin}(y) + \lambda c_\omega(y))), \quad E_0 = \text{Exp}. \quad (12.6)$$

Each $E_\lambda : (\mathbb{K}, +) \rightarrow (\mathbb{K}^\times, \cdot)$ is a surjective group homomorphism which

- (a) extends $\exp_{\mathbf{No}}$ and the ordinary complex exponential, and agrees with Exp on all finite surcomplex arguments;
- (b) satisfies $E_\lambda(0) = 1$, $|E_\lambda(x + iy)| = \exp_{\mathbf{No}}(x)$ and, for $h \in \mathfrak{m}$, $E_\lambda(b + h) = E_\lambda(b) \exp_{\mathfrak{m}}(h)$, hence $E'_\lambda = E_\lambda$ and germ-analyticity at every point of $\mathbb{K}$;
- (c) is locally coherent in finite translated charts: $E_\lambda(b + w) = E_\lambda(b) (e^w)^\#$ for $w \in \mathcal{O}$, a surcomplex constant times a canonical classical lift;
- (d) satisfies $E_\lambda(i\omega) = e^{i\lambda}$, so that $E_0(i\omega) = 1$ while $E_\pi(i\omega) = -1$, and $E_\lambda \neq E_\mu$ whenever $\lambda \neq \mu$.

Proof. The map $y \mapsto \text{Fin}(y) + \lambda c_\omega(y)$ is additive and is the identity on finite y , so (a) and the group law follow from Theorem 12.4. The correction factor $E_{\text{fin}}(i\lambda c_\omega(y))$ lies on the ordinary unit circle, giving the modulus identity. For an infinitesimal increment $h = p + iq$ one has $c_\omega(q) = 0$ and $\text{Fin}(q) = q$, so the correction factor is unchanged and the local expansion of Theorem 12.4 applies verbatim; this is (b). For (c), on the translated finite tube $b + \mathcal{O}$ the same computation gives $E_\lambda(b + w) = E_\lambda(b)(e^w)^\#$, a product of a constant of $\mathbb{K}$ with the lift of the ordinary exponential, hence a coherent section in that normalized chart. For (d), $c_\omega(\omega) = 1$ and $\text{Fin}(\omega) = 0$; taking $y = r\omega$ with $r = \pi/(\lambda - \mu)$ separates E_λ from E_μ . Surjectivity follows from the preimages of Proposition 12.2, whose imaginary parts are finite and hence untouched by the twist. $\square$

Theorem 12.7 is a diagnostic statement. It says that

$$E' = E, \quad E(0) = 1, \quad \text{agreement with } \exp_{\mathbf{No}} \text{ and } \exp_{\mathbb{C}}, \quad |E(x + iy)| = \exp_{\mathbf{No}}(x)$$

do *not* determine an exponential's values at infinite imaginary arguments. The additional requirement that the phase annihilate the purely infinite imaginary summand — that is, c_ω and all lower monomials contribute nothing — is what selects Exp from the family. This is a normalization, not a theorem.

Corollary 12.8 (A structured unit-modulus witness). *For $\lambda \notin 2\pi\mathbb{Z}$ the quotient E_λ/Exp is a nonconstant function on all of $\mathbb{K}$ with values in the ordinary unit circle; it is germ-analytic everywhere and every one of its positive-order derivatives vanishes identically.*

Proof. By (12.6) the quotient is $z \mapsto E_{\text{fin}}(i\lambda c_\omega(\text{Im } z))$, which is unchanged by any finite increment of z , hence locally constant on every translated tube $b + \mathcal{O}$ and therefore germ-analytic with vanishing derivatives. It is nonconstant by Theorem 12.7(d). $\square$

This is a more structured version of the locally constant counterexample of Section 5: local analytic Taylor data alone impose no continuation between infinitely separated regions of $\mathbb{K}$. In particular Corollary 12.8 is exactly why no theorem of Section 11 may be applied to a function that is merely germ-analytic on $\mathbb{K}$.

Theorem 12.9 (The period obstruction). *Let $E : (\mathbb{K}, +) \rightarrow (\mathbb{K}^\times, \cdot)$ be a group homomorphism which extends the strip exponential — that is, which agrees with $\exp_{\mathbf{No}}$ on $\mathbf{No}$ and with E_{fin} on finite imaginary arguments — and which satisfies $E(\bar{z}) = \overline{E(z)}$. Then for every infinite real surreal y there is a finite real surreal v with*

$$i(y - v) \in \ker E. \tag{12.7}$$

Consequently no such global extension has kernel exactly $2\pi i\mathbb{Z}$.

Proof. The homomorphism property and conjugation symmetry give $E(iy)\overline{E(iy)} = E(iy)E(-iy) = E(0) = 1$, so $E(iy)$ lies on the surcomplex unit circle. By the polar form (12.1), the whole surcomplex unit circle is parametrized by *finite* real angles: choose a finite v with $E(iy) = E_{\text{fin}}(iv) = E(iv)$, the last equality by the hypothesis that E agrees with E_{fin} on finite imaginary arguments. Then $E(i(y-v)) = 1$, and $y-v$ is infinite, so this period is not an ordinary integer multiple of 2π . $\square$

Theorem 12.9 is the general statement behind the kernel computations: the other constructions compute the kernel of one particular exponential, whereas this argument uses only the homomorphism property, conjugation symmetry and the finite-angle parametrization. The obstruction is algebraic, not topological. Extending the ordinary complex exponential to all of $\mathbb{K}$ while keeping conjugation and the finite-angle values necessarily changes the period structure.

12.4 The global fine-analytic logarithm

Fix once and for all the ordinary principal argument $\text{Arg} : \{u \in \mathbb{C} : |u| = 1\} \rightarrow (-\pi, \pi]$. This is a choice of values on ordinary unit complex numbers only; it is not a claim that the ordinary principal argument is continuous across its cut.

Definition 12.10 (The global logarithm). For $z \in \mathbb{K}^\times$ put $r = |z|$, $u = z/r$ and $u_0 = \text{st}(u) \in \mathbb{C}$, so $|u_0| = 1$ and $u/u_0 \in 1 + \mathfrak{m}$. Define

$$\mathcal{L}(z) = \log_{\mathbf{No}}(r) + i \text{Arg}(u_0) + \log_{\mathfrak{m}}\left(\frac{u}{u_0}\right). \quad (12.8)$$

The branch is chosen by $u_0 = \text{st}(z/|z|)$, the standard part of the unit direction — a quantity that is constant on a sufficiently small fine-neighbourhood of each nonzero point, since an infinitesimal relative perturbation does not move a standard part. That is what makes (12.8) analytic rather than merely defined.

Theorem 12.11 (A logarithm on the whole punctured class plane). *The function $\mathcal{L} : \mathbb{K}^\times \rightarrow \mathbb{K}$ is germ-analytic at every point, has finite imaginary part, and satisfies*

$$\mathcal{L}'(z) = \frac{1}{z}, \quad E_\lambda(\mathcal{L}(z)) = z \quad \text{for every ordinary real } \lambda.$$

More precisely, whenever $h/z \in \mathfrak{m}$,

$$\mathcal{L}(z+h) - \mathcal{L}(z) = \log_{\mathfrak{m}}\left(1 + \frac{h}{z}\right) = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n z^n} h^n. \quad (12.9)$$

Its restriction to ordinary nonzero complex points has the values of the ordinary principal logarithm, including the selected value $i\pi$ of the argument on the negative real axis.

Proof. Set $\eta = h/z \in \mathfrak{m}$. The unit direction of $z(1+\eta)$ has the same standard part u_0 as that of z , so the argument term in (12.8) is unchanged. The modulus becomes $r|1+\eta|$ and the unit direction becomes $u(1+\eta)/|1+\eta|$, so by the group laws for $\log_{\mathbf{No}}$ and $\log_{\mathfrak{m}}$,

$$\mathcal{L}(z+h) - \mathcal{L}(z) = \log_{\mathbf{No}}|1+\eta| + \log_{\mathfrak{m}}\left(\frac{1+\eta}{|1+\eta|}\right) = \log_{\mathfrak{m}}(1+\eta),$$

which is (12.9); the displayed series is Hahn summable and has linear coefficient $1/z$, giving germ-analyticity and the derivative. The last term of (12.8) is purely imaginary (by Proposition 12.1, since $|u/u_0| = 1$ exactly) and infinitesimal, and $\text{Arg}(u_0)$ is an ordinary real, so the imaginary part of $\mathcal{L}(z)$ is finite. The right-inverse identity is then the surjectivity computation of Theorem 12.7, because $\mathcal{L}(z)$ has finite imaginary part and every E_λ agrees with Exp whenever the imaginary part is finite, even when the real part is infinite. At an ordinary nonzero complex z one has $u = u_0$ and r ordinary, so (12.8) reduces to $\log|z| + i \text{Arg}(z/|z|)$. $\square$

Corollary 12.12 ($\mathcal{L}$ is not a coherent primitive of $1/z$). $\mathcal{L}$ is not the evaluation of any element of $\mathcal{H}(\mathbb{C}^\times)$ on $(\mathbb{C}^\times)^\#$.

Proof. If it were, then $\mathcal{L}$ would be a coherent primitive of $1/z$ on $(\mathbb{C}^\times)^\#$, and the coherent fundamental theorem of Section 8 would force $\oint_{|z|=1} dz/z = 0$. But that coefficientwise integral is $2\pi i$. Equivalently, the ordinary coefficient data of $\mathcal{L}$ have a jump across the selected argument cut and do not glue holomorphically around the circle. $\square$

This is the property that must travel with $\mathcal{L}$ wherever it goes. It is also what makes the architecture of Sections 8 and 9 forced rather than merely cautious: $\mathcal{L}$ is a globally germ-analytic G on $\mathbb{K}^\times$ with $G' = 1/z$, so by the contour no-go theorem of Section 5 no integral along the ordinary unit circle can simultaneously return the classical value $2\pi i$ for $1/z$ and annihilate G' for every such G . The coefficientwise definition of the contour, and the restriction of the fundamental theorem to *coherent* primitives, are consequences of that, not stylistic conservatism.

Remark 12.13 (No paradox at the negative real axis). It is tempting to object that $\mathcal{L}$ cannot be analytic at a negative real point, where the ordinary principal logarithm is discontinuous. The objection confuses two topologies. An infinitesimal perturbation of a fixed negative real point leaves the standard unit direction $u_0 = -1$ unchanged, so it does not cross the chosen standard-part cut; the branch is fine-locally constant. Ordinary points approaching the axis from below do *not* form a fine-convergent family — by the discreteness theorem of Sections 3–4 no nontrivial set-indexed family converges in the fine topology at all — so no fine-continuity requirement is violated. The classical topological obstruction to a global logarithm and the fine-local differential calculation concern different structures. The obstruction survives in the coherent category, where it is Corollary 12.12; it disappears in the more permissive germ-analytic category, which is Theorem 12.11.

12.5 The principal branch on the cut plane, and the $2\pi i$ discrepancy

A second, genuinely different function is also called “the” logarithm in the sources. It is retained here under its own name.

Theorem 12.14 (Principal logarithm on the cut plane). On $\mathbb{K} \setminus \mathbf{No}_{\leq 0}$ define

$$\text{Log } z = \log_{\mathbf{No}} |z| + i \text{Arg}_c(z), \quad -\pi < \text{Arg}_c(z) < \pi,$$

where $\text{Arg}_c(z)$ is the unique finite angle in $(-\pi, \pi)$ supplied by the polar form (12.1). Then Log is germ-analytic, $\text{Exp}(\text{Log } z) = z$, $\text{Log}'(z) = 1/z$, and Exp restricts to a bijection of the strip $\{x + iy : x \in \mathbf{No}, -\pi < y < \pi\}$ onto the cut plane, with inverse Log .

Proof. Finite-angle uniqueness in Proposition 12.2 gives the bijection and the inverse identities. For fixed z off the cut and sufficiently small h , the candidate value $\text{Log } z + \log_m(1 + h/z)$ still has imaginary part inside $(-\pi, \pi)$, and its image under Exp is $z + h$; uniqueness in the strip therefore gives $\text{Log}(z + h) = \text{Log } z + \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} (h/z)^n$, which proves germ-analyticity and the derivative. The required shrinking of h is legitimate even when the angle is only infinitesimally far from an endpoint of the strip, since a smaller fine radius is always available. $\square$

Proposition 12.15 ($\mathcal{L}$ and Log are different functions). $\mathcal{L}$ and Log agree wherever $\text{Arg}_c(z)$ and $\text{Arg}(\text{st}(z/|z|))$ agree, and differ by exactly $2\pi i$ on the lower half of the monad of every negative ordinary real. Concretely, let $\varepsilon > 0$ be a real infinitesimal and $z = -1 - i\varepsilon$. Then

$$\mathcal{L}(z) = i(\pi + \varepsilon) + O_{\text{formal}}(\varepsilon^2), \quad \text{Log}(z) = i(-\pi + \varepsilon) + O_{\text{formal}}(\varepsilon^2), \quad \mathcal{L}(z) - \text{Log}(z) = 2\pi i.$$

Proof. Here $|z| = 1 + \varepsilon^2/2 + \dots$ is infinitesimally near 1, so the real parts of both expressions are infinitesimal and equal. For $\mathcal{L}$: the unit direction is $u = (-1 - i\varepsilon)/|z|$ with $\text{st}(u) = u_0 = -1$, and $\text{Arg}(-1) = +\pi$ under the convention $(-\pi, \pi]$; the correction term is $\log_{\mathfrak{m}}(u/u_0) = i\varepsilon + O_{\text{formal}}(\varepsilon^2)$. For Log : the finite angle of z in $(-\pi, \pi)$ is $-\pi + \varepsilon + O_{\text{formal}}(\varepsilon^2)$, since z lies infinitesimally below the negative real axis. Subtracting gives $2\pi i$ exactly, the infinitesimal corrections cancelling. On the upper half-monad, $z = -1 + i\varepsilon$, both give $i(\pi - \varepsilon) + \dots$ and the two agree. $\square$

Both branches are legitimate, and neither is wrong; they are different functions and must not be identified. This document adopts $\mathcal{L}$ as the primary logarithm, for three reasons. Its domain is all of $\mathbb{K}^\times$, not $\mathbb{K}$ minus a class-sized ray. It is germ-analytic everywhere, because its branch is selected by the fine-locally constant quantity $\text{st}(z/|z|)$, and Log is likewise fine-analytic at every point of its cut-plane domain. The discontinuity concerns extending the chosen principal values across a deleted negative-axis point: the values from its two sides differ by $2\pi i$. Proposition 12.15 does not show a discontinuity at any point where Log is defined. And the contour no-go theorem of Section 5 requires a global primitive of $1/z$ on all of $\mathbb{K}^\times$, which Log does not supply.

12.6 Local logarithm charts

Theorem 12.16 (Logarithm charts on multiplicative monads). *Let $b \in \mathbb{K}^\times$ and choose $\lambda \in \mathbb{K}$ with $\text{Exp}(\lambda) = b$. For $w \in b(1 + \mathfrak{m})$ define*

$$\text{Log}_\lambda(w) = \lambda + \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \left(\frac{w - b}{b} \right)^n = \lambda + \log_{\mathfrak{m}}\left(\frac{w}{b}\right).$$

This is a bijection $b(1 + \mathfrak{m}) \rightarrow \lambda + \mathfrak{m}$ inverse to the restriction of Exp , and $\text{Log}'_\lambda(w) = 1/w$. Any two admissible choices of λ differ by an element of $2\pi i \mathbf{Oz}$, and the corresponding charts differ by that constant.

Proof. The formal identities $\log(e^h) = h$ and $e^{\log(1+h)} = 1 + h$ evaluate at infinitesimals by Proposition 12.1; the derivative is the evaluated derivative of the formal logarithm followed by the affine chain rule. The last assertion is the kernel computation of Theorem 12.4. $\square$

These charts describe branch choices without requiring a fine-continuous principal argument on all of $\mathbb{K}^\times$ or a path-connected covering-space structure. A principal argument can be chosen as a function, but its conventional values do not extend continuously across the negative real axis; independently, Sections 3–4 excludes nonconstant ordinary-parameter fine-continuous paths. Note that the ambiguity is by an element of $2\pi i \mathbf{Oz}$, not merely by an ordinary integer multiple of $2\pi i$.

12.7 A value-distribution theorem at an explicit essential singularity

Put $P(z) = \text{Exp}(1/z)$ for $z \in \mathbb{K}^\times$. By Theorem 12.4 and the chain rule P is germ-analytic on $\mathbb{K}^\times$ with $P'(z) = -z^{-2}P(z)$, and it never takes the value zero.

Theorem 12.17 (Every nonzero value at every surreal scale). *For every $w \in \mathbb{K}^\times$ and every positive surreal δ there is a proper class of distinct z with*

$$0 < |z| < \delta, \quad \text{Exp}(1/z) = w.$$

Consequently P has at zero neither a removable singularity nor a pole in the fine topology.

Proof. Choose λ with $\text{Exp}(\lambda) = w$, possible by Theorem 12.4. For $q \in \mathbf{Oz}$ with $\lambda + 2\pi i q \neq 0$ put

$$z_q = \frac{1}{\lambda + 2\pi i q}.$$

These are distinct, and $P(z_q) = \text{Exp}(\lambda + 2\pi iq) = w$ because $2\pi i\mathbf{Oz} = \ker \text{Exp}$. The omnific integers are cofinal in $\mathbf{No}^{>0}$ — indeed $\mathbf{Oz}$ contains every ordinal — so for all sufficiently large positive $q \in \mathbf{Oz}$,

$$|\lambda + 2\pi iq| \geq 2\pi q - |\lambda| > \delta^{-1},$$

whence $0 < |z_q| < \delta$. A proper-class tail of such q supplies the required solutions. Taking $w = 1$ and $w = 2$ shows that P has no fine limit at zero, since every punctured fine-neighbourhood of zero contains both values; taking $w = 1$ also excludes a pole, which would require $|P|$ to exceed every fixed surreal bound near zero. $\square$

Remark 12.18 (Scope, and the phase dependence of the twisted variants). Theorem 12.17 is an exact value-distribution theorem for one specified function. It is not a general Picard theorem for germ-analytic functions, and no such theorem is proved or claimed anywhere in this article: a general statement of that breadth would be incompatible with the locally constant witnesses of Section 5 and with Corollary 12.8. Nor does the phrase “essential singularity” here assert a global Laurent classification; it records exactly the two proved failures, of removability and of a pole.

The value-distribution conclusion also holds for every twist E_λ . The canonical periods $2\pi i\mathbf{Oz}$ need not remain periods of a twist, but a cofinal family of periods does remain:

$$E_\lambda(i\omega^\alpha) = 1 \quad (\alpha \geq 2 \text{ an ordinal}),$$

because ω^α is purely infinite and has zero coefficient at ω . These positive monomials are cofinal in $\mathbf{No}$, and every tail is a proper class. Given $w \neq 0$, choose ℓ with $E_\lambda(\ell) = w$. The distinct points $z_\alpha = (\ell + i\omega^\alpha)^{-1}$ then have $E_\lambda(1/z_\alpha) = w$, and all sufficiently large α give $0 < |z_\alpha| < \delta$, by the same triangle estimate as above. Thus the earlier claim that the canonical phase was necessary for this conclusion was incorrect.

The kernels are not generally nested. For example $i\omega$ belongs to $\ker \text{Exp}$ but not to $\ker E_\pi$, while $i(\omega - \pi)$ belongs to $\ker E_\pi$ but not to $\ker \text{Exp}$. The functions still differ at specified points: at $z = -i/\omega$ their values are respectively 1 and -1 . Agreement on ordinary arguments and the ordinary Laurent series does not fix those values, even though all the twists share the stated value-distribution property.

Finally, a summability boundary that belongs beside the theorem. The canonical lift $(e^{1/Z})^\#$ is a coherent section on $(\mathbb{C}^\times)^\#$, but that halo excludes the entire monad of zero. At a nonzero infinitesimal z the formal expression $\sum_{n \geq 0} z^{-n}/n!$ is *not* Hahn summable: its term valuations $-n v(z)$ form a strictly decreasing sequence. So the ordinary Laurent series does not itself define any extension into that monad, and $P = \text{Exp}(1/z)$ is an extension chosen by the exponential, not derived from the series. On ordinary annuli avoiding zero, P does agree with $(e^{1/Z})^\#$, so its coefficientwise contour integral on an ordinary positively oriented circle is $2\pi i$, the ordinary Laurent coefficient of Z^{-1} being one. That is consistent with, but not determined by, the behaviour of P on the infinitesimal monads near the singularity.

13 Affine charts at all scales, and the rigidity theorems

The field $\mathbb{K}$ carries a proper class of multiplicative scales, and nothing in the coherent theory of $\mathcal{H}(U)$ privileges the scale 1. This part first transports that theory to an arbitrary centre and an arbitrary nonzero scale, then asks what survives if one insists on a coherent description at *every* scale at once. The answer is a pair of classification theorems: all-scale coherence admits exactly the polynomials, and its meromorphic counterpart admits exactly the rational functions. Every statement in this part is a statement about the *coherent* class $\mathcal{H}(U)$ and about class functions assembled from coherent charts; none of it is asserted for germ-analytic functions, whose class contains the locally constant counterexample of the locally constant witness of Section 5 and therefore admits no identity theorem to run the argument with.

13.1 Coherent affine charts at an arbitrary centre and scale

Definition 13.1 (Scaled coherent chart). Let $b \in \mathbb{K}$, let $r \in \mathbb{K}^\times$, and let $U \subseteq \mathbb{C}$ be an ordinary domain. The *scaled chart* with centre b and scale r is the fine-open class

$$\mathcal{U}(b, r; U) = b + r U^\# = \{b + rw : w \in U^\#\}.$$

A function f on $\mathcal{U}(b, r; U)$ is *coherent in the coordinate* $(z - b)/r$ if there is $F \in \mathcal{H}(U)$ with

$$f(z) = F^\# \left(\frac{z - b}{r} \right) \quad (z \in \mathcal{U}(b, r; U)).$$

The affine coordinate is part of the specified structure, not a normalization that may afterwards be suppressed. The centre b may have infinite real or imaginary part, and $|r|$ may be infinitesimal, finite or infinite.

Theorem 13.2 (Affine transport of the coherent theory). *Let f be coherent in the coordinate $(z - b)/r$, with $F \in \mathcal{H}(U)$ as in Definition 13.1. Then:*

(i) *Derivatives rescale: $f^{(n)}(b + rw) = r^{-n} (F^{(n)})^\#(w)$ for every ordinary $n \geq 0$ and every $w \in U^\#$.*

(ii) *For an ordinary piecewise- C^1 path γ in U , the chart contour functional is defined by*

$$\int_{b+r\gamma}^H f(z) dz := r \int_\gamma^H F(w) dw. \quad (13.1)$$

It is $\mathbb{K}$ -linear, additive under concatenation, orientation reversing, and invariant under ordinary reparameterization of γ .

(iii) *The identity, open-mapping, local maximum-modulus, preparation, division and zero-counting theorems transport verbatim from $U^\#$ to $\mathcal{U}(b, r; U)$, and Cauchy's theorem, the Cauchy integral formula and the residue theorem hold for (13.1).*

(iv) *If f is a coherent meromorphic quotient in the chart and $p = b + r\lambda$ with $\lambda \in U^\#$, then*

$$\text{Res}_p(f(z) dz) = r \text{Res}_\lambda(F^\#(w) dw). \quad (13.2)$$

Zero and pole multiplicities and the normalized logarithmic-derivative index are unchanged by the affine coordinate.

Proof. (i) is the finite chain rule applied inside the coefficientwise derivative, which introduces no new Hahn exponents. (ii) is a definition together with the corresponding properties of the coefficientwise contour functional, 4.5. For (iii), $w \mapsto b + rw$ is a fine homeomorphism with analytic inverse and nonzero linear term, so it preserves local multiplicities; pulling a preparation factorization back through it gives the factorization in the chart coordinate with the same finite number of roots, and the Cauchy and residue formulas follow from (13.1) by substitution.

For (iv), write $F(\lambda + X) = \sum_n a_n X^n$ as a formal Laurent germ with finite principal part. Then $f(p + Y) = \sum_n a_n r^{-n} Y^n$, whose Y^{-1} coefficient is $r a_{-1}$. In the logarithmic derivative the factor r^{-1} produced by differentiation cancels the factor r carried by the differential $dz = r dw$, so the index computed by $\oint^H F'/F$ is exactly the ordinary integer of the argument principle, 10.10. $\square$

Remark 13.3 (What a scaled contour is and is not). The symbol $b + r\gamma$ in (13.1) records an affine contour *datum*: an ordinary path γ together with a centre and a scale. It is not an assertion that $b + r\gamma$ is a nonconstant fine-continuous image of a real interval, and by the discreteness theorem 4.2 no such image exists. This caveat does not weaken when the centre or the scale changes.

Remark 13.4 (No unrestricted chart independence). Ordinary complex affine coordinate changes preserve the coherent class by coefficientwise composition. Changes involving surreal centres and scales may relate very different ordinary base domains, and no unrestricted gluing or chart-independence theorem is asserted here. Compatibility of two charts means equality of their *evaluated* functions on the overlap; every contour computation below names the chart in which it is performed. When two admissible descriptions yield the same pointwise pulled-back integrand, uniqueness of normal forms identifies their coefficient functions and their integrals therefore agree.

Example 13.5 (Charts far from the ordinary plane). Take $b = \omega + i\omega^2$ and $r = \omega^{-\omega}$. The resulting chart is an exceedingly small neighbourhood of an infinite point, and its zero clusters, preparation polynomials and residues are as accessible as those in the halo of the ordinary unit disk $\mathbb{D}^\#$. Taking $r = \omega^\omega$ instead gives a chart at an infinitely large scale. Nothing in Theorem 13.2 distinguishes these cases from $b = 0, r = 1$.

13.2 The chart-change criterion, and the source of the obstruction

The transport of Theorem 13.2 is along a *fixed* affine map. A finer question is when a normalized change of chart carries a coherent section to a coherent section. The criterion is explicit, and its failure is exactly what separates local coherence from coherence at all scales.

Proposition 13.6 (Chart-change criterion). *Let $F = \sum_{s \in S} f_s t^s \in \mathcal{H}(U)$, and consider the normalized coordinate change $z = \alpha + \beta w$ with $\alpha, \beta \in \mathcal{O}$ finite. Write $\alpha = \alpha_0 + \delta_\alpha$ and $\beta = \beta_0 + \delta_\beta$ with $\alpha_0 = \text{st}(\alpha)$, $\beta_0 = \text{st}(\beta)$ in $\mathbb{C}$ and $\delta_\alpha, \delta_\beta \in \mathfrak{m}$. Let $V \subseteq \mathbb{C}$ be an ordinary domain with $\alpha_0 + \beta_0 V \subseteq U$. Then*

$$F(\alpha + \beta w) = \sum_{s \in S} \sum_{n \geq 0} \frac{f_s^{(n)}(\alpha_0 + \beta_0 w)}{n!} (\delta_\alpha + \delta_\beta w)^n t^s \quad (13.3)$$

defines a section of $\mathcal{H}(V)$, whose evaluation on $V^\#$ agrees with $w \mapsto F^\#(\alpha + \beta w)$. The degenerate case $\beta_0 = 0$ is permitted, provided $\alpha_0 \in U$.

Proof. For fixed s and n the function $w \mapsto f_s^{(n)}(\alpha_0 + \beta_0 w)$ is ordinary holomorphic on V , and in the degenerate case $\beta_0 = 0$ it is the constant $f_s^{(n)}(\alpha_0)$, which is legitimate because $\alpha_0 \in U$. The support of the coefficient of t^s in (13.3) is contained in $s + \langle \text{supp } \delta_\alpha \cup \text{supp } \delta_\beta \rangle$, a translate of the Neumann [3] monoid of one fixed well-ordered set of positive exponents. By Neumann's support lemma 3.2 that union is well ordered and each exponent receives only finitely many contributions, so the double family is Hahn summable and its sum is a section over the one ordinary domain V . Agreement of the evaluations is the exact monad-wide Taylor expansion of 7.8. $\square$

Remark 13.7 (An infinite normalized coefficient breaks the criterion). Proposition 13.6 requires α and β *finite*. If the normalized scale coefficient β is infinite there is no standard part $\beta_0 \in \mathbb{C}$ to serve as the ordinary inner map, the remainder δ_β is not infinitesimal, and the substitution is no longer justified by (13.3). For example, substituting an infinite β into the exponential Taylor series gives decreasing valuations $nv(\beta)$. Polynomial coefficient functions can still be substituted at infinite scales; failure of the criterion is not a claim that every such substitution fails. This is the precise mechanism by which a function can be coherent in every finite translated chart and yet fail to be coherent at all scales. It is the structural reason behind Theorem 13.12, and it is why the canonical exponential Exp of 12.4, which is locally coherent, is excluded by that theorem.

13.3 Cauchy estimates at an arbitrary coherent scale

Corollary 13.8 (Cauchy estimates at an arbitrary coherent scale). *Let $a \in \mathbb{K}$, $r \in \mathbf{No}_{>0}$, and suppose $f(a + rw) = F^\#(w)$ for some $F \in \mathcal{H}(B(0, 2))$. If there is a single $M \in \mathbf{No}$ with*

$$|f(a + re^{i\theta})| \leq M \quad \text{for every ordinary real } \theta,$$

then

$$|f^{(n)}(a)| \leq \frac{n! M}{r^n} \quad (n \geq 0). \quad (13.4)$$

Proof. By Theorem 13.2(i), $F^{(n)}(0) = r^n f^{(n)}(a)$. Apply the pointwise surreal Cauchy estimate 8.5 on the ordinary unit circle to F ; its hypothesis is a single *pointwise* surreal bound on the path, and its bound M is allowed to be surreal because the underlying ML inequality 8.3 is proved from the positivity of the coefficientwise integral rather than from a coefficientwise majorant. $\square$

Remark 13.9 (Which estimate this is). Corollary 13.8 uses the *pointwise* estimate 8.5, resting on 8.3, which bounds the surreal *modulus* of a derivative. It is not the coefficientwise majorant $B_r(F) = \sum_s (\max |f_s|) t^s$ of 8.12, and it does not assert that a single surreal bound controls every coefficient function: that assertion is false, as $F(w) = w(1 + it)$ shows, since $|F^\#| = \sqrt{1 + t^2}$ on the unit circle admits the pointwise bound $1 + t^2$ while its coefficientwise majorant is $1 + t > 1 + t^2$. The two estimates are about different quantities and both are retained.

Remark 13.10 (Some small chart, but not every large one). Every germ-analytic germ admits *at least one* coherent affine chart, by the scale-embedding bridge 7.25: after a single monomial rescaling $\rho = t^\lambda$ chosen by block separation, each Hahn coefficient of the rescaled series is a finite sum and hence an ordinary polynomial, so the rescaled datum is coherent on all of $\mathbb{C}$. There is *no* implication that the same germ has a chart of every larger radius. Making that distinction precise is exactly the content of the next subsection.

13.4 All-scale coherence forces polynomiality

Definition 13.11 (All-scale Hahn-entire). A class function $f : \mathbb{K} \rightarrow \mathbb{K}$ is *all-scale Hahn-entire* if for every $r \in \mathbf{No}_{>0}$ there exists $F_r \in \mathcal{H}(B(0, 2))$ with

$$f(rw) = F_r^\#(w) \quad (w \in B(0, 2)^\#). \quad (13.5)$$

Only charts centred at zero are required, and the well-ordered supports $S_r = \text{supp } F_r$ may depend on r ; *no* common support across scales is imposed.

An equivalent formulation replaces $B(0, 2)$ by the ordinary unit disk $\mathbb{D}$ and quantifies over $R \in \mathbf{No}^{>0}$; the two conditions define the same class of functions, and the proof below is the same argument with different constants. These charts cover $\mathbb{K}$ and make f fine-locally analytic; but although each individual chart is set-supported, the requirement ranges over a proper class of radii, and that is what makes it restrictive.

Theorem 13.12 (All-scale polynomial rigidity). *A class function $f : \mathbb{K} \rightarrow \mathbb{K}$ is all-scale Hahn-entire if and only if it is a polynomial in z with coefficients in $\mathbb{K}$ and ordinary finite degree.*

Proof. ($\Leftarrow$) Let $P(z) = \sum_{n=0}^N a_n z^n$ with $a_n \in \mathbb{K}$. For a fixed r , expanding each $a_n r^n$ into normal form uses finitely many well-ordered supports, whose union is well ordered, and the coefficient function attached to each exponent is an ordinary polynomial in w of degree at most N . Hence $P(rw)$ is the evaluation of a member of $\mathcal{H}(B(0, 2))$ for every r .

($\Rightarrow$) Put $a_n = f^{(n)}(0)/n! \in \mathbb{K}$ for $n \geq 0$; the chart at $r = 1$ supplies these derivatives, and $(a_n)_{n \in \mathbb{N}}$ is a set. Fix r and let S_r be the support of F_r . Since coefficientwise differentiation of an ordinary holomorphic coefficient function introduces no new Hahn exponents, the n th Taylor coefficient of F_r at 0, which by the chain rule equals $a_n r^n$, has normal-form support contained in S_r ; in particular $v(a_n r^n) \in S_r$ whenever $a_n \neq 0$.

Suppose infinitely many a_n are nonzero. The valuations $\{v(a_n) : a_n \neq 0\}$ form a set of surreals, so by the surreal cut property there is $\Delta > 0$ with

$$|v(a_n)| < \Delta/4 \quad \text{whenever } a_n \neq 0.$$

Take the scale $r = t^{-\Delta}$, so $v(r) = -\Delta$. For two indices $m > n$ with $a_m, a_n \neq 0$,

$$v(a_m r^m) - v(a_n r^n) = v(a_m) - v(a_n) - (m - n)\Delta < \frac{\Delta}{2} - \Delta < 0.$$

Thus the valuations of the nonzero $a_n r^n$ form an infinite strictly decreasing sequence of elements of the single set S_r , contradicting its well ordering.

Hence $a_n = 0$ for all $n > N$, for some ordinary finite N . Put $P(z) = \sum_{n=0}^N a_n z^n$. For each r the coherent section $F_r - P(r \cdot)$ lies in $\mathcal{H}(B(0, 2))$ and has all derivatives zero at 0, so by the coherent identity theorem 7.18 it is the zero section and $f(rw) = P(rw)$ on $B(0, 2)^\#$. Given any $z \in \mathbb{K}$, choose $r \in \mathbf{No}_{>0}$ with $r > 2|z|$; then z/r is finite with standard part of modulus at most $1/2$, hence lies in $B(0, 2)^\#$, and therefore $f(z) = P(z)$. $\square$

Remark 13.13 (The mechanism is support-theoretic, not Archimedean). Theorem 13.12 is not Liouville's theorem with a larger word for infinity, and its proof contains no growth estimate. Two proper-class facts do the work: the surreal cut property supplies *one* exponent Δ dominating a whole set of coefficient valuations, and $\mathbb{K}$ then supplies a scale $r = t^{-\Delta}$ large enough to reverse the order of infinitely many nonzero Taylor coefficients. A single fixed set-sized Hahn field K_Γ need contain no such Δ or r , so the theorem is a statement about $\mathbb{K}$ and must not be asserted inside a chosen set-sized workspace.

Corollary 13.14 (Global classical consequences in the rigid category). *Let f be all-scale Hahn-entire.*

- (i) *If $|f(z)| \leq M$ for all $z \in \mathbb{K}$, for some $M \in \mathbf{No}$, then f is constant.*
- (ii) *If $|f(z)| \leq C(1 + |z|)^N$ for all $z \in \mathbb{K}$, with $C \in \mathbf{No}_{>0}$ and N an ordinary nonnegative integer, then $\deg f \leq N$.*
- (iii) *If f is nonconstant it is surjective onto $\mathbb{K}$.*
- (iv) *If f is injective it is affine with nonzero slope.*
- (v) *There is no all-scale Hahn-entire solution of $f' = f$, $f(0) = 1$.*

Proof. By Theorem 13.12, f is a polynomial $\sum_{n \leq d} a_n z^n$ with $a_d \neq 0$. For (ii), if $d > N$ choose a positive real-coordinate surreal x large enough that the leading term dominates the sum of the lower terms and $|a_d| x^{d-N} > 2^{N+1} C$; then $|f(x)| \geq |a_d| x^d / 2 > C(1 + x)^N$, a contradiction. (i) is the case $N = 0$ after enlarging M to a nonnegative bound. (iii) is algebraic closedness of $\mathbb{K}$ applied to $f(z) - c$. For (iv), a polynomial of degree at least two has a critical point, since its nonconstant derivative has a root in the algebraically closed $\mathbb{K}$; the finite-ramification normal form 6.19 at that point gives more than one preimage of sufficiently close noncritical values, so f is not injective, and degree ≤ 1 with nonzero leading coefficient is exactly the affine case. (v) is a comparison of degrees: no nonzero polynomial equals its own derivative. $\square$

Remark 13.15 (This is not the Liouville hierarchy). Corollary 13.14(i) must not be confused with any of the three Liouville hypotheses tabulated in §15.3. Hypothesis (a) there — a single surreal bound on a coherent section over the finite halo $\mathbb{C}^\#$ — is *vacuous*, by 11.7. The hypothesis here is different in two ways: the bound is required on all of the class $\mathbb{K}$, not on the finite halo, and the function is required to be coherent at *every* scale, not on one. It is the all-scale hypothesis, through Theorem 13.12, that produces constancy, not the bound by itself.

Remark 13.16 (All-scale coherence is a rigidity condition, not a definition). Condition (13.5) should not be offered as the definition of “entire surcomplex analyticity.” It excludes the canonical exponential Exp , whose Taylor coefficients at zero are all nonzero, and by Remark 13.7 the exclusion is caused by an infinite normalized affine coefficient rather than by any defect of Exp as an analytic object: Exp and each twist E_λ of 12.7 is germ-analytic everywhere and coherent in every finite translated chart. What (13.5) does is classify exactly what is possible when one demands the coherence of this article on every scale simultaneously.

13.5 Pole-free fractions and all-scale rational rigidity

Lemma 13.17 (A pole-free coherent fraction is coherent holomorphic). *Let $U \subseteq \mathbb{C}$ be a connected ordinary domain, let $F, G \in \mathcal{H}(U)$ with $G \neq 0$, and suppose the evaluated fraction $F^\# / G^\#$ has no actual poles in $U^\#$, its removable values being filled in. Then it is the evaluation of a member of $\mathcal{H}(U)$.*

Proof. Let $c \in U$. If the leading coefficient function $g_{v_H(G)}$ of G does not vanish at c , then G is a unit of $\mathcal{H}(U')$ on a small ordinary disk $U' \ni c$ by the coherent unit criterion, and $F/G \in \mathcal{H}(U')$ directly.

Otherwise $g_{v_H(G)}$ vanishes at c , and Hahn–Weierstrass preparation 9.3 on a small ordinary disk about c gives, in the local coordinate $u = z - c$,

$$G(c + u) = t^\lambda q_0(u) P(u) Q(u),$$

where q_0 and Q are coherent units, $\lambda = v_H(G)$, and P is monic of degree $m = \text{ord}_c(g_\lambda)$ with infinitesimal lower coefficients. Absorb $t^\lambda q_0 Q$ into the numerator, obtaining a coherent N , and apply the division theorem 9.7:

$$\frac{N}{P} = J + \frac{R}{P}, \quad J \in \mathcal{H}(U'), \quad \deg R < m.$$

Every root of P is infinitesimal, since a monic polynomial with infinitesimal lower coefficients has only infinitesimal roots: a root of valuation ≤ 0 would leave the monic term uncancellable. As J is coherent holomorphic and the original fraction has no pole in $U'^\#$, the rational function $R/P \in \mathbb{K}(u)$ has no pole at any root of P . Hence $P \mid R$ in $\mathbb{K}[u]$, and $\deg R < m = \deg P$ forces $R = 0$. So $F^\# / G^\# = J^\#$ on $U'^\#$.

These local representations agree as evaluated functions on overlaps, so the coherent gluing theorem 7.20 joins them into a single member of $\mathcal{H}(U)$. The gluing step is what supplies a *common global* support; it is not being assumed from an arbitrary union of local supports, which need not be well ordered. $\square$

Definition 13.18 (All-scale Hahn-meromorphic). A function $f : \mathbb{K} \rightarrow \mathbb{P}^1(\mathbb{K})$, not identically ∞ , is *all-scale Hahn-meromorphic* if for every $r \in \mathbf{No}_{>0}$ the pullback $w \mapsto f(rw)$ is represented on $B(0, 2)^\#$ by a fraction $F_r^\# / G_r^\#$ with $F_r, G_r \in \mathcal{H}(B(0, 2))$ and $G_r \neq 0$. Fractions are understood as local meromorphic germs, including their uniquely filled removable values and their pole values in $\mathbb{P}^1(\mathbb{K})$.

Theorem 13.19 (All-scale rational rigidity). *The all-scale Hahn-meromorphic functions on $\mathbb{K}$ are precisely the rational functions with surcomplex coefficients.*

Proof. First, f has only finitely many poles in the whole class plane. Otherwise choose a countable set of distinct poles $a_1, a_2, \dots$ and, by the surreal cut property, a positive surreal r exceeding $n|a_j|$ for all ordinary positive integers n and all j ; then every a_j/r is infinitesimal, hence lies in the monad of 0 inside $B(0, 2)^\#$. The radius- r representation writes $f(rw)$ as a coherent fraction $F_r^\# / G_r^\#$ there. By the zero-cluster theorem 9.9 applied to the leading coefficient function of G_r , the denominator has exactly $\text{ord}_0(g_{v_H(G_r)})$ zeros in that monad, a finite number. So $f(r \cdot)$ cannot have infinitely many distinct poles there, a contradiction.

Let the actual poles be $a_1, \dots, a_k$ with orders $m_1, \dots, m_k$, and set $Q(z) = \prod_{j=1}^k (z - a_j)^{m_j} \in \mathbb{K}[z]$. Fill the removable values of $g = Qf$. In every scale chart, g is a coherent fraction with no actual poles, hence by Lemma 13.17 is the evaluation of a member of $\mathcal{H}(B(0, 2))$. So g is all-scale Hahn-entire, and Theorem 13.12 makes $g = P$ a polynomial. Therefore $f = P/Q$ is rational.

Conversely, the numerator and denominator of a rational function over $\mathbb{K}$ become polynomials with surcomplex coefficients after every affine scaling, and by the converse half of Theorem 13.12 these supply coherent fractions at every scale. $\square$

Remark 13.20. The proof has a global divisor reading: all-scale coherence forces the pole divisor of f on the class plane to be finite, and multiplying by the corresponding polynomial removes it. The conclusion resembles the classical rationality of meromorphic functions on the Riemann sphere, but the hypothesis is scale coherence on a proper-class plane, not compactness of a sphere; no compactness, completeness or normal-family argument is used anywhere in Theorems 13.12 and 13.19.

14 Worked examples with exact arithmetic

Each example below is chosen to separate a hypothesis proved earlier: it exhibits a function satisfying all but one clause of some theorem, or it computes exactly a quantity that an approximate argument would blur. Every displayed series is a Hahn sum in the sense of 3.1, never a limit of partial sums. The finite verifications are collected in §14.10.

14.1 Classically divergent series: what is germ-analytic and what is coherent

Example 14.1 (A germ-analytic function with zero classical radius). Let $\eta \in \mathfrak{m}$ and put

$$A(\eta) = \sum_{n \geq 0} (n!)^2 \eta^n.$$

The family $((n!)^2 \eta^n)_{n \geq 0}$ is Hahn summable: the supports lie in $\langle \text{supp } \eta \rangle$, which is well ordered, and Neumann's all-lengths clause gives only finitely many contributing word lengths at each exponent. This is stronger than merely observing that the leading valuations increase. The case $\eta = 0$ is immediate. Hence A is a *germ-analytic* function on $\mathfrak{m}$, and its formally differentiated series is its actual fine ε - δ derivative, by the germ-algebra isomorphism 6.9. At every nonzero ordinary complex argument h the displayed classical power series diverges, the ratio of successive absolute values being $(n+1)^2|h|$. The germ-analytic class is therefore strictly larger than the class of Taylor lifts of ordinary analytic germs. No growth condition on the coefficients was used, and none is available to be used.

Example 14.2 (A germ with no coherent representative in the original chart). Let $E(X) = \sum_{n \geq 0} n! X^n$. Its germ at 0 is germ-analytic, by the same computation. But there is *no* connected ordinary neighbourhood U of 0 and *no* $F = \sum_{\gamma \in S} f_\gamma t^\gamma \in \mathcal{H}(U)$ whose evaluated germ at 0 is this germ. Indeed, equality of analytic germs would give

$$\sum_{\gamma \in S} t^\gamma \frac{f_\gamma^{(n)}(0)}{n!} = n! \quad (n \geq 0),$$

and uniqueness of Hahn coefficients forces $f_\gamma^{(n)}(0) = 0$ for every n and every $\gamma \neq 0$. Ordinary analyticity and connectedness of U then make every such f_γ identically zero, so the exponent-zero coefficient function would be an ordinary holomorphic function on a neighbourhood of 0 with Taylor series $\sum n! X^n$, of radius zero. No such function exists.

Remark 14.3 (The two statements are about different charts). Example 14.2 and the scale-embedding bridge 7.25 look contradictory and are not. The bridge asserts that after *one* monomial rescaling every formal series over $\mathbb{K}$ becomes a coherent family on all of $\mathbb{C}$; concretely,

$$E(tz) = \sum_{n \geq 0} n! t^n z^n$$

is a section of $\mathcal{H}(\mathbb{C})$ whose Hahn coefficient at the exponent n is the *monomial* $n! z^n$, an entire function. Example 14.2 concerns the *original* chart, where the exponent-zero coefficient function would have to carry all the growth itself. The rescaling changes which function is being represented. Scale is part of the analytic data, and by Remark 13.10 and Theorem 13.12 having some sufficiently small coherent chart carries no implication of a chart of every larger radius.

14.2 An exact infinitesimal Lambert-type root

Example 14.4 (Lambert-type solution, valid for every infinitesimal). Let $\eta \in \mathfrak{m}$ and consider the coherent equation

$$z - \eta(e^z)^\# = 0,$$

where e^z denotes the ordinary entire coefficient function. The leading coefficient function is z , with a simple zero at the origin and no other zero in a small disk, so by the zero-cluster theorem 9.9 the equation has exactly one root in the halo $\mathfrak{m}$, counted with multiplicity. Lagrange–Bürmann inversion 6.29 gives that root explicitly:

$$z = \sum_{n \geq 1} \frac{n^{n-1}}{n!} \eta^n = \eta + \eta^2 + \frac{3}{2}\eta^3 + \frac{8}{3}\eta^4 + \frac{125}{24}\eta^5 + \frac{54}{5}\eta^6 + \cdots. \quad (14.1)$$

The sum is Hahn summable and solves the equation exactly, by formal substitution, for *every* surcomplex infinitesimal η , regardless of the complexity of its positive support. The identity (14.1) was verified symbolically through degree 40. It asserts nothing about possible infinite roots of an equation formed with a globally chosen exponential of 12.7.

14.3 Preparation at a double root

Example 14.5 (First preparation coefficients for $w^2 - t(e^w)^\#$). Take $A(w) = w^2 - t(e^w)^\#$ with $t = \omega^{-1}$, so the leading coefficient function is w^2 and $m = 2$. Put

$$B(w) = \frac{e^w - 1 - w}{w^2},$$

with its removable value $1/2$ at $w = 0$. The first homogeneous terms of the preparation recursion 9.3, graded by positive-support word degree in t , are

$$p^{[1]} = -t(1 + w), \quad q^{[1]} = -tB(w),$$

and

$$p^{[2]} = -t^2\left(\frac{1}{2} + \frac{2}{3}w\right),$$

because $p^{[1]}q^{[1]} = t^2(1 + w)B(w)$, whose constant and linear ordinary Taylor coefficients are $1/2$ and $2/3$. The prepared monic polynomial therefore begins

$$P(w) = w^2 - t(1 + w) - t^2\left(\frac{1}{2} + \frac{2}{3}w\right) + \cdots,$$

and its two roots in $\mathbb{K}$ are precisely the two roots of $w^2 - t(e^w)^\#$ in $\mathfrak{m}$, counted with multiplicity. The recursion computes them without assuming an a priori square-root splitting of the perturbation; 77 exact coefficients were computed and checked.

14.4 A cubic, by both constructive algorithms

Example 14.6 (Perturbed double zero, an escaping third root, and contour moments). Let $t = \omega^{-1}$ and

$$F(Z) = Z^2 - t + tZ^3 \in \mathcal{H}(\mathbb{C}), \quad (14.2)$$

a coherent section with support $\{0, 1\}$ and polynomial coefficient functions. Three quantities must be kept apart here, and the example exists to separate them: the algebraic degree of F over $\mathbb{K}$, which is 3; the ordinary zero count of the leading coefficient function $f_0(Z) = Z^2$ on the disk $|Z| < 1/2$, which is 2; and the number of surcomplex zeros of $F^\#$ in the halo of that disk, which by the zero-cluster theorem 9.9 is exactly 2.

(a) *The preparation algorithm.* Preparation 9.3 gives $F = UW$ with W monic of degree 2. The homogeneous pieces, all obtainable by polynomial division by Z^2 alone, are

$$\begin{aligned} p_1 = -t, \quad q_1 = tZ, \quad p_2 = t^2Z, \quad q_3 = -t^3, \\ p_4 = -t^4, \quad p_5 = 2t^5Z, \quad q_6 = -2t^6, \quad p_7 = -3t^7, \end{aligned}$$

the unlisted earlier pieces being zero. Assembling them,

$$W(Z) = Z^2 + (t^2 + 2t^5 + 7t^8 + 30t^{11} + \dots)Z - (t + t^4 + 3t^7 + 12t^{10} + \dots), \quad (14.3)$$

$$U(Z) = 1 + tZ - t^3 - 2t^6 + \dots. \quad (14.4)$$

Substituting $Z = sw$ with $s = t^{1/2}$ turns the root equation into $w^2(1 + s^3w) = 1$, whose two finite branches give

$$\xi_\pm = \pm t^{1/2} - \frac{1}{2}t^2 \pm \frac{5}{8}t^{7/2} - t^5 + \dots. \quad (14.5)$$

The third root of (14.2) over $\mathbb{K}$ is infinite: the sum of all three roots is $-t^{-1}$, so

$$\xi_\infty = -t^{-1} + t^2 + 2t^5 + \dots.$$

A higher-degree term of positive valuation can therefore add a root outside the finite halo without changing the zero count inside it.

(b) *The contour-moment algorithm.* Let Γ be the ordinary circle $|w| = 1/2$, positively oriented. The exact identities are

$$\frac{1}{2\pi i} \oint_{\Gamma}^H \frac{F'}{F} dw = 2, \quad \frac{1}{2\pi i} \oint_{\Gamma}^H w \frac{F'}{F} dw = \xi_+ + \xi_- = -t^2 - 2t^5 - \dots,$$

the first an ordinary integer with no infinitesimal error term, as 10.10 requires. The higher moments $s_k = \frac{1}{2\pi i} \oint_{\Gamma}^H w^k F'/F dw = \sum_j \xi_j^k$ together with Newton's identities 9.17 recover every coefficient of W in (14.3). Thus the contour detects the infinitesimal displacement of the cluster's centre even though both roots have standard part zero, and the two algorithms produce the same W by independent routes.

14.5 Scales beyond every finite power

Example 14.7 (An exponent beyond all integral powers of t). Let $t = \omega^{-1}$ and $F(Z) = Z^2 - t - t^\omega$. Since $t^\omega/t = t^{\omega-1} \in \mathfrak{m}$, the binomial family is Hahn summable and the roots are exact:

$$\xi_\pm = \pm t^{1/2} \sum_{n \geq 0} \binom{1/2}{n} t^{n(\omega-1)}, \quad (14.6)$$

whose square is $t(1 + t^{\omega-1}) = t + t^{\omega}$. The quantity t^{ω} is smaller than every positive ordinary integral power of t . No truncation of (14.6) at a finite t -order captures its role, and neither does any *sequence* of such truncations: by 4.2 a set-indexed convergent net in the fine topology is eventually equal to its limit, so there is no limiting process to appeal to. The support formulation includes t^{ω} without any change to the proofs of preparation or zero counting. A variant with the same moral is $F(Z) = Z^2 - t - t^{\omega}Z$, whose roots are

$$\lambda_{\pm} = \frac{t^{\omega}}{2} \pm t^{1/2} \left(1 + \frac{t^{2\omega-1}}{4}\right)^{1/2},$$

the square root being expanded at 1; the root count is two regardless of the scale separation.

Example 14.8 (Two roots in one monad at incomparable scales). Let $\tau = \omega^{-1}$, $\sigma = \omega^{-\omega}$, so $\sigma \prec \tau^n$ for every positive integer n , and let $P(z) = z^2 - \tau z - \sigma$. Put $q = \sigma/\tau^2 \in \mathfrak{m}$. The quadratic formula and the binomial expansion give

$$z_+ = \tau(1 + q - q^2 + 2q^3 - 5q^4 + \cdots), \quad z_- = -\tau(q - q^2 + 2q^3 - 5q^4 + \cdots),$$

so that $v(z_+) = 1$ and $v(z_-) = \omega - 1$. Both roots lie in the *same* monad, the monad of 0, yet one is smaller than every finite power of the scale of the other. The zero-cluster theorem accommodates this with no Archimedean or rank-one restriction on the value class.

Example 14.9 (Two genuinely separated surreal scales, to second order). Let

$$\lambda = \omega^{-1/2}, \quad s = \omega^{-\omega}, \quad F(z) = z^2 - \lambda^2 - s(e^z)^{\#},$$

so the perturbation scale s is smaller than every positive ordinary power of λ . The leading coefficient function z^2 gives exactly two roots in $\mathfrak{m}$. Set $z = \lambda w$ and divide by λ^2 : the normalized leading coefficient function is $w^2 - 1$, so one root lies in the monad of $w = 1$ and one in the monad of $w = -1$. Writing $r = \lambda$ or $r = -\lambda$,

$$z_r = r + s \frac{e^r}{2r} + s^2 \frac{e^{2r}(2r-1)}{8r^3} \pmod{s^3}, \quad (14.7)$$

where e^r is the infinitesimal Taylor exponential. To verify this, put $z = r + sa + s^2b$ and use $r^2 = \lambda^2$; the coefficients of s and s^2 in $F(z)$ are

$$2ra - e^r, \quad a^2 + 2rb - e^r a,$$

which vanish exactly for $a = e^r/(2r)$ and $b = e^{2r}(2r-1)/(8r^3)$. Higher coefficients are determined the same way.

Remark 14.10 ((14.7) is an s -adic statement). The congruence in (14.7) takes place in $\mathbb{C}((\lambda))[[s]]$, embedded in $\mathbb{K}$ by the ordered exponents $n\omega + k/2$. It says that the difference lies in the ideal generated by s^3 . It is *not* an assertion that the remainder is bounded by a fixed real multiple of s^3 , and no such Archimedean reading is available: there is no norm on $\mathcal{H}(U)$ anywhere in this article.

14.6 Poles off the ordinary plane, and residues

Example 14.11 (A pole that moves off the ordinary complex plane). Let $\varepsilon \in \mathfrak{m} \setminus \{0\}$ and $H(z) = 1/(z - \varepsilon)$. This function has exactly one actual pole, at the surcomplex point ε , and $\text{Res}_{\varepsilon}(H(z) dz) = 1$. On any ordinary circle about 0,

$$H(z) = \sum_{n \geq 0} \varepsilon^n z^{-n-1},$$

a Hahn summable family whose coefficient functions have poles at the *ordinary* point 0 of arbitrarily large order; only the term $n = 0$ contributes an ordinary residue there. Hence

$$\operatorname{Res}_0^H H = 1, \quad \frac{1}{2\pi i} \int_{|z|=\rho}^H \frac{dz}{z - \varepsilon} = 1 \quad (\rho > 0 \text{ ordinary}).$$

But the *local* residue at the exact point 0 is

$$\operatorname{Res}_0(H(z) dz) = 0,$$

because on the fine neighborhood $\varepsilon \mathbf{m}$ one has the expansion $H(\zeta) = -\varepsilon^{-1} \sum_{n \geq 0} (\zeta/\varepsilon)^n$, with no pole at $\zeta = 0$ at all. So $\operatorname{Res}_0^H H = 1 = \operatorname{Res}_\varepsilon H$ while $\operatorname{Res}_0 H = 0$: the cluster residue at an ordinary centre and the local residue at that same point are numerically different. This is the simplest instance of Bridge 1, 10.8 — the cluster residue equals the finite sum of actual residues in the monad — and it is the standing reason the three residue symbols may never be merged. Note also that the criterion of Bridge 2, 10.11, fails here exactly as it must: the coefficient attached to ε^n has a pole of order $n + 1$ at 0, so pole orders are not uniformly bounded in the Hahn exponent, and Res_0^H is under no obligation to agree with Res_0 .

Example 14.12 (Split poles with infinite residues and a finite sum). Let $t = \omega^{-1}$ and

$$H(z) = \frac{(\exp)^\#(z)}{z^2 - t},$$

with the standard unit circle positively oriented. The poles in the finite halo are $p_\pm = \pm\sqrt{t}$, with

$$\operatorname{Res}_{p_+} H = \frac{(\exp)^\#(\sqrt{t})}{2\sqrt{t}}, \quad \operatorname{Res}_{p_-} H = -\frac{(\exp)^\#(-\sqrt{t})}{2\sqrt{t}}.$$

Each of these is infinite in modulus, of valuation $-1/2$, yet their sum is finite:

$$\operatorname{Res}_{p_+} H + \operatorname{Res}_{p_-} H = \frac{(\sinh)^\#(\sqrt{t})}{\sqrt{t}} = \sum_{n \geq 0} \frac{t^n}{(2n+1)!}.$$

Therefore

$$\oint_{|z|=1}^H \frac{e^z}{z^2 - t} dz = 2\pi i \sum_{n \geq 0} \frac{t^n}{(2n+1)!}, \quad (14.8)$$

where in the integrand e^z denotes the ordinary coefficient function. Expanding instead $(z^2 - t)^{-1} = \sum_{n \geq 0} t^n z^{-2n-2}$ on the contour and taking ordinary residues coefficientwise gives the same answer. The example therefore compares two genuinely different organizations of one exact result: split poles at half-integral scales, summed as $\operatorname{Res}_\cdot$, and an integral-power Hahn expansion at the ordinary centre, summed as Res_0^H . Their agreement is Bridge 1, not a coincidence, and the cancellation of the two infinite summands is exact rather than asymptotic.

14.7 A Cauchy formula at an infinite centre

Example 14.13 (Infinite centre, hyper-small radius, surcomplex coefficient). Choose $a = \omega + i\omega^2$ and $s = \omega^{-\omega}$, and let

$$F(a + su) = (\sin)^\#(u) + t^{\sqrt{2}} (e^u)^\#, \quad u \in B(0, 2)^\#,$$

a coherent datum in the chart of Definition 13.1 with support $\{0, \sqrt{2}\}$. For every infinitesimal η , the affine circle $\Gamma = a + s\{u : |u| = 1\}$ satisfies

$$F(a + s\eta) = \frac{1}{2\pi i} \int_{\Gamma}^H \frac{F(z)}{z - (a + s\eta)} dz,$$

the right-hand side being the coefficientwise integral (13.1) in the stated chart. The example combines an infinite centre, a radius smaller than every finite power of ω^{-1} , and a genuinely surcomplex coefficient $t^{\sqrt{2}}$, none of which the theory treats as exceptional. A companion at an ordinary centre with a displaced evaluation point is

$$\frac{1}{2\pi i} \int_{|z|=1}^H \frac{e^z + t/(2-z)}{z - a'} dz = e^{1/3} \sum_{n \geq 0} \frac{\varepsilon^n}{n!} + \frac{t}{5/3 - \varepsilon}, \quad a' = \frac{1}{3} + \varepsilon, \quad \varepsilon \in \mathfrak{m},$$

exact because the joint support is well ordered and each exponent receives only finitely many contributions.

14.8 No growth hypothesis on the coefficient family

Example 14.14 (Arbitrary factorial growth in the scale index). Let

$$F(Z) = Z^2 - t + \sum_{n \geq 2} (n!)^2 t^n e^{nZ}.$$

This is a member of $\mathcal{H}(\mathbb{C})$: its support is a subset of $\mathbb{N}$, hence well ordered with finite multiplicity at each exponent, and every coefficient function is entire. On any ordinary circle centred at zero the leading coefficient function Z^2 is nonzero, so $F^\#$ has exactly two zeros in the corresponding halo, counted with multiplicity, and its preparation polynomial and contour moments exist as Hahn-summed surcomplex coefficients. No positive ordinary radius of convergence in a numerical parameter replacing t is required, and no boundedness of the family $((n!)^2)_n$ is used.

Contrast this with Example 14.2. Growth in the *scale* index γ is entirely unrestricted, because the summability condition is about supports and not about magnitudes. Growth in the ordinary *Taylor* index of a single coefficient function is exactly what the obstruction of Example 14.2 forbids, because each f_γ must be an ordinary holomorphic function on a common ordinary domain.

14.9 The permitted and forbidden support boundary

Example 14.15 (An arbitrary well-ordered family of perturbation scales). Let $E \subset \mathbf{No}^{>0}$ be any well-ordered set, let $(c_\gamma)_{\gamma \in E}$ be arbitrary ordinary complex constants, and let $m \geq 1$. Then

$$F(z) = z^m + \sum_{\gamma \in E} t^\gamma e^{c_\gamma z}$$

is a coherent section on the finite plane, and $F^\#$ has exactly m roots in $\mathfrak{m}$, counted with multiplicity. No boundedness condition on the family (c_γ) is needed or available. The positivity and well ordering of E , however, are essential: a decreasing sequence of exponents tending to 0 is not a permitted Hahn support. Thus $\{1/n : n \geq 1\}$ is excluded, while the increasing set $\{1 - 1/n : n \geq 2\}$ is a perfectly legal support. The separating elementary example is the one recorded with 3.1: $\sum_{n \geq 0} 2^{-n}$ is *not* Hahn summable, because infinitely many of its terms sit at the single exponent 0; its ordinary complex sum must be computed first and used as a single coefficient.

14.10 What the verification script checks, and what it does not

The nine preserved companion scripts together perform the finite verifications cited above, in exact rational and symbolic arithmetic, with no floating point. It checks: the Lambert-type series (14.1) and its substitution into the defining equation through degree 40; the preparation identity of Example 14.5 in a finite bivariate truncation, with 77 exact coefficients; the preparation recursion for the cubic (14.2), the cancellation of the product error $F - UW$ to the requested

order, the root expansions (14.5), and the low-order contour moments; the multiscale binomial identity (14.6); the second-order two-scale root residual (14.7); and the split-pole residue cancellation (14.8).

Remark 14.16 (Standing disclaimer on the finite checks). These calculations test signs, factors and coefficients in finite truncations of displayed formulas. They certify nothing about the class-theoretic foundations, nothing about the summability arguments, and nothing about any unrestricted theorem statement; in particular they are not a machine-checked development, and no proof assistant has been used. The all-order and general-support arguments are the proofs in the body, not the script.

Remark 14.17 (What is constructive, and in what sense). Preparation and reversion are constructive whenever the input supplies the coefficient functions together with an effective presentation of the relevant supports. A finite computation is organized by positive-support word degree: to reach degree N one runs the preparation recursion for $1 \leq n \leq N$, and at each degree only finite products and ordinary Taylor division operators occur; with finitely many formal perturbation generators, the degree- n coefficient is a finite sum of analytic expressions indexed by binary trees. For a completely arbitrary surreal support the existence theorem is *not* a promise that all coefficients can be enumerated by a finite program. Moreover, in higher rank a bound on the valuation need not leave only finitely many terms, so an algorithm must specify a finite support region or an explicit truncation scheme, rather than merely saying “compute below precision t^δ .”

15 Boundaries: what is and is not established

The qualifications collected here are results, not defensive padding. Each of them is the exact hypothesis that separates a true statement from a false one, and in several cases a counterexample in the body shows that dropping it produces a false theorem. They are stated per-theorem in the body and consolidated here so that no reader has to reconstruct the pattern.

15.1 Which class each classical principle lives in

In the table, *germ-analytic* means a Hahn summable power series at one point with arbitrary surcomplex coefficients; *coherent* means a member of $\mathcal{H}(U)$ with the stated ordinary domain conditions; *lift* means the canonical classical lift $f^\#$ of an ordinary holomorphic f , that is, a coherent section supported at the exponent 0; and *all-scale* means Definition 13.11.

Classical principle	Class in which it is true	The qualification that cannot be dropped
Power-series realization and Taylor calculus	Germ-analytic	A sufficiently small scale; Hahn sums, not sequential convergence. No radius-of-convergence formula exists from ordinary coefficient magnitudes.
Cauchy–Riemann equations	Germ-analytic	The equations on a fine neighbourhood. Existence of two partial derivatives at a point does not imply complex differentiability.
Inverse, implicit and ramification theorems	Germ-analytic; coherent local inverse in one variable	Invertible Jacobian for the germ inverse/implicit theorem; finite order for ramification. The coherent one-variable inverse is obtained after a scale change. No multivariable coherent implicit theorem is proved here.

Classical principle	Class in which it is true	The qualification that cannot be dropped
Open mapping and local maximum modulus	Germ-analytic locally; coherent globally	Valid at arbitrary surcomplex points, via the monomial normalization radius $r = t^\rho$. In global form both fail for merely germ-analytic functions, witnessed by L of the locally constant witness of Section 5.
Identity theorem	Coherent, U connected	Fails in the germ-analytic class: L vanishes on a nonempty fine-open class and is not zero. Accumulation is of <i>standard parts</i> in $\mathbb{C}$; no set-sized fine-convergent sequence is claimed.
Cauchy, Morera, primitives	Coherent, with $\int^H$	Integrals are coefficientwise, on ordinary contours or in transported affine charts. Uniqueness of a primitive up to a constant holds <i>inside the coherent category only</i> .
Preparation, division, Rouché	Coherent	Leading coefficient function holomorphic near $\bar{\Omega}$ and nonzero on $\Gamma = \partial\Omega$. Rouché's strict inequality is an ordinary complex margin, or carries an ordinary real margin $q < 1$.
Argument principle, exact zero counts, Bridge 1	Quotients of coherent families	Nonvanishing boundary reduction; finite zero clusters. The index is an ordinary integer with <i>no</i> infinitesimal error term.
Coefficientwise residue theorem	$\mathcal{M}(U)$: Hahn series with meromorphic coefficients	Poles may move with the exponent and infinitely many ordinary centres may contribute. <i>No</i> conclusion about actual surcomplex poles transfers to this class.
Liouville	See §15.3	Three inequivalent hypotheses with three different conclusions; a single surreal bound on $\mathbb{C}^\#$ is vacuous.
Removable singularities	Coherent	Coefficientwise removal is necessary and sufficient; a scalar bound is insufficient (t/Z , $t(e^{1/z})^\#$). A punctured halo deletes a whole <i>monad</i> , not a point.
Schwarz lemma and Schwarz–Pick	Lifts $f^\#$	False for general coherent self-maps of $\mathbb{D}^\#$: $(1+t)z$ fixes 0, maps $\mathbb{D}^\#$ into itself, and has $ F'(0) = 1+t > 1$. The same-monad case needs the divided difference.
Riemann mapping	Lifts $f^\#$	Halos of ordinary simply connected proper domains only. The existence input is the <i>ordinary</i> Riemann mapping theorem; no uniformization of arbitrary surcomplex open classes is claimed.
Montel / normal families	Coherent with <i>countable</i> support	Convergence is a topology on coefficient data, not pointwise all-radii convergence of values. Countability is not cosmetic.
Global logarithm	Germ-analytic on $\mathbb{K}^\times$	Exists globally, with branch chosen by a fine-locally constant quantity; it is <i>not</i> a coherent primitive of $1/z$ on $(\mathbb{C}^\times)^\#$.
Classification of entire and meromorphic functions	All-scale coherent	Exactly the polynomials, respectively the rational functions. This rigid category excludes Exp and every twist E_λ .

15.2 The analytic categories, and what each one buys

Category	Defining structure	Conclusions and restrictions
Differentiable	A bare ε - δ complex difference quotient over all positive surreal radii; <i>no</i> power-series requirement	Uniqueness of the derivative and the sum, product and chain rules, and the counterexamples. No regularity theorem asserting a power-series representation from bare differentiability is proved here.
Germ-analytic	A Hahn summable power series at one point with arbitrary surcomplex coefficients, realized near the centre; no coefficient growth condition	The full formal Taylor calculus, including classically divergent series; the germ algebra at any point is $\mathbb{K}[[X]]$ as a differential $\mathbb{K}$ -algebra. Local identity, openness at nonconstant germs and primitive uniqueness hold; the corresponding unrestricted global propagation fails.
Canonical lift $f^\#$	Taylor extension of one ordinary holomorphic f ; a coherent section supported at exponent 0	Classical continuation; the sharp Schwarz lemma and Schwarz–Pick theorem, which are false one level up. Creates no infinitesimally displaced zeros, so surcomplex coefficients are necessary for a genuine perturbation.
Coherent, $\mathcal{H}(U)$	One common connected ordinary domain carrying one ordinary holomorphic coefficient function per exponent, with well-ordered set support	Faithful evaluation on $U^\#$, identity, Cauchy and residue calculus, preparation and division, exact zero counts, open mapping and local inverse theorems, coherent primitives. This is not sheaf-coherence in the sense of complex geometry.
All-scale coherent	A coherent representative on a fixed ordinary disk after <i>every</i> positive surreal dilation, supports free to vary with the scale	Exactly the finite-degree polynomials (13.12); exactly the rational functions in the meromorphic version (13.19). A rigidity condition, not a proposed definition of “entire”.
Canonical global exponential and its twists	The real surreal exponential plus a normal-form phase convention on the purely infinite imaginary part	A surjective germ-analytic homomorphism with $\ker \text{Exp} = 2\pi i \mathbf{Oz} = i(\mathbf{J} + 2\pi\mathbb{Z})$; locally coherent in finite translated charts, but not all-scale coherent, by Remark 13.7.

The inclusions are

$$\{f^\#\} \subsetneq \mathcal{H}(U) \subsetneq \{\text{germ-analytic}\}|_{U^\#} \subseteq \{\text{differentiable}\}.$$

Only the first two strict inclusions are established here: positive support gives a section that is not a lift, and Example 14.2 gives the obstruction to coherence. The locally constant witness q of Example 5.7 gives a function on a whole halo. Strictness of the last inclusion is not proved; L is germ-analytic and cannot witness it. The restriction of an all-scale Hahn-entire function to any bounded halo is coherent there, and that containment is strict as well, since Exp restricted to a finite halo is coherent while Exp is not all-scale coherent.

15.3 The three hierarchies that must never be flattened

Three function classes. The identity theorem, the global maximum principle, Liouville’s theorem, uniqueness of primitives and the open mapping theorem have coherent versions

under the stated domain hypotheses; Liouville in particular requires bounded entire coefficient functions. Their unrestricted global forms fail in the germ-analytic class. The sharp Schwarz and Schwarz–Pick inequalities hold for lifts $f^\#$ and *fail* in $\mathcal{H}(\mathbb{D})$. Results in the germ-analytic class do not descend to $\mathcal{H}(U)$: $\sum n! \varepsilon^n$ is a legitimate germ (Example 14.2) and is not the restriction of any member of $\mathcal{H}(U)$ on an ordinary neighbourhood of 0 with those derivatives. The one bridge that has been proved, and that must be invoked explicitly whenever a result is moved between the two derivatives, is that for a coherent section the coefficientwise derivative equals the fine ε – δ derivative.

Three residues. Res is formal, the coefficient of X^{-1} in $\mathbb{K}((X))$ with a finite principal part, at any surcomplex centre. Res_a^H is the cluster residue at an *ordinary* centre a , the Hahn sum of the ordinary residues of the coefficient functions. Res_b is the actual local residue at a surcomplex point b . They are not interchangeable: Example 14.11 gives $\text{Res}_0^H = 1$ and $\text{Res}_0 = 0$ for the same function at the same point. Bridge 1 identifies Res_a^H with the finite sum of the Res. over the actual poles in $a + \mathfrak{m}$, for quotients of coherent families only; Bridge 2 identifies Res_a^H with Res at the exact point a when pole orders are uniformly bounded in the Hahn exponent. This is a sufficient condition, not a necessary condition for numerical equality. On the larger class $\mathcal{M}(U)$ there need be no actual pole at all, so Res. has no meaning there and the finite-sum conclusion must not be imported.

Three Liouville hypotheses. (a) A single *surreal* bound on the finite halo is vacuous: every nonzero $F \in \mathcal{H}(\mathbb{C})$ already satisfies $|F^\#| < t^{v_H(F)-1}$ there. (b) A single *ordinary real* bound is exactly characterized and does *not* give constancy: $|F^\#(z)| \leq M$ on the finite plane for an ordinary real M if and only if $F = c + R$ with $c \in \mathbb{C}$ and R of positive support, so the algebra of real-bounded macroscopically entire functions is precisely $\mathbb{C} \oplus \mathcal{H}_{>0}(\mathbb{C})$. The function $F = tz$ is real-bounded by 1 on the whole finite plane and is not constant; it is $0 + tz$, exactly as that characterization predicts. (c) *Coefficientwise* bounds give constancy, and coefficientwise polynomial growth of degree d gives a polynomial of degree at most d . Importing (c)’s conclusion under hypothesis (b) or (a) is false. Conversely the constant $F = \omega$ satisfies coefficientwise boundedness and has no ordinary-real bound; (b) and the boundedness case of (c) are incomparable. Note finally that the counterexamples tz and εZ are a different obstruction from the locally constant L : they are coherent, on the halo of the ordinary plane, whereas L is germ-analytic on all of $\mathbb{K}$ and restricts to the coherent constant 1 on each finite halo.

Two Cauchy estimates, and two logarithms. One must not read the warning that a single coarse surreal bound does not control every *coefficient function* — which is true, and witnessed by $F(w) = w(1 + it)$ — as forbidding the ML inequality, which bounds the surreal *modulus* of a derivative and is proved from the positivity of the coefficientwise integral. The two are about different quantities and both are retained; see Remark 13.9. Similarly, the global fine-analytic logarithm on $\mathbb{K}^\times$ and the principal logarithm on the cut plane $\mathbb{K} \setminus \mathbf{No}_{\leq 0}$ are *different functions*, differing by exactly $2\pi i$ on the lower half of the monad of every negative ordinary real; both are legitimate branches and both are named separately.

15.4 A practical proof checklist

A proposed surcomplex analytic argument should identify, explicitly:

- (1) its coefficient field, and whether the work happens inside a set-sized K_Γ or in all of $\mathbb{K}$;
- (2) its set support, and the well-ordering and finiteness clauses that make it one;
- (3) its summability convention, and where the finiteness clause is used;

- (4) the topology appearing in any limit, and whether that limit is a Hahn sum rather than a limit of partial sums;
- (5) the coherence mechanism connecting its local formulas, or the explicit absence of one;
- (6) for a contour argument, whether integration is ordinary, coefficientwise, definable, or a separately constructed surreal integral;
- (7) when counting zeros, whether multiplicities refer to ordinary leading coefficient functions, to actual surcomplex roots, or to formal polynomial factors — the prepared setting connects all three exactly;
- (8) which of Res , Res_a^H , Res_b is meant;
- (9) which of the three Liouville hypotheses is being assumed;
- (10) which of the two logarithms is in use.

15.5 Assertions deliberately not made

The following distinctions delimit the proved statements; none of the stronger assertions they exclude is being assumed.

- There is no radius-of-convergence formula based on ordinary complex coefficient magnitudes for arbitrary surcomplex coefficients.
- A Hahn sum is not a fine-topological limit of its ordinary partial sums.
- Agreement at ordinary points does not determine a germ-analytic function.
- A real-parameter contour is not a fine-continuous path: every continuous map $[0, 1] \rightarrow \mathbb{K}$ is constant, so an ordinary circle is not a path in $\mathbb{K}$.
- A bound in the ordered field does not bound every coefficient function. (This does not forbid the ML inequality; see §15.3.)
- The canonical single-valued exponential on all of $\mathbb{K}$ is constructed using a specified infinite-phase convention, but $E' = E$, $E(0) = 1$, agreement with $\exp_{\mathbf{No}}$ and $\exp_{\mathbb{C}}$, and the modulus law do not determine the infinite imaginary phase.
- No unrestricted global contour integral for arbitrary class functions is constructed, and the coefficientwise functional $\int^H$ is not proposed as the unique notion of surreal integration.
- No universal analytic continuation across scales is claimed, and no unrestricted gluing or chart-independence theorem across surreal centres and scales is asserted (Remark 13.4).
- No general theory of essential singularities or Picard theorem is proved for arbitrary germ-analytic functions. The specific value-distribution result for $\text{Exp}(1/z)$ in 12.17 also holds for the twists $E_\lambda(1/z)$, as proved in Remark 12.18.
- No composition law compatible with the Berarducci [9]–Mantova [11] field derivation is claimed on $\mathbf{No}$; and the variable derivative D_z , which fixes every surreal coefficient, is provably incompatible with a field derivation normalized by $\partial\omega = 1$ (6.13).
- Nothing asserts that a maximum of the modulus is *attained* on a closed surcomplex disk. The missing compactness is precisely why the maximum principle yields no extremal-value theorem.

- Nothing asserts the existence of a nontrivial set-sized fine-convergent sequence; the discreteness theorem forbids it.
- The discreteness, realization and rigidity theorems are theorems about $\mathbb{K}$, not about an arbitrary fixed set-sized Hahn subfield with only its internal radii: their proofs use positive surreal bounds that need not belong to such a subfield. Nor does the fine topology induced on a nontrivially valued set-sized Hahn subfield coincide with that subfield's own valuation topology (the trivially valued case is discrete for both).
- No uniformization of arbitrary surcomplex open classes is claimed; the Riemann mapping corollary takes the ordinary theorem as input.

15.6 A concrete summability boundary at an essential singularity

It is worth exhibiting exactly where the theory stops, rather than merely saying that it does.

The canonical lift $(e^{1/z})^\#$ is coherent on $(\mathbb{C}^\times)^\#$, but that domain excludes the entire monad of 0. At a nonzero infinitesimal z the formal expression

$$\sum_{n \geq 0} \frac{z^{-n}}{n!}$$

is *not* Hahn summable: its term valuations $-n v(z)$ form a strictly decreasing sequence, so their union is not well ordered. Thus the ordinary Laurent series does not itself define any extension into that monad.

One may instead choose a global exponential E_α of 12.7 and form $E_\alpha(1/z)$. This is a germ-analytic extension on all of $\mathbb{K}^\times$, but its values depend on the phase convention: at $z = -i/\omega$ one has $1/z = i\omega$, and the values for $\alpha = 0$ and $\alpha = \pi$ differ, since $E_0(i\omega) = 1$ while $E_\pi(i\omega) = -1$. Restriction to ordinary nonzero complex points and the ordinary Laurent series do not remove this ambiguity.

Accordingly, what has been proved is a finite-order meromorphic theory and a coherent contour theory, not a universal classification of essential singularities on all surreal scales. A Casorati–Weierstrass or Picard statement must specify its domain, its topology and its extension operator before it is even a well-posed target. The value-distribution theorem 12.17 is an exact statement about a specified exponential and, by Remark 12.18, its explicit twists. It is not a general Picard theorem.

15.7 Research directions, with their prerequisites

These are proposals for further development, not theorems proved here, and not claims that their broad subjects are unstudied.

Non-affine coherent atlases. The affine chart construction of §13.1 is completely explicit. The next task is to specify classes of nonlinear chart transitions preserving local Hahn support control, and to prove that their coefficientwise contour functionals agree on admissible overlaps. The overlap conditions must be formulated *before* asserting any theory of surcomplex Riemann surfaces: arbitrary topological gluing permits the locally constant counterexamples and is too weak. The difficult point is not local differentiation, which already works, but compatibility between different summation descriptions over a common ordinary base. Theorem 13.12 shows the direction is forced: a global transcendental theory must weaken all-scale coherence, because retaining it admits only polynomials even when the support is allowed to vary with the scale.

Relative preparation in several variables. The local realization, Cauchy–Riemann, inverse and implicit arguments already work in finitely many variables. The coherent inverse theorem proved here is one-dimensional; a multivariable coherent implicit-function theorem requires an additional statement and proof over a specified common ordinary parameter domain (Remark 6.22). A relative preparation theorem, with one distinguished polynomial variable and coherent parameter domains, requires a coefficientwise division operator that preserves the chosen parameter domain, together with one common parameter domain for all coefficient functions. The one-variable Hermite division used here is not by itself a proof of that generalization; the positive-support recursion identifies exactly where those inputs would enter.

Broader meromorphic and cohomological theories. The residue theory covers quotients of coherent families with a controlled boundary reduction, and $\mathcal{M}(U)$ with moving poles. Allowing arbitrary meromorphic coefficient families, essential singularities or infinitely many poles requires explicit locally finite pole and support conditions. Cohomological statements would require an admissible-cover or coefficient-domain formalism; the ordinary connectedness of the fine topology supplies nothing, since $\mathbb{K}$ is totally disconnected.

Essential singularities and resummation. A natural problem is to determine which resurgent complex functions admit sectorial surcomplex extensions with compatible residue and monodromy data. Allowing two-sided formal expansions requires new summability hypotheses: a support harmless in one variable may cease to be well ordered after evaluation, as §15.6 shows. Equality of formal Taylor coefficients is not sufficient asymptotic data.

Normal families and approximation. These should be formulated in a coefficient topology, or in an explicitly fixed Hahn workspace. In the full fine topology every set-indexed convergent net is eventually equal to its limit, so an ordinary sequential Montel theorem would be the wrong statement; the coefficientwise Montel theorem 11.21 is the correctly shaped one, and its countable-support hypothesis is not cosmetic. Quantitative control of coefficient growth and of support changes is the appropriate starting point.

Effective support presentations. For a finitely generated positive support, the preparation recursion and the contour moments give implementable exact algorithms up to a specified finite support cutoff. Termination at a requested cut is a separate algorithmic question, and in higher rank a valuation bound need not leave finitely many terms. General surreal coefficients are not finite input data, and the unrestricted construction should not be called an executable algorithm without first specifying an effective subfield and a support language.

15.8 Status of the arguments, and relation to existing work

The following are used as established inputs and are cited rather than claimed: the surreal cut property and the normal-form theorem; real closedness of **No** and the algebraic closedness of $\mathbb{K}$ together with the Hahn-field results that supply it; Neumann’s support lemma; Alling [2]’s foundations of analysis over surreal number fields and the restricted analytic structure of van den Dries and Ehrlich [12]; the existence, group law and kernel of the canonical surcomplex exponential, due to Ehrlich and Kaplan [20, Theorem 11.1]; and ordinary complex analysis on the ordinary domains, which enters only through the coefficient functions and the canonical lifts, never through an unstated transfer principle to all of $\mathbb{K}$.

The coefficientwise contour functional is not asserted to coincide with every integration operator in the surreal integration literature. Where two constructions agree coefficientwise on a common Hahn expansion, their agreement can be checked directly; outside that overlap an additional comparison theorem would be needed, and none is proved here. Likewise, work

on transseries as germs of surreal functions, and on Taylor expansions over generalized power series, addresses composition and derivation questions that should not be conflated with the fact that a formal series in an independent variable has a sufficiently small surreal realization.

This article is a mathematical research exposition with proofs. It has not been externally refereed, and neither it nor its foundational inputs have been machine-checked in a proof assistant; the finite checks of §14.10 have been executed and are not a substitute for that. No exhaustive novelty or priority determination has been made for any formulation, and none is asserted.

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This list is the union of the reference lists of the nine source manuscripts: 94 bibliography entries naming 44 distinct works. It is reproduced in full so that the merged document does not lose the literature those manuscripts consulted, and the comment on each entry records which of the nine cited it. Only some are cited in the text above; the others were consulted by one or more of the sources for background. The historical provenance record is `MERGE_NOTES.md`; it is not a current certification of every bibliographic or mathematical comparison made by those sources.

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